FUNCTIONS, POLYNOMIALS, AND SEQUENCES

VERSION 20180315 EDITED BY HOJOO LEE

Dedicated to Paul Erdős (1913 – 1996) and Jedi

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Luke: But I need your help. I've come back to complete the training.
Yoda: **No more training, do you require. Already know you that which you need.**
Luke: Then I am a Jedi.
Yoda: **Not yet. One thing remains: Vader. You must... confront... Vader.**

Star Wars: Episode VI – Return of the Jedi (1983)

Notations

$\mathbb{Z}$	the set of integers
$\mathbb{N}$	the set of positive integers
$\mathbb{N}_0$	the set of nonnegative integers
$\mathbb{Q}$	the set of rational numbers
$\mathbb{Q}^+$	the set of positive rational numbers
$\mathbb{R}$	the set of real numbers
$\mathbb{R}^+$	the set of positive real numbers
$\mathbb{C}$	the set of complex numbers

Resources

- **American Mathematical Monthly** https://www.maa.org/press/periodicals
- **Art of Problem Solving** https://artofproblemsolving.com/community
- **Asian Pacific Mathematics Olympiad** http://www.apmo-official.org
- **Crux Mathematicorum** https://cms.math.ca/crux
- **International Mathematical Olympiad** https://www.imo-official.org
- **IMO Compendium** http://www.imomath.com
- **Középiskolai Matematikai és Fizikai Lapok** https://www.komal.hu
- **Mathematics Magazine** https://www.maa.org/press/periodicals
- **William Lowell Putnam Mathematics Competition** http://kskedlaya.org/putnam-archive

1. Polynomials in $\mathbb{Z}[x]$ and $\mathbb{Q}[x]$

Logic will get you from A to B. Imagination will take you everywhere. – Albert Einstein

Z 1. Prove that the polynomial $f(x) = x^5 - x + a$, where a is an integer number which is not divisible by 5, cannot be written as a product of two polynomials with lower degree.

Bulgarian Mathematical Olympiad 1969

Z 2. Given the polynomial $f(x) = x^n + a_1x^{n-1} + a_2x^{n-2} + \cdots + a_{n-1}x + a_n \in \mathbb{Z}[x]$, there exist four distinct integers a, b, c and d such that
$$f(a) = f(b) = f(c) = f(d) = 5.$$
Show that there is no integer k such that $f(k) = 8$.

Canadian Mathematical Olympiad 1970

Z 3. Let P be a non-constant polynomial with integer coefficients. If $n(P)$ is the number of distinct integers k such that $(P(k))^2 = 1$, prove that Prove that $n(P) - \deg(P) \le 2.$, where $\deg(P)$ denotes the degree of the polynomial P.

IMO 1974 Problem 6

Z 4. Let $P(x)$ be a polynomial of degree n such that
$$P(k) = \frac{k}{k+1}$$
for all $k = 0, 1, 2, \cdots, n$. Determine $P(n+1)$.

United States of America Mathematical Olympiad 1975

Z 5. The polynomial $1976(x + x^2 + \cdots + x^n)$ is decomposed into a sum of polynomials of the form
$$a_1x + a_2x^2 + \cdots + a_nx^n,$$
where $a_1, a_2, \cdots, a_n$ are distinct positive integers not greater than n. Find all values of n for which such a decomposition is possible.

IMO Shortlist 1976

Z 6. Show that for $n > 2$ the polynomial
$$x^n - x^{n-1} - x^{n-2} - \cdots - x - 1$$
is irreducible over the rationals.

Amer. Math. Monthly 1983 Problem E 3008
Proposed by Roger Cuculière

Z 7. Can the polynomials $x^5 - x - 1$ and $x^2 + ax + b$, where a and b are rational numbers, have common complex roots?

Bulgarian Mathematical Olympiad 1983 (Fourth Round)

Z 8. Let p and $q \neq 0$ be integers. Show that there exists an interval $\mathbf{I}$ of length $\frac{1}{q}$ and a polynomial P with integral coefficients such that

$$\left|P(x) - \frac{p}{q}\right| < \frac{1}{q^2}$$

for all $x \in \mathbf{I}$.

IMO Shortlist 1983

Z 9. Let $f(x)$ be a non-constant polynomial with integer coefficients and n, k be natural numbers. Show that there exist n consecutive natural numbers a, $a+1$, $\cdots$, $a+n1$ such that the numbers $f(a)$, $f(a+1)$, $\cdots$, $f(a+n1)$ all have at least k prime factors.

Bulgarian Mathematical Olympiad 1985 (Fourth Round)

Z 10. For any polynomial $P(x) = a_0 + a_1 x + \cdots + a_k x^k$ with integer coefficients, the number of odd coefficients is denoted by $\omega(P)$. For $i = 0, 1, 2, \cdots$ let $Q_i(x) = (1+x)^i$. Prove that if $i_1, i_2, \cdots, i_n$ are integers such that $0 \leq i_1 < i_2 < \cdots < i_n$, then

$$\omega(Q_{i_1} + Q_{i_2} + \cdots + Q_{i_n}) \geq \omega(Q_{i_1}).$$

IMO 1985 Problem 3

Z 11. Let k be the smallest positive integer for which there exist distinct integers m_1, m_2, m_3, m_4, m_5 such that the polynomial

$$p(x) = (x - m_1)(x - m_2)(x - m_3)(x - m_4)(x - m_5)$$

has exactly k nonzero coefficients. Find, with proof, a set of integers m_1, m_2, m_3, m_4, m_5 for which this minimum k is achieved.

William Lowell Putnam Competition 1985

Z 12. Let $f(x) = x^n + a_1 x^{n-1} + \cdots + a_n \in \mathbb{R}[x]$ be a polynomial with $n \geq 3$ real roots. Suppose that $\frac{a_{n-1}}{a_n} > n+1$ and that $a_{n2} = 0$. Prove that at least one root of $f(x)$ lies in the open interval $\left(-\frac{1}{2}, \frac{1}{n+1}\right)$.

Bulgarian Mathematical Olympiad 1987 (Fourth Round)

Z 13. Let n be a positive integer. Find the number of odd coefficients of the polynomial

$$u_n(x) = \left(x^2 + x + 1\right)^n.$$

IMO Shortlist 1988

Z 14. Suppose P is a given polynomial with integer coefficients and degree greater than one. Let

$$W = \{w \mid w \text{ is an irrational real number but } P(w) \text{ is rational}\}.$$

Prove that W is dense on the real line.

Amer. Math. Monthly 1989 Problem E 3336
Proposed by Nick MacKinnon

Z 15. Let $\overline{a_n a_{n-1} \cdots a_1 a_0}$ be the decimal representation of a prime positive integer such that $n > 1$ and $a_n > 1$. Prove that the polynomial $P(x) = a_n x^n + \cdots + a_1 x + a_0$ cannot be written as a product of two non-constant integer polynomials.

Balkan Mathematical Olympiad 1989

Z 16. Let p be a real number and $f(x) = x^p - x + p$.

(1) Prove that every root $\alpha \in \mathbb{C}$ of $f(x)$ satisfies the estimation $|\alpha| < p^{\frac{1}{p-1}}$.
(2) Prove that if p is a prime number, then $f(x)$ cannot be written as the product of two non-constant polynomials with integer coefficients.

Bulgarian Mathematical Olympiad 1989 (Fourth Round)

Z 17. Let $n = p_1 p_2 \cdots p_s$, where $p_1, \cdots, p_s$ are distinct odd prime numbers.

(1) Prove that
$$F_n(x) = \prod_{\{p_{i_1},\cdots,p_{i_k}\}\subset\{p_1,\cdots,p_s\}} \left(x^{\frac{n}{p_{i_1}\cdots p_{i_k}}} - 1\right)^{(-1)^k}$$
can be written as a polynomial in x with integer coefficients.
(2) Prove that if p is a prime divisor of $F_n(2)$, then either $p \,|\, n$ or $n \,|\, p-1$.

Bulgarian Mathematical Olympiad 1990 (Fourth Round)

Z 18. Let $p(x)$ be a cubic polynomial with rational coefficients. q_1, q_2, q_3, ... is a sequence of rationals such that $q_n = p(q_{n+1})$ for all positive n. Show that for some k, we have $q_{n+k} = q_n$ for all positive n.

IMO Shortlist 1990

Z 19. Find all pairs of cubic polynomials $x^3 + ax^2 + bx + c$ and $x^3 + bx^2 + ax + c$, where a and b are positive integers and c is a nonzero integer, so that they have three integer roots each, exactly one of which is common.

Crux Mathematicorum 1994 Problem 1992
Proposed by K. R. S. Sastry

Z 20. Let $f(x) = x^8 + 4x^6 + 2x^4 + 28x^2 + 1$. Let $p > 3$ be a prime and suppose there exists an integer z such that p divides $f(z)$. Prove that there exist integers $z_1, z_2, \cdots, z_8$ such that if
$$g(x) = (x - z_1)(x - z_2) \cdot \cdots \cdot (x - z_8),$$
then all coefficients of $f(x) - g(x)$ are divisible by p.

IMO Shortlist 1992

Z 21. Let n be an integer greater than 1 and let $a_1, a_2, \cdots, a_n$ be different integers. Prove that the polynomial
$$f(x) = (x - a_1)(x - a_2) \cdots (x - a_n) - 1$$
is not divisible by any monic polynomial of positive degree less than n and with integer coefficients.

Nordic Mathematical Contest 1992

Z 22. Let $n > 1$ be an integer and let $f(x) = x^n + 5x^{n-1} + 3$. Prove that there do not exist polynomials $g(x)$, $h(x)$ each having integer coefficients and degree at least one, such that $f(x) = g(x)h(x)$.

IMO 1993 Problem 1

Z 23. Let $f(x)$ and $g(x)$ be two polynomials with real coefficients such that for infinitely many rational values x, $\frac{f(x)}{g(x)}$ is rational. Prove that $\frac{f(x)}{g(x)}$ can be written as the ratio of two polynomials with rational coefficients.

Iranian Mathematical Olympiad 1993

Z 24. Let $f(x)$ be a polynomial having rational coefficients and degree at least 2. Suppose that $a_1, a_2, a_3, \cdots$ is a sequence of rational numbers such that
$$f(a_{n+1}) = a_n$$
for all $n \geq 1$. Prove that there exists $k \geq 1$ such that
$$a_{n+k} = a_n$$
for all $n \geq 1$.

Amer. Math. Monthly 1994 Problem 10369
Proposed by Bjorn Poonen

Z 25. Given distinct prime numbers p and q and a natural number $n \geq 3$, find all $a \in \mathbb{Z}$ such that the polynomial $f(x) = x^n + ax^{n-1} + pq$ can be factored into 2 integral polynomials of degree at least 1.

China Team Selection Test 1994

Z 26. Let n be a positive integer. Show that the only integral polynomials of degree less than n that are real and nonnegative at all n-th roots of unity and have constant term 1 are of the form
$$1 + x^d + x^{2d} + \cdots + x^{n-d}$$
with $d|n$, or
$$1 - x^d + x^{2d} - \cdots - x^{n-d}$$
with $2d|n$.

Amer. Math. Monthly 1995 Problem 10492
Proposed by William Duke

Z 27. If $F(x)$ and $G(x)$ are polynomials with integer coefficients such that
$$\frac{F(k)}{G(k)}$$
is an integer for all $k \in \mathbb{N}$, prove that the polynomial $G(x)$ divides the polynomial $F(x)$.

Crux Mathematicorum 1995 Klamkin Quickies 9

Z 28. Find all polynomials f such that $f(p)$ is a power of two for every prime p.

Crux Mathematicorum 1995 Problem 2087
Proposed by TobyGee

Z 29. Let $m_1, m_2, ..., m_n$ be mutually distinct integers. Prove that there exists a polynomial $f(x) \in \mathbb{Z}[x]$ of degree n satisfying the following conditions.

- $f(m_i) = -1$ for all $i = 1, 2, ..., n$.
- $f(x)$ is irreducible.

Taiwanese Mathematical Olympiad 1995

Z 30. Suppose $q_0, q_1, q_2, \cdots$ is an infinite sequence of integers satisfying the following conditions.

- $m-n$ divides $q_m - q_n$ for $m > n \geq 0$.
- There exists a polynomial P such that $|q_n| < P(n)$ for all n.

Prove that there exists a polynomial Q such that $q_n = Q(n)$ for all n.

United States of America Mathematical Olympiad 1995

Z 31. Find all integers k such that for infinitely many integers $n \geq 3$ the polynomial
$$P(x) = x^{n+1} + kx^n - 870x^2 + 1945x + 1995$$
is reducible in $\mathbb{Z}[x]$.

Vietnam Team Selection Test 1995

Z 32. Let $P \in \mathbb{Q}[x]$ be a polynomial such that $P^{-1}(\mathbb{Q}) \subset \mathbb{Q}$. Show that the polynomial P is linear.

Iranian Mathematical Olympiad (Final Round) 1996

Z 33. Prove that the equation
$$x^2 + y^2 + z^2 + 3(x+y+z) + 5 = 0$$
has no solution in rational numbers.

Bulgarian Mathematical Olympiad 1997

Z 34. Find all positive integers k for which the following statement is true: If $F(x) \in \mathbb{Z}[x]$ is a polynomial satisfying the condition $0 \leq F(c) \leq k$ for each $c \in \{0, 1, \cdots, k+1\}$, then $F(0) = F(1) = \cdots = F(k+1)$.

IMO Shortlist 1997

Z 35. Let p be a prime number and f an integer polynomial of degree d such that $f(0) = 0$, $f(1) = 1$ and $f(n)$ is congruent to 0 or 1 modulo p for every integer n. Prove that $d \geq p-1$.

IMO Shortlist 1997

Z 36. Let $P(x) \in \mathbb{Z}[x]$ be a polynomial such that $P(a)P(b) = -(a-b)^2$ for some integers a and b. Prove that $P(a) + P(b) = 0$.

Iranian Mathematical Olympiad 1997

Z 37. Let m and n be positive integers. Show that the Maclaurin series
$$f(x) = \frac{2}{\sqrt{3}} \sqrt{\frac{1-mx}{nx^3}} \sin\left(\frac{1}{3} \arcsin\left(\frac{3\sqrt{3}}{2} \sqrt{\frac{nx^3}{(1-mx)^3}}\right)\right)$$
has integer coefficients.

Mathematics Magazine 1997 Problem 1523
Proposed by Emeric Deutsch

Z 38. Let $P(x), Q(x)$ be monic irreducible polynomials with rational coefficients. Suppose that $P(x)$ and $Q(x)$ have roots α and β respectively, such that $\alpha + \beta$ is rational. Prove that $P(x)^2 - Q(x)^2$ has a rational root.

Romania Team Selection Test 1997
Proposed by Bogdan Enescu

Z 39. Prove that for any integer n, there exists a unique polynomial Q with coefficients in $\{0, 1, \cdots, 9\}$ such that $Q(-2) = Q(-5) = n$.

United States of America Mathematical Olympiad 1997

Z 40. Prove that for all positive integers n, the polynomial

$$p_n(x) = x^n - 2$$

is irreducible in $\mathbb{Q}[x]$.

Wai Ling Yee, Powers of Two, Crux Mathematicorum (24) 1998, no. 1, 33 – 36.

Z 41. The coefficients of a polynomial p are integers whose absolute values are not greater than 1998. Given that $p(2000)$ is a prime number, prove that p cannot be written as a product of two polynomials, each of positive degree and of integer coefficients.

Középiskolai Matematikai és Fizikai Lapok, January 1998, N. 161

Z 42. Find all monic polynomials $P(x) \in \mathbb{Z}[x]$ such that there exist infinitely irrational numbers a such that $P(a)$ is a positive integer.

Vietnam Team Selection Test 1998

Z 43. Prove that for every natural number n there exists a polynomial $p(x) \in \mathbb{Z}[x]$ such that $p(1)$, $p(2)$, $\cdots$, $p(n)$ are distinct powers of 2.

Iranian Mathematical Olympiad 1999

Z 44. Prove that, for any positive integer n, there exists a polynomial p of degree at most n whose coefficients are all integers such that, $p(x)$ is divisible by 2^n for every even integer x, and $p(x) - 1$ is divisible by 2^n for every odd integer x.

Középiskolai Matematikai és Fizikai Lapok, September 1999, A. 216
Proposed by Hajnal Péter

Z 45. Give an example of a nonzero polynomial of integral coefficients which admits $\cos 18^\circ$ as a root.

Középiskolai Matematikai és Fizikai Lapok, December 1999, B. 3327

Z 46. Let a, n be integer numbers, p a prime number such that $p > |a|+1$. Prove that the polynomial $f(x) = x^n + ax + p$ cannot be represented as a product of two integer polynomials.

Romania Team Selection Test 1999
Proposed by Laurentiu Panaitopol

Z 47. Prove that for every positive integer a one can find a positive integer n and polynomials $R(x), Q(x) \in \mathbb{Z}[x]$ such that

$$x^{n-1} + x^{n-2} + x^{n-3} + \cdots + x + 1 = [1 + ax + x^2 R(x)]Q(x).$$

Turkish Mathematical Olympiad 2000

Z 48. Let $f(x)$ be a polynomial of integer coefficients, p and q coprime integers such that $q \neq 0$. Prove that if $\frac{p}{q}$ is a root of the polynomial, then $f(k0)$ is divisible by $p - kq$ for every integer k. Is the converse of the statement also true?

Középiskolai Matematikai és Fizikai Lapok, December 2001, B. 3507

Z 49. For each integer m, consider the polynomial
$$P_m(x) = x^4 - (2m+4)x^2 + (m-2)^2.$$
For what values of m, is $P_m(x)$ the product of two non-constant polynomials with integer coefficients?

William Lowell Putnam Competition 2001

Z 50. Two players alternately replace the stars in the expression
$$\star x^{2000} + \star x^{1999} + \cdots + \star x + 1$$
by real numbers. The player who makes the last move loses if the resulting polynomial has a real root t with $|t| < 1$, and wins otherwise. Give a winning strategy for one of the players.

Vietnam Team Selection Test 2000

Z 51. Let n be a positive integer and $f(x) = a_m x^m + \cdots + a_1 x + a_0$, with $m \geq 2$, a polynomial with integer coefficients such that:

- $a_2, a_3 \cdots a_m$ are divisible by all prime factors of n,
- a_1 and n are relatively prime.

Prove that for any positive integer k, there exists a positive integer c, such that $f(c)$ is divisible by n^k.

Romania Team Selection Test 2001

Z 52. Let $p(x) \in \mathbb{Q}[x]$ be a polynomial of degree at least 2. Suppose that there exists a sequence $r_1, r_2, \cdots$ of rational numbers such that
$$r_n = p(r_{n+1})$$
for all $n \in \mathbb{N}$. Prove that the sequence $r_1, r_2, \cdots$ is periodic.

Hungary-Israel Binational Mathematical Competition 2002

Z 53. Let P be a cubic polynomial given by $P(x) = ax^3 + bx^2 + cx + d$, where a, b, c, d are integers and $a \neq 0$. Suppose that $xP(x) = yP(y)$ for infinitely many pairs x, y of integers with $x \neq y$. Prove that the equation $P(x) = 0$ has an integer root.

IMO Shortlist 2002 A3

Z 54. Is there a non-constant polynomial of integer coefficients, such that its value is a power of 2 at every positive integer?

Középiskolai Matematikai és Fizikai Lapok, May 2002, B. 3558
Inspired by a problem of the National Competition, 2002

Z 55. Find all polynomials $P(x) \in \mathbb{Z}[x]$ such that the polynomial
$$Q(x) = \left(x^2 + 6x + 10\right) P(x)^2 - 1$$
is the square of a polynomial with integer coefficients.

Vietnam Team Selection Test 2002

Z 56. Let $P(x)$ and $Q(x)$ be integer polynomials of degree p and q respectively. Assume that $P(x)$ divides $Q(x)$ and all their coefficients are either 1 or 2002. Show that $p+1$ is a divisor of $q+1$.

Romania Team Selection Test 2002
Proposed by Mihai Cipu

Z 57. Let $f(x)$ and $g(x)$ be polynomials with non-negative integer coefficients, and let m be the largest coefficient of f. Suppose that there exist natural numbers $a < b$ such that $f(a) = g(a)$ and $f(b) = g(b)$. Show that if $b > m$, then $f = g$.

All-Russian Mathematical Olympiad 2003

Z 58. The side lengths of a triangle are the roots of a cubic polynomial with rational coefficients. Prove that the altitudes of this triangle are roots of a polynomial of sixth degree with rational coefficients.

All-Russian Mathematical Olympiad 2003

Z 59. Determine all polynomials $P(x) \in \mathbb{Z}[x]$ such that, for any positive integer n, the equation $P(x) = 2^n$ has an integer root.

Bulgarian Mathematical Olympiad 2003

Z 60. Let $p(x)$ be a polynomial with integer coefficients and let n be an integer. Suppose that there is a positive integer k for which $f^{(k)}(n) = n$, where $f^{(k)}(x)$ is the polynomial obtained as the composition of k polynomials f. Prove that $p(p(n)) = n$.

Italy Team Selection Test 2003

Z 61. Is there a 2003rd-degree polynomial $f(x)$ of integer coefficients, such that the values of $f(n)$, $f(f(n))$, $f(f(f(n)))$, $\cdots$ are pairwise relative primes for every integer n?

Középiskolai Matematikai és Fizikai Lapok, January 2003, B. 3609

Z 62. We define the polynomial
$$\mathbf{s}_n(x) = \sum_{d|n} \frac{n}{d} x^d.$$
Define the sequence of polynomials $\mathbf{p}_0(x) = 1$, $\mathbf{p}_1(x)$, $\mathbf{p}_2(x)$, $\cdots$, by the recursion
$$\mathbf{p}_n(x) = \frac{1}{n} \sum_{d|n} \mathbf{s}_k(x)\mathbf{p}_{n-k}(x).$$
Prove that the coefficients of the polynomial $\mathbf{p}_n(x)$ are all integers.

Középiskolai Matematikai és Fizikai Lapok, January 2003, A. 310

Z 63. Find all a, b, m, $n \in \mathbb{Z}$ with $m > n > 1$ for which the polynomial $f(X) = X^n + aX + b$ divides the polynomial $g(X) = X^m + aX + b$.

Romania Team Selection Test 2003
Proposed by Laurentiu Panaitopol

Z 64. Find all polynomials $p(x) \in \mathbb{Z}[x]$ such that $\gcd(p(m), p(n)) = 1$ whenever $\gcd(m, n) = 1$.

Iranian Mathematical Olympiad 2004

Z 65. Let p be an odd prime, and let $P(x) = a_0 + a_1 x + \cdots + a_{p-1}x^{p-1}$ be a polynomial of degree $p-1$ with integral coefficients such that $p \nmid P(b) - P(a)$ whenever a and b are integers such that $p \nmid b - a$. Prove that $p \mid a_{p-1}$.

Mathematics Magazine 2004 Problem 1688
Proposed by Mihai Manea

Z 66. Let $P(x) = c_n x^n + c_{n-1}x^{n-1} + \cdots + c_0$ be a polynomial with integer coefficients. Suppose that r is a rational number such that $P(r) = 0$. Show that the n numbers

$$c_n r,\ c_n r^2 + c_{n-1} r,\ c_n r^3 + c_{n-1} r^2 + c_{n-2} r, \cdots,\ c_n r^n + c_{n-1} r^{n-1} + \cdots + c_1 r$$

are integers.

William Lowell Putnam Competition 2004

Z 67. For any positive integer n, prove that there exists a polynomial P of degree n such that all coefficients of this polynomial P are integers, and such that the numbers $P(0)$, $P(1)$, $P(2)$, $\cdots$, $P(n)$ are pairwise distinct powers of 2.

Germany Team Selection Test 2005

Z 68. Let the polynomial $P(x) = x^3 + 19x^2 + 94x + a$ where $a \in \mathbb{N}$. If p a prime number, prove that no more than three numbers of the numbers $P(0), P(1), \cdots, P(p-1)$ are divisible by p.

Greece Team Selection Test 2005

Z 69. Find all pairs of integers a, b for which there exists a polynomial $P(x) \in \mathbb{Z}[X]$ such that

$$(x^2 + ax + b)\, P(x) = x^n + c_{n-1}x^{n-1} + \cdots + c_1 x + c_0,$$

where $c_0, c_1, \cdots, c_{n-1} \in \{-1, 1\}$.

IMO Shortlist 2005 A1

Z 70. Let $P(x) = a_n x^n + a_{n-1}x^{n-1} + \cdots + a_0 \in \mathbb{Z}[x]$, where $a_n > 0$ and $n \geq 2$. Prove that there exists a positive integer m such that $P(m!)$ is a composite number.

IMO Shortlist 2005 N7

Z 71. Let α be a fixed irrational number and let P be a polynomial with integer coefficients and with $\deg(P) > 1$. Prove that there are infinitely many pairs of integers such that $P(m) = \lfloor \alpha n \rfloor$.

Mathematics Magazine 2005 Problem 1734
Proposed by H. A. ShahAli

Z 72. Let A be the set of all polynomials $f(x)$ of order 3 with integer coefficients and cubic coefficient 1, so that for every $f(x)$ there exists a prime number p which does not divide 2004 and a number q which is coprime to p and 2004, so that $f(p) = 2004$ and $f(q) = 0$. Prove that there exists a infinite subset $B \subset A$, so that the function graphs of the members of B are identical except of translations.

Mediterranean Mathematics Olympiad 2005

Z 73. Assume that the polynomial $(x+1)^n - 1$ is divisible by some polynomial

$$P(x) = x^k + c_{k-1}x^{k-1} + c_{k-2}x^{k-2} + \ldots + c_1 x + c_0,$$

whose degree k is even and whose coefficients c_{k-1}, c_{k-2}, ..., c_1, c_0 all are odd integers. Show that $k + 1 \mid n$.

All-Russian Mathematical Olympiad 2006

Z 74. Find all second degree polynomial $d(x) = x^2 + ax + b$ with integer coefficients, so that there exists a polynomial $p(x) \in \mathbb{Z}[x]$ and a non-zero integer coefficient polynomial $q(x)$ such that

$$(p(x))^2 - d(x)\,(q(x))^2 = 1.$$

China Team Selection Test 2006

Z 75. Let $P(x)$ be a polynomial of degree $n > 1$ with integer coefficients and let k be a positive integer. Consider the polynomial $Q(x) = P(P(\cdots P(P(x))\cdots))$, where P occurs k times. Prove that there are at most n integers t such that $Q(t) = t$.

IMO Shortlist 2006 N4, IMO 2006 Problem 5

Z 76. Let $P(x) \in \mathbb{Z}[x]$ be a polynomial of degree $n > 1$. Let k be a positive integer. Consider the polynomial

$$Q(x) = P(P(\cdots P(P(x))\cdots)),$$

where P occurs k times. Prove that there are at most n integers t such that $Q(t) = t$.

IMO 2006 Problem 5
Proposed by Dan Schwarz (Romania)

Z 77. Does there exist a a sequence $a_0, a_1, a_2, \cdots$ in $\mathbb{N}$ satisfying the following conditions?

- $\gcd(a_i, a_j) = 1$ for $i \neq j$.
- The polynomial $\sum_{i=0}^{n} a_i x^i$ is irreducible in $\mathbb{Z}[x]$.

Iran Team Selection Test 2007

Z 78. Find all pairs $f(x)$, $g(x)$ of polynomials with integer coefficients such that

$$f(g(x)) = x^{2007} + 2x + 1.$$

Középiskolai Matematikai és Fizikai Lapok, May 2007, A. 429
Proposed by Katalin Gyarmati

Z 79.

(1) Find all infinite arithmetic progressions formed with positive integers such that there exists a number $N \in \mathbb{N}$, such that for any prime p, $p > N$, the p-th term of the progression is also prime.
(2) Find all polynomials $f(x) \in \mathbb{Z}[x]$, such that there exist $N \in \mathbb{N}$, such that for any prime p, $p > N$, $|f(p)|$ is also prime.

Romania Team Selection Test 2007

Z 80. Let $f(x) = x^n + a_{n-1}x^{n-1} + \cdots + a_1 x + a_0 \in \mathbb{Z}[x]$ be a polynomial of degree $n \geq 3$ such that $a_k + a_{n-k}$ is even for all $k \in \{1, \cdots, n-1\}$ and a_0 is even. Suppose that $f(x) = g(x)h(x) \in \mathbb{Z}[x]$ for some polynomials with $\deg g \leq \deg h$ and that all the coefficients of $h(x)$ are odd. Prove that $f(x)$ has an integer root.

Romania Team Selection Test 2007

Z 81. For a polynomial $P(x) \in \mathbb{Z}[x]$, $r(2i-1)$ (for $i = 1, 2, 3, \cdots, 512$) is the remainder obtained when $P(2i-1)$ is divided by 1024. The sequence

$$r(1),\ r(3),\ \cdots, r(1023)$$

is called the remainder sequence of $P(x)$. A remainder sequence is called complete if it is a permutation of

$$1,\ 3,\ 5,\ \cdots,\ 1023.$$

Prove that there are no more than 2^{35} different complete remainder sequences.

USA Team Selection Test 2007

Z 82. Let n be an integer greater than 1, and let f_n be the polynomial given by

$$\left(\begin{array}{c} f_n(x) = \sum_{i=0}^{n} .n \\ i(-x)^{n-i} \prod_{j=0}^{i-1} (x+j) \end{array} \right)$$

Find the degree of f_n.

Amer. Math. Monthly 2008 Problem 11403
Proposed by Yaming Yu

Z 83. Find all polynomials p of one variable with integer coefficients such that if a and b are natural numbers such that $a+b$ is a perfect square, then $p(a) + p(b)$ is also a perfect square.

Iran Team Selection Test 2008

Z 84. Let $P(x)$ be a polynomial with integer coefficients such that $P\left(n^2\right) = 0$ for some integers $n \neq 0$. Prove that $P\left(a^2\right) \neq 1$ for all rational numbers $a \neq 0$.

Japanese Mathematical Olympiad 2008

Z 85. Let n be a positive integer, and let $a_1, a_2, \cdots, a_n$ be pairwise different integers. Prove that the polynomial

$$(x-a_1)\cdots(x-a_n)+1$$

cannot be expressed as a product of two polynomials of integer coefficients whose degrees are at least one.

Középiskolai Matematikai és Fizikai Lapok, September 2008, B. 4111

Z 86. Let n be a positive integer. Given an integer coefficient polynomial $f(x)$, define its signature modulo n to be the (ordered) sequence $f(1), \cdots, f(n)$ modulo n. Of the n^n such n-term sequences of integers modulo n, how many are the signature of some polynomial $f(x)$ if

(1) n is a positive integer not divisible by the square of a prime.
(2) n is a positive integer not divisible by the cube of a prime.

USA Team Selection Test 2008

Z 87. Let p be a prime number. Let $h(x)$ be a polynomial with integer coefficients such that

$$h(0), h(1), \cdots, h\left(p^2-1\right)$$

are distinct modulo p^2. Show that

$$h(0), h(1), \cdots, h\left(p^3-1\right)$$

are distinct modulo p^3.

William Lowell Putnam Competition 2008

Z 88. Does there exist a non-constant polynomial which always takes non-square integers at integer values of the variable?

Középiskolai Matematikai és Fizikai Lapok, October 2009, A. 489

Z 89. Show that there is no polynomial of rational coefficients that takes a non-integer value at exactly one integer. Is there a polynomial of this property with real coefficients?

Középiskolai Matematikai és Fizikai Lapok, October 2009, B. 4211
Proposed by P. Maga

Z 90. On a faded piece of paper it is possible to read the following:

$$(x^2+x+a)(x^{15}-\cdots)=x^{17}+x^{13}+x^5-90x^4+x-90.$$

Some parts have got lost, partly the constant term of the first factor of the left side, partly the majority of the summands of the second factor. It would be possible to restore the polynomial forming the other factor, but we restrict ourselves to asking the following question:

What is the value of the constant term a?

We assume that all polynomials in the statement have only integer coefficients.

Nordic Mathematics Contest 2009

Z 91. Let n be a even positive integer and let $P(x)$ be a monic polynomial with integral coefficients of degree n with n real solutions, not necessarily distinct. Let y be a positive real number such that $P(x)>0$ every $x<y$. Show that

$$P(0)^{\frac{1}{n}}-P(y)^{\frac{1}{n}}\geq y.$$

Italy Team Selection Test 2009

Z 92. Let $P(x), Q(x), R(x)\in\mathbb{Z}[x]$ be polynomials of degree at least 1 such that

$$P\left(Q(x)\right)=Q\left(R(x)\right)=R\left(P(x)\right).$$

Show that the polynomials $P(x)$, $Q(x)$, $R(x)$ are equal.

Polish Mathematical Olympiad 2009

Z 93. Let n and k be positive integers. Find all monic polynomials $f\in\mathbb{Z}[x]$, of degree n, such that $f(a)$ divides $f\left(2a^k\right)$ for $a\in\mathbb{Z}$ with $f(a)\neq 0$.

Romania Team Selection Test 2009

Z 94. Determine all prime numbers p such that there exists $Q(x)\in\mathbb{Z}[x]$ such that the polynomial

$$1+p+Q(x^1)Q(x^2)\cdots\ Q(x^{2p-2})$$

has an integral root.

Turkey Team Selection Test 2009

Z 95. Fix a prime number $p>5$. Let a,b,c be integers no two of which have their difference divisible by p. Let i,j,k be nonnegative integers such that $i+j+k$ is divisible by $p-1$. Suppose that for all integers x, the quantity

$$(x-a)(x-b)(x-c)\left[\,(x-a)^i\,(x-b)^j\,(x-c)^k\,-1\,\right]$$

is divisible by p. Prove that each of i,j,k must be divisible by $p-1$.

USA Team Selection Test 2009
Proposed by Kiran Kedlaya and Peter Shor

Z 96. Let a, b be integers, and let $P(x) = ax^3 + bx$. For any positive integer n we say that the pair (a, b) is n-good if $n \mid P(m) - P(k)$ implies $n \mid m - k$ for all integers m, k. We say that (a, b) is *very good* if (a, b) is n-good for infinitely many positive integers n.

(1) Find a pair (a, b) which is 51-good, but not very good.
(2) Show that all 2010-good pairs are very good.

IMO Shortlist 2010 N4
Proposed by Okan Tekman (Turkey)

Z 97. Let $p(x)$ be a polynomial with integer coefficients, and let w be a complex number of unit absolute value. Prove that if the number $c = p(w)$ is purely real, then there is a polynomial $q(x)$ with integer coefficients for which

$$c = q\left(2 + \frac{1}{w}\right).$$

Középiskolai Matematikai és Fizikai Lapok, January 2010, A. 498

Z 98. For what primes p does there exist a cubic polynomial f with integer coefficients with the property that p does not divide the leading coefficient of f and $f(1), f(2), \cdots, f(p)$ give pairwise distinct remainders modulo p?

Középiskolai Matematikai és Fizikai Lapok, September 2010, A. 513
Proposed by Péter Maga

Z 99. Let a and b denote positive integers. Given that p is a polynomial of integer coefficients that has both a value divisible by a and a value divisible by b at integer points. Prove that there is an integer point where the value of p is also divisible by the least common multiple of a and b.

Középiskolai Matematikai és Fizikai Lapok, September 2010, B. 4290

Z 100. Determine whether there exists a polynomial $f(x_1, x_2)$ with two variables, with integer coefficients, and two points $A = (a_1, a_2)$ and $B = (b_1, b_2)$ in the plane, satisfying the following conditions.

- A is an integer point (i.e a_1 and a_2 are integers).
- $|a_1 - b_1| + |a_2 - b_2| = 2010$.
- $f(n_1, n_2) > f(a_1, a_2)$ for all integer points (n_1, n_2) in the plane other than A.
- $f(x_1, x_2) > f(b_1, b_2)$ for all integer points (x_1, x_2) in the plane other than B.

Romanian Masters In Mathematics 2010
Proposed by Massimo Gobbino (Italy)

Z 101. Let p be a prime number, let $n_1, n_2, \cdots, n_p$ be positive integer numbers, and let d be the greatest common divisor of the numbers $n_1, n_2, \cdots, n_p$. Prove that the polynomial

$$\frac{x^{n_1} + x^{n_2} + \cdots + x^{n_p} - p}{x^d - 1}$$

is irreducible in $\mathbb{Q}[x]$.

Romania Team Selection Test 2010

Z 102. A nonconstant polynomial f with integral coefficients has the property that, for each prime p, there exist a prime q and a positive integer m such that $f(p) = q^m$. Prove that $f = x^n$ for some positive integer n.

Romania Team Selection Test 2010

Z 103. Let P be a polynomial with integer coefficients such that $P(0) = 0$ and

$$\gcd(P(0), P(1), P(2), \cdots) = 1.$$

Show there are infinitely many n such that

$$\gcd(P(n) - P(0), P(n+1) - P(1), P(n+2) - P(2), \cdots) = n.$$

USA Team Selection Test 2010

Z 104. For each integer k with $k \geq 2$, find all nonconstant $f \in \mathbb{Z}[x]$ such that for every prime p, $f(p)$ has no nontrivial kth power divisor.

Amer. Math. Monthly 2011 Problem 11563
Proposed by Vlad Matei

Z 105. Let n be a positive even integer and let p be prime. Show that the polynomial

$$f(x) = p + x + \cdots + x^n$$

is irreducible over $\mathbb{Q}$.

Amer. Math. Monthly 2011 Problem 11577
Proposed by Pál Peéter Dályay

Z 106. Let $P(x), Q(x) \in \mathbb{Z}[x]$ be polynomials such that $P(x) \mid Q(x)^2 + 1$.

(1) Prove that there exists polynomials $A(x), B(x) \in \mathbb{Q}[x]$ and a number $c \in \mathbb{Q}$ such that $P(x) = c(A(x)^2 + B(x)^2)$.
(2) Suppose that $P(x) \in \mathbb{Z}[x]$ is a monic polynomial. Prove that there exists two polynomials $A(x), B(x) \in \mathbb{Z}[x]$ such that $P(x) = A(x)^2 + B(x)^2$.

Iranian Mathematical Olympiad 2011
Proposed by Mohammad Gharakhani

Z 107. Let $f(x) = x^n + a_{n-1}x^{n-1} + \cdots + a_1 x + a_0 \in \mathbb{Z}[x]$, and let $d_1, \cdots, d_n$ be pairwise distinct integers. Suppose that, for infinitely many prime numbers p, there exists an integer k_p for which

$$f(k_p + d_1) \equiv \cdots \equiv f(k_p + d_n) \pmod{p}.$$

Prove that there exists an integer k_0 such that

$$f(k_0 + d_1) = \cdots = f(k_0 + d_n) = 0.$$

Középiskolai Matematikai és Fizikai Lapok, January 2011, A. 525

Z 108. Find a monic polynomial $f(x)$ with integer coefficients such that $f(x) = 0$ has no integer solutions but $f(x) \equiv 0 \pmod{p}$ has a solution for every prime p.

Mathematics Magazine 2011 Problem Q 1017
Proposed by Allan Berele, Jeffery Bergen

Z 109. Prove that there are no polynomials $f_1(x), f_2(x), f_3(x), f_4(x) \in \mathbb{Q}[x]$ satisfying
$$f_1(x)^2 + f_2(x)^2 + f_3(x)^2 + f_4(x)^2 = x^2 + 7.$$

Polish Mathematical Olympiad 2011

Z 110. Let f and g be two nonzero polynomials with integer coefficients and $\deg f > \deg g$. Suppose that for infinitely many primes p the polynomial $pf + g$ has a rational root. Prove that f has a rational root.

IMO Shortlist 2012 A4, Romania Team Selection Test 2013

Z 111. Find all polynomials f of integer coefficients such that for every positive integer n the prime factors of the numbers n and $f(n)$ coincide.

Középiskolai Matematikai és Fizikai Lapok, October 2012, B. 4481

Z 112. Is there an at least second-degree polynomial of integer coefficients, such that the numbers n, $f(n)$, $f(f(n))$, $\cdots$ are pairwise relative primes for all positive integers n?

Középiskolai Matematikai és Fizikai Lapok, December 2012, B. 4500

Z 113. Find all polynomials $P(x) \in \mathbb{Z}[x]$ such that for all positive integers n satisfies $P(n!) = |P(n)|!$.

Turkish Mathematical Olympiad 2012

Z 114. Consider two three-variable polynomials
$$P_n(x, y, z) = (x-y)^{2n}(y-z)^{2n} + (y-z)^{2n}(z-x)^{2n} + (z-x)^{2n}(x-y)^{2n}$$
and
$$Q_n(x, y, z) = [(x-y)^{2n} + (y-z)^{2n} + (z-x)^{2n}]^{2n}.$$
Determine all positive integers n such that the quotient $\frac{Q_n(x,y,z)}{P_n(x,y,z)}$ is a three-variable polynomial with rational coefficients.

USA Team Selection Test 2012

Z 115. Determine whether the polynomial
$$P(x) = (x^2 - 2x + 5)(x^2 - 4x + 20) + 1$$
is irreducible over $\mathbb{Z}[x]$.

Greece Team Selection Test 2013

Z 116. Let p be a fixed positive prime. For any polynomial $f(x)$ with integer coefficients, let $f^\star(x) = f((1+x)^p - 1)$. Find all polynomials g with integer coefficients for which the following holds: for every positive integer n there is a polynomial f with integer coefficients such that the coefficients of $g - (f - f^\star)$ with degree less than n are all divisible by p^n.

Középiskolai Matematikai és Fizikai Lapok, February 2013, A. 582
Proposed by Péter Maga

Z 117. Find all integers a for which there exists a polynomial $p(x) \in \mathbb{Z}[x]$ such that

$$p\left(a^{\frac{2}{3}} + a^{\frac{1}{3}}\right) = a^{\frac{2}{3}} - a^{\frac{1}{3}}.$$

Középiskolai Matematikai és Fizikai Lapok, May 2013, A. 590
Based on a problem of the Vojtech Jarník competition

Z 118. Determine if there exists a (three-variable) polynomial $P(x, y, z)$ with integer coefficients satisfying the property that a positive integer n is not a perfect square if and only if there is a triple (x, y, z) of positive integers such that $P(x, y, z) = n$.

USA Team Selection Test 2013

Z 119. Determine all polynomials $f \in \mathbb{Z}[x]$ with the property that for any two distinct primes p and q, $f(p)$ and $f(q)$ are relatively prime

Indian National Mathematical Olympiad Program 2014

Z 120. Find all polynomials $P(x) \in \mathbb{Z}[x]$ such that the set

$$P(\mathbb{Z}) = \{p(k) \mid k \in \mathbb{Z}\}$$

contains a geometric progression.

Iran Team Selection Test 2014

Z 121. There is given a prime number p and two positive integers k and n. Determine the smallest nonnegative integer d for which there exists a polynomial $f(x_1, \cdots, x_n)$ on n variables, with degree d and having integer coefficients that satisfies the following property that, for arbitrary $a_1, \cdots, a_n \in \{0, 1\}$, p divides $f(a_1, \cdots, a_n)$ if and only if p^k divides $a_1 + \cdots + a_n$.

Középiskolai Matematikai és Fizikai Lapok, February 2014, A. 610

Z 122. Let n be a positive integer. Determine the smallest possible value of

$$|p(1)|^2 + |p(2)|^2 + \cdots + |p(n+3)|^2$$

over all monic polynomials p with degree n.

Középiskolai Matematikai és Fizikai Lapok, March 2014, A. 611

Z 123. Let p be an odd prime number. Determine all pairs of polynomials f and g in $\mathbb{Z}[x]$ such that

$$f(g(x)) = \sum_{k=0}^{p-1} x^k$$

Romania Team Selection Test 2014

Z 124. Let $P_n(x) = 1 + 2x + 3x^2 + \cdots + nx^{n-1}$. Prove that the polynomials $P_j(x)$ and $P_k(x)$ are relatively prime for all positive integers j and k with $j \neq k$.

William Lowell Putnam Competition 2014

Z 125. Let $P(x) = x^4 - x^3 - 3x^2 - x + 1$. Prove that there are infinitely many positive integers n such that $P(3^n)$ is not a prime.

Mediterranean Mathematics Olympiad 2015

Z 126. Let n be a positive integer and let $k_0, k_1, \cdots, k_{2n}$ be nonzero integers such that $k_0 + k_1 + \cdots + k_{2n} \neq 0$. Is it always possible to a permutation $(a_0, a_1, \cdots, a_{2n})$ of $(k_0, k_1, \cdots, k_{2n})$ so that the equation

$$a_{2n}x^{2n} + a_{2n-1}x^{2n-1} + \cdots + a_0 = 0$$

has no integer solutions?

All-Russian Mathematical Olympiad 2016

Z 127. Find all polynomials $P(x) \in \mathbb{Z}[x]$ such that $P(P(n) + n)$ is a prime number for infinitely many integers n.

Canadian Mathematical Olympiad 2016

Z 128. The equation

$$(x-1)(x-2)\cdots(x-2016) = (x-1)(x-2)\cdots(x-2016)$$

is written on the board, with 2016 linear factors on each side. What is the least possible value of k for which it is possible to erase exactly k of these 4032 linear factors so that at least one factor remains on each side and the resulting equation has no real solutions?

IMO Shortlist 2016 A6, IMO 2016 Problem 5

Z 129. For any $k \in \mathbb{N}$, denote the sum of digits of k in its decimal representation by $S(k)$. Find all polynomials $P(x) \in \mathbb{Z}[x]$ such that for any positive integer $n \geq 2016$, the integer $P(n)$ is positive and $S(P(n)) = P(S(n))$.

IMO Shortlist 2016 N1
Proposed by Warut Suksompong (Thailand)

Z 130. Find all polynomials $P(x)$ of odd degree d and with integer coefficients satisfying the following property: for each positive integer n, there exists n positive integers $x_1, x_2, \cdots, x_n$ such that $\frac{1}{2} < \frac{P(x_i)}{P(x_j)} < 2$ and $\frac{P(x_i)}{P(x_j)}$ is the d-th power of a rational number for every pair of indices i and j with $1 \leq i, j \leq n$

IMO Shortlist 2016 N8

Z 131. Suppose that p is a polynomial of integer coefficients, not identically 0, such that $p(n)$ is divisible by 2016 for all integers n. What is the minimum possible value of the sum of the absolute values of the coefficients of p?

Középiskolai Matematikai és Fizikai Lapok, November 2016, B. 4827

Z 132. Let p be a given prime number. Find all non-negative integers n for which polynomial

$$P(x) = x^4 - 2(n+p)x^2 + (n-p)^2$$

may be rewritten as product of two quadratic polynomials $P_1(x), P_2(x) \in \mathbb{Z}[x]$.

Polish Mathematical Olympiad 2016

Z 133. Let p be a prime. Let K_p be the set of all polynomials with coefficients from the set $\{0, 1, \cdots, p-1\}$ and degree less than p. Assume that for all pairs of polynomials $P, Q \in K_p$ such that $P(Q(n)) \equiv n \pmod{p}$ for all integers n, the degrees of P and Q are equal. Determine all primes p with this condition.

Turkey Team Selection Test 2016

Z 134. Determine all positive integers n satisfying the following condition: for every monic polynomial P of degree at most n with integer coefficients, there exists a positive integer $k \leq n$ and $k+1$ distinct integers $x_1, x_2, \cdots, x_{k+1}$ such that

$$P(x_1) + P(x_2) + \cdots + P(x_k) = P(x_{k+1}).$$

Romanian Masters In Mathematics 2017

Whatever is worth saying, can be stated in fifty words or less.

– Stanislaw Ulam

2. Polynomials in $\mathbb{R}[x]$ and $\mathbb{C}[x]$

In the broad light of day mathematicians check their equations and their proofs, leaving no stone unturned in their search for rigour. But, at night, under the full moon, they dream, they float among the stars and wonder at the miracle of the heavens. They are inspired. Without dreams there is no art, no mathematics, no life.

– Michael Atiyah, *The art of Mathematics*, Notices Amer. Math. Soc. 57 (2010), no. 1, p. 8.

P 1. Let $P(z) \in \mathbb{C}[x]$ be a polynomial of degree n satisfying the following conditions.

- $P(0) = 1$,
- There exist a constant $M > 0$ such that $|P(z)| \leq M$ for all $z \in \mathbb{C}$ with $|z| \leq 1$.

Prove that every root of $P(z)$ in the closed unit disc has multiplicity at most $c\sqrt{n}$, where $c = c(M) > 0$ is a constant depending only on M.

Miklós Schweitzer Competition 1972
Proposed by G. Halász

P 2. Let $n > 0$ be an integer. Find all polynomials P, in two variables, with the following properties:

- $P(tx, ty) = t^n P(x, y)$ for all $x, y \in \mathbb{R}$.
- $P(a+b, c) + P(b+c, a) + P(c+a, b) = 0$ for all $a, b, c \in \mathbb{R}$.
- $P(1, 0) = 1$.

IMO 1975 Problem 6

P 3. Let $P_1(x) = x^2 - 2$ and $P_j(x) = P_1(P_{j-1}(x))$ for $j = 2, 3, \cdots$. Show that, for any positive integer n, the roots of the equation $P_n(x) = x$ are real and distinct.

IMO 1976 Problem 2

P 4. Let P be a polynomial with real coefficients such that $P(x) > 0$ if $x > 0$. Prove that there exist polynomials Q and R with nonnegative coefficients such that $P(x) = \frac{Q(x)}{R(x)}$ if $x > 0$.

IMO Shortlist 1976

P 5. If $P(x)$, $Q(x)$, $R(x)$, and $S(x)$ are polynomials such that

$$P\left(x^5\right) + xQ\left(x^5\right) + x^2 R\left(x^5\right) = (x^4 + x^3 + x^2 + x + 1)S(x),$$

prove that $x - 1$ is a factor of $P(x)$.

United States of America Mathematical Olympiad 1976

P 6. Consider the polynomial $W(x) = (x-a)^k Q(x)$, where $a \neq 0$, $Q(x)$ is a nonzero polynomial, and k a positive integer. Prove that the polynomial $W(s)$ has at least $k+1$ nonzero coefficients.

Polish Mathematical Olympiad 1977

P 7. Find all polynomials $f(x) \in \mathbb{R}[x]$ such that
$$f(x)f(2x^2) = f(2x^3 + x).$$

IMO Shortlist 1979

P 8. Determine all positive integers n such that there exists a polynomial $P(x)$ of degree $3n$ satisfying
$$\begin{aligned} P(0) &= P(3) = \cdots = P(3n) = 2, \\ P(1) &= P(4) = \cdots = P(3n-2) = 1, \\ P(2) &= P(5) = \cdots = P(3n-1) = 0, \\ P(3n+1) &= 730. \end{aligned}$$

United States of America Mathematical Olympiad 1984

P 9. A sequence of polynomials $P_m(x, y, z), m = 0, 1, 2, \cdots$, in x, y, and z is defined by $P_0(x, y, z) = 1$ and by
$$P_m(x, y, z) = (x + z)(y + z)P_{m-1}(x, y, z + 1) - z^2 P_{m-1}(x, y, z)$$
for $m > 0$. Prove that each $P_m(x, y, z)$ is symmetric, in other words, is unaltered by any permutation of x, y, z.

IMO Shortlist 1985

P 10. Suppose that $f(x)$ and $g(x)$ are nonzero real polynomials satisfying
$$f\left(x^2 + x + 1\right) = g(x)f(x).$$
Show that $f(x)$ has even degree.

Mathematics Magazine 1986 Problem Q 718
Proposed by Bjorn Poonen

P 11. Show that the polynomial
$$x^6 - x^5 + x^4 - x^3 + x^2 - x + \frac{3}{4}$$
has no real roots.

Swedish Mathematics Olympiad (Final Round) 1986

P 12. Let $f(x, y, z) = x^2 + y^2 + z^2 + xyz$. Let $p(x, y, z), q(x, y, z), r(x, y, z) \in \mathbb{R}[x, y, z]$ be polynomials such that
$$f\left(p(x, y, z), q(x, y, z), r(x, y, z)\right) = f(x, y, z).$$
Prove or disprove the assertion that the sequence p, q, r consists of some permutation of $\pm x, \pm y, \pm z$, where the number of minus signs is 0 or 2.

William Lowell Putnam Competition 1986

P 13. Determine all polynomials $P_n(x) = x^n + a_1 x^{n-1} + \cdots + a_{n-1}x + a_n \in \mathbb{R}[x]$ such that $P_n(x)$ has as its n zeros precisely the numbers $a_1, \cdots, a_n$ (counted in their respective multiplicities).

Austrian Mathematical Olympiad 1987

P 14. Let $P(X) = a_n X^n + a_{n-1}X^{n-1} + \cdots + a_1 X + a_0$ be a real polynomial of degree n, where n is an even number. Suppose that

- $a_0 > 0$,
- $a_n > 0$,
- $a_1^2 + a_2^2 + \cdots + a_{n-1}^2 \leq \frac{4}{n-1}\min(a_0^2, a_n^2)$.

Prove that $P(x) \geq 0$ for all real values x.

Romania Team Selection Test 1987

P 15. Let $P(X, Y) = X^2 + 2aXY + Y^2$ be a real polynomial where $|a| \geq 1$. For a given positive integer n, $n \geq 2$ consider the system of equations:

$$P(x_1, x_2) = P(x_2, x_3) = \cdots = P(x_{n-1}, x_n) = P(x_n, x_1) = 0.$$

We call two solutions $(x_1, x_2, \cdots, x_n)$ and $(y_1, y_2, \cdots, y_n)$ of the system to be *equivalent* if there exists a real number $\lambda \neq 0$, $x_1 = \lambda y_1$, $\cdots$, $x_n = \lambda y_n$. How many nonequivalent solutions does the system have?

Romania Team Selection Test 1987

P 16. Find all polynomials of two variables $P(x, y)$ which satisfy

$$P(a, b)P(c, d) = P(ac + bd, ad + bc).$$

Balkan Mathematical Olympiad 1988

P 17. Find all real values of the parameters p and q for which the polynomial

$$f(x) = x^4 - \frac{8p^2}{q}x^3 + 4qx^2 - 3px + p^2$$

has four positive roots. For such p and q, find the roots.

Bulgarian Mathematical Olympiad 1988

P 18. Let S be the boundary of the unit square $[0, 1] \times [0, 1]$ in $\mathbb{R}^2$. Suppose f is a continuous real-valued function on S such that $f(x, 0)$ and $f(x, 1)$ are polynomial functions of x on $[0, 1]$ and such that $f(0, y)$ and $f(l, y)$ are polynomial functions of y on $[0, 1]$. Prove that f is the restriction to S of a polynomial function of x and y.

Amer. Math. Monthly 1990 Problem E 3400
Proposed by Burt J. Totaro

P 19. A nonzero polynomial $P(x) = a_0 + a_1x + \cdots + a_nx^n$ is called n-palindromic if $a_j = a_{n-j}$ for $j \in \{0, 1, \cdots, n\}$. Note that a_n may equal to 0, so that $P(x)$ need not have degree n. It is plain that if P and Q are n-palindromic then $F = \frac{P}{Q}$ satisfies

$$F\left(\frac{1}{x}\right) = F(x)$$

for all x such that $F\left[\frac{1}{x}\right)$ and $F(x)$ are defined. Prove or disprove that, conversely, every rational function $F \neq 0$ satisfying $F\left[\frac{1}{x}\right) = F(x)$ is the quotient of two n-palindromic polynomials for some n.

Crux Mathematicorum 1990 Problem 1582
Proposed by Marcin E. Kuczma

P 20. Express

$$x_1{}^4 + x_2{}^4 + x_3{}^4 + x_4{}^4 - 4x_1x_2x_3x_4$$

as a sum of squares of rational functions with real coefficients. (By the A.M.-G.M. inequality, this polynomial is nonnegative for all real values of its variables, and so by a theorem of Hilbert it can be so expressed.)

Crux Mathematicorum 1990 Problem 1594
Proposed by Murray S. Klamkin

P 21. Is there an infinite sequence $a_0, a_1, a_2, \cdots$ of nonzero real numbers such that, for all $n \in \mathbb{N}$, the polynomial

$$p_n(x) = a_0 + a_1 x + a_2 x^2 + \cdots + a_n x^n$$

has exactly n distinct real roots?

William Lowell Putnam Competition 1990

P 22. Let $P(x) \in \mathbb{R}[x]$ be a polynomial with $P(x) > 0$ for all $x \in [0, 1]$. Prove that there exist polynomials $P_0(x)$, $P_1(x)$, and $P_2(x)$ satisfying the following conditions.

- For each $i \in \{0, 1, 2\}$, $P_i(x) > 0$ for all $x \in \mathbb{R}$.
- $P(x) = P_0(x) + xP_1(x) + (l - x)P_2(x)$ for all $x \in \mathbb{R}$.

Austrian Polish Mathematical Competition 1991

P 23. Let $f(x) = x^n + a_1 x^{n-1} + \cdots + a_n \in \mathbb{R}[x]$ be a polynomial with degree $n \geq 2$ having real roots $b_1, b_2, \cdots, b_n$. Prove that

$$f(x+1) \geq \frac{2n^2}{\frac{1}{x-b_1} + \frac{1}{x-b_2} + \cdots + \frac{1}{x-b_n}}$$

for all $x \geq max\{b_1, b_2, \cdots, b_n\}$.

China Team Selection Test 1991

P 24. Let $f(x) \in \mathbb{Z}[x]$ be a polynomial satisfying the following conditions.

- $f(0) = 11$
- $f(x_1) = f(x_2) = ... = f(x_n) = 2002$ for some distinct integers $x_1, x_2, \cdots, x_n$.

Find the largest possible value of n.

Hungary-Israel Binational Mathematical Competition 1991

P 25. Let $f(x) \in \mathbb{Z}[x]$ be a monic polynomial of degree 1991. Define

$$g(x) = [\, f(x) \,]^2 - 9.$$

Show that the number of distinct integer solutions of $g(x) = 0$ cannot exceed 1995.

IMO Shortlist 1991

P 26. Find all polynomials $f(x) \in \mathbb{R}[x]$ such that

$$f\left(x^2\right) = [\, f(x) \,]^2.$$

Irish Mathematical Olympiad 1991

P 27. Find all polynomials $f(x) \in \mathbb{R}[x]$ satisfying the following conditions.

- All the coefficients belong to the set $\{-1, 0, 1\}$.
- All the roots of the equation $f(x) = 0$ are real.

Irish Mathematical Olympiad 1991

P 28. A polynomial $f(x) = x^3 + ax^2 + bx + c \in \mathbb{R}[x]$ is such that $b < 0$ and $ab = 9c$. Prove that the polynomial f has three different real roots.

Baltic Way 1992

P 29. Let $f(x) = x^8 + 4x^6 + 2x^4 + 28x^2 + 1$. Let $p > 3$ be a prime and suppose there exists an integer z such that p divides $f(z)$. Prove that there exist integers $z_1, z_2, \cdots, z_8$ such that if

$$g(x) = (x - z_1)(x - z_2)\cdots(x - z_8),$$

then all coefficients of $f(x) - g(x)$ are divisible by p.

IMO Shortlist 1992

P 30. Let $P(z)$ be a polynomial with complex coefficients which is of degree 1992 and has distinct zeros. Prove that there exist complex numbers a_1, a_2, $\cdots$, a_{1992} such that $P(z)$ divides the polynomial

$$\left(\cdots\left((z - a_1)^2 - a_2\right)^2 \cdots - a_{1991}\right)^2 - a_{1992}.$$

United States of America Mathematical Olympiad 1992

P 31. Let a polynomial $f(x)$ be given with real coefficients and has degree greater or equal than 1. Show that for every real number $c > 0$, there exists a positive integer n_0 satisfying the following condition: if polynomial $P(x)$ of degree greater or equal than n_0 with real coefficients and has leading coefficient equal to 1 then the number of integers x for which $|f(P(x))| \le c$ is not greater than degree of $P(x)$.

Vietnam Team Selection Test 1992

P 32. Let $9 < n_1 < n_2 < \cdots < n_s < 1992$ be positive integers and

$$P(x) = 1 + x^2 + x^9 + x^{n_1} + \cdots + x^{n_s} + x^{1992}.$$

Prove that if x_0 is real root of $P(x)$ then $x_0 \le \frac{1-\sqrt{5}}{2}$.

Vietnamese Mathematical Olympiad 1992

P 33. Let $\{P_m(z) \mid m = 1, 2, 3, \cdots\}$ be the sequence of polynomials defined by $P_1(z) = z - 1$, $P_2(z) = z^2 - z - 1$, and $P_m(z) = zP_{m-1}(z) - P_{m-2}(z)$ for $m > 2$. Show that the roots of $P_m(z)$ are $2\cos\left(\frac{(2k-1)\pi}{2m+1}\right)$ for $1 \le k \le m$.

Amer. Math. Monthly 1993 Problem 10300
Proposed by Eric H. Mason

P 34. Let $f(x) = a_n x^n + a_{n-1}x^{n-1} + \cdots + a_0$ and $g(x) = c_{n+1}x^{n+1} + c_n x^n + \cdots + c_0$ be non-zero polynomials in $\mathbb{R}[x]$ such that $g(x) = (x + r)f(x)$ for some constant $r \in \mathbb{R}$. If $a = \max(|a_n|, \cdots, |a_0|)$ and $c = \max(|c_{n+1}|, \cdots, |c_0|)$, prove that $\frac{a}{c} \le n + 1$.

Asian Pacific Mathematics Olympiad 1993

P 35. Suppose that the polynomial $x^4 + ax^3 + bx^2 + cx + d \in \mathbb{R}[x]$ has four positive roots. Prove that

$$abc \ge a^2 d + 5c^2.$$

Crux Mathematicorum 1993 Problem 1825
Proposed by Marcin E. Kuczma

P 36. Can the equation $f(g(h(x))) = 0$, where f, g, h are quadratic polynomials in $\mathbb{R}[x]$, have the solutions $1, 2, 3, 4, 5, 6, 7, 8$?

All-Russian Mathematical Olympiad 1995
Proposed by S. Tokarev

P 37. Let $P(x)$ and $Q(x)$ be monic polynomials in $\mathbb{R}[x]$. Prove that the sum of the squares of the coefficients of the polynomial $P(x)Q(x)$ is not smaller than the sum of the squares of the constant terms of $P(x)$ and $Q(x)$.

All-Russian Mathematical Olympiad 1995
Proposed by A. Galochkin, O. Ljashko

P 38.

- Show that the polynomial
$$2\left(x^7 + y^7 + z^7\right) - 7xyz\left(x^4 + y^4 + z^4\right)$$
has $x + y + z$ as a factor.
- Is the remaining factor irreducible over $\mathbb{C}[x]$?

Crux Mathematicorum 1995 Problem 2014
Proposed by Murray S. Klamkin

P 39. Say that a sequence $\mathbf{q} = q_0, q_1, q_2, \cdots$ of integers has the *divisible differences property* if
$$n - m \mid q_n - q_m$$
for all n and m.

(1) Show that if $\mathbf{q}$ has the divisible differences property and $\limsup |q_n|^{\frac{1}{n}} < e - 1$, then there is a polynomial $Q(x)$ such that $q_n = Q(n)$.
(2) Show that there is a sequence $\mathbf{q}$ that has the divisible differences property and satisfies $\limsup |q_n|^{\frac{1}{n}} < e$, for which q_n is not given in polynomial in n.
(3) Is it true that $\limsup |q_n|^{\frac{1}{n}} \geq e$ for all non-polynomial $\mathbf{q}$ with the divisible differences property?

Amer. Math. Monthly 1996 Problem 10553
Proposed by Bjore Poonen, Jim Prop, Richard Stong

P 40.

- Suppose that $f(t) \in \mathbb{R}[t]$ is a polynomial that maps rationals to rationals and irrationals to irrationals. Show that $f(t) = at + b$ with a and b rational.
- Does the same conclusion hold under the weaker assumption that $f \to \mathbb{R} \to \mathbb{R}$ is an algebraic function (i.e., if there is a polynomial $P(x, y) \in \mathbb{R}[x, y]$ such that $P(t, f(t))$ is identically zero)?

Amer. Math. Monthly 1996 Problem 10555
Proposed by Jeffrey C. Lagarias, Eric M. Rains

P 41. Let $a_1, a_2, \cdots, a_n$ be non-negative reals, not all zero.

(1) Show that the polynomial $p(x) = x^n - a_1x^{n-1} + \cdots - a_{n-1}x - a_n$ has precisely one positive real root R.
(2) Let $A = \sum_{i=1}^n a_i$ and $B = \sum_{i=1}^n ia_i$. Show that $A^A \leq R^B$.

IMO Shortlist 1996

P 42. Let $P(x) = ax^3 + bx^2 + cx + d \in \mathbb{R}[x]$ be a polynomial such that
$$|P(x)| \leq 1$$
for all $x \in [-1, 1]$. Prove that
$$|a| + |b| + |c| + |d| \leq 7.$$

IMO Shortlist 1996

P 43. Let n be an even positive integer. Prove that there exists a positive integer k such that
$$k = f(x)(x+1)^n + g(x)\left(x^n + 1\right)$$
for some polynomials $f(x), g(x) \in \mathbb{Z}[x]$. If k_0 denotes the least such k, determine k_0 as a function of n.

IMO Shortlist 1996

P 44. Find all quadruplets of polynomials $p_1(x)$, $p_2(x)$, $p_3(x)$, $p_4(x) \in \mathbb{R}[x]$ possessing the property that
$$p_1(x)p_2(y) - p_3(z)p_4(t) = 1$$
holds for all $x, y, z, t \in \mathbb{Z}$ such that $xy - zt = 1$.

St Petersburg Mathematical Olympiad 1996

P 45. Fix a positive real number v. Find all polynomials $P(x)$ with nonnegative real coefficients such that

(1) $P(0) = 0$, $P(1) = 1$, and $P(x) \leq x^v$ for all $x > 0$.
(2) $P(0) = 0$, $P(1) = 1$, and $P(x) \geq x^v$ for all $x > 0$.

Amer. Math. Monthly 1997 Problem 10613
Proposed by J. Flanigan

P 46. Suppose that $n \geq 3$ is an odd natural number. Show that the only polynomial $P \in \mathbb{R}[x]$ satisfying the functional equation
$$P(x+1)^n = P(x)^n + \sum_{k=0}^{n} \binom{n}{k} x^k.$$

Crux Mathematicorum 1997 Problem 2247
Proposed by Walther Janous

P 47. Let $P(x) \in \mathbb{R}[X]$ be a polynomial such that
$$P(x) > 0$$
for all $x \geq 0$. Prove that there exists a positive integer n such that
$$(1+x)^n P(x)$$
is a polynomial with nonnegative coefficients.

IMO Shortlist 1997

P 48. Does there exist a polynomial $f(x, y, z)$ of real coefficients such that, $f(x, y, z)$ is positive if and only if $|x|$, $|y|$ and $|z|$ are the sides of a triangle?

Középiskolai Matematikai és Fizikai Lapok, September 1997, N. 145

P 49. Is it possible that the range of a polynomial of real coefficients in two variables is the open interval $(0, \infty)$?

Középiskolai Matematikai és Fizikai Lapok, January 1998, F. 3210

P 50. Find all values of m for which polynomial $x^m + y^m + z^m - (x+y+z)^m$ is divisible by $(y+z)(z+x)(x+y)$.

Középiskolai Matematikai és Fizikai Lapok, April 1998, F. 3227

P 51. A polynomial $p \in \mathbb{R}[x]$ has degree at most $n - 10\sqrt{n}$ for some $n \in \mathbb{N}$. Prove that

$$|\, p(0)\, | \le \frac{1}{10} \sum_{k=1}^{n} \binom{n}{k} |\, p(k)\, |.$$

Középiskolai Matematikai és Fizikai Lapok, December 1998, N. 190

P 52. Prove that, for any sufficiently large integer n, there exists a polynomial l $p \in \mathbb{R}[x]$ of degree at most $n - \frac{1}{10}\sqrt{n}$ for some $n \in \mathbb{N}$, such that

$$|\, p(0)\, | > \frac{1}{10} \sum_{k=1}^{n} \binom{n}{k} |\, p(k)\, |.$$

Középiskolai Matematikai és Fizikai Lapok, November 1998, N. 193

P 53. Show that for any positive integer n the polynomial $f(x) = (x^2+x)^{2^n} + 1$ cannot be decomposed into the product of two integer non-constant polynomials.

Romania Team Selection Test 1998
Proposed by Marius Cavachi

P 54. Let n be a positive integer and $\mathcal{P}_n$ be the set of integer polynomials of the form $a_0 + a_1 x + \cdots + a_n x^n$ where $|a_i| \le 2$ for $i = 0, 1, \cdots, n$. Find, for each positive integer k, the number of elements of the set $A_n(k) = \{f(k) | f \in \mathcal{P}_n\}$.

Romania Team Selection Test 1998
Proposed by Marian Andronache

P 55. Find all positive integers k for which the following statement is true: If $F(x)$ is a polynomial with integer coefficients satisfying the condition $0 \le F(c) \le k$ for each $c \in \{0, 1, \cdots, k+1\}$, then $F(0) = F(1) = \cdots = F(k+1)$.

Romania Team Selection Test 1998

P 56. Let $f : \mathbb{R} \to \mathbb{R}$ be a real function such that, for each $c > 0$, there exists a polynomial $P(x)$ (maybe dependent on c) such that

$$|f(x) - P(x)| \le c\, x^{1998}$$

for all $x \in \mathbb{R}$. Prove that $f(x) \in \mathbb{R}[x]$.

Vietnam Team Selection Test 1998

P 57. Find all positive integers n such that there exists a $P \in \mathbb{R}[x]$ such that

$$P\left(x^{1998} - x^{-1998}\right) = x^n - x^{-n}$$

for all $x \in \mathbb{R} - \{0\}$.

Vietnamese Mathematical Olympiad 1998

P 58. Determine all polynomials $f(x) \in \mathbb{R}[x]$ such that
$$f\left(x^2-2x\right) = (f(x-2))^2.$$

Hungary-Israel Binational Mathematical Competition 1993

P 59. Prove that, for infinitely many positive integers n, there exists a polynomial p of degree n with real coefficients such that $p(1)$, $p(2)$, $\cdots$, $p(n+2)$ are different whole powers of 2.

Középiskolai Matematikai és Fizikai Lapok, February 1999, N. 200

P 60. Let $p(x) \in \mathbb{R}[x]$ be a polynomial that is nonnegative for all real x. Prove that for some k, there are polynomials $f_1(x), \cdots, f_k(x)$ such that
$$p(x) = \sum_{j=1}^{k} (f_j(x))^2.$$

William Lowell Putnam Competition 1999

P 61. Let $P, Q \in \mathbb{C}[x]$ be two monic polynomials such that
$$P(P(x)) = Q(Q(x)).$$
Prove that $P = Q$.

Romania Team Selection Test 2000
Proposed by Marius Cavachi

P 62. Let $P(x) \in \mathbb{R}[x]$ be a nonzero polynomial such that
$$P\left(x^2-1\right) = P(x)P(-x).$$
Determine the maximum possible number of real roots of $P(x)$.

Vietnamese Mathematical Olympiad 2000

P 63. The polynomial $P(x) = x^3 + ax^2 + bx + d \in \mathbb{R}[x]$ has three distinct real roots. The polynomial
$$P\left(x^2+x+2001\right)$$
has no real roots. Prove that
$$P(2001) > \frac{1}{64}.$$

All-Russian Mathematical Olympiad 2001

P 64. The rational numbers in the interval $(0,1)$ are mapped, by a polynomial p of real coefficients, onto the set of all rational numbers of an other interval. Prove that the degree of p is not greater than 1.

Középiskolai Matematikai és Fizikai Lapok, April 2001, A. 264
Proposed by E. Fried

P 65. Find all polynomials $P(x) \in \mathbb{R}[x]$ such that

$$P(x)P\left(2x^2-1\right) = P\left(x^2\right)P\left(2x-1\right).$$

Romania Team Selection Test 2001

P 66.

(1) Consider the polynomial $P(X) = X^5 \in \mathbb{R}[X]$. Show that for every $\alpha \in \mathbb{R} - \{0\}$, the polynomial $P(X+\alpha) - P(X)$ has no real roots.
(2) Let $P(X) \in \mathbb{R}[X]$ be a polynomial of degree $n \geq 2$, with real and distinct roots. Show that there exists $\alpha \in \mathbb{Q}^*$ such that the polynomial $P(X+\alpha) - P(X)$ has only real roots.

Romanian Mathematical Olympiad 2001

P 67. The polynomials P, Q, R with real coefficients, one of which is degree 2 and two of degree 3, satisfy the equality

$$P^2 + Q^2 = R^2.$$

Prove that one of the polynomials of degree 3 has three real roots.

All-Russian Mathematical Olympiad 2002

P 68.

- Is there an infinite sequence $p_1(x)$, $p_2(x) = x^2 - 2$, $p_3(x)$, $\cdots$ of polynomials satisfying that the degree of $p_k(x)$ is exactly k, and that $p_i\left(p_j(x)\right) = p_j\left(p_i(x)\right)$ for all pairs (i, j) of positive integers?
- Is there an infinite sequence $p_1(x)$, $p_2(x) = x^2 - 3$, $p_3(x)$, $\cdots$ of polynomials satisfying that the degree of $p_k(x)$ is exactly k, and that $p_i\left(p_j(x)\right) = p_j\left(p_i(x)\right)$ for all pairs (i, j) of positive integers?

Középiskolai Matematikai és Fizikai Lapok, April 2003, B. 3641

P 69. Let $n \in \mathbb{N}$. Construct a polynomial p of degree n such that, for all $x \in \left[0, \frac{1}{2}\right]$,

$$\left| p(x) - \frac{1}{1-x} \right| < \frac{4}{\left(1+\sqrt{2}\right)^{2n+2}}.$$

Középiskolai Matematikai és Fizikai Lapok, April 2003, A. 318

P 70. Define $p(x) = 4x^3 - 2x^2 - 15x + 9$, $q(x) = 12x^3 + 6x^2 - 7x + 1$. Show that each polynomial has just three distinct real roots. Let A be the largest root of $p(x)$ and B the largest root of $q(x)$. Show that $A^2 + 3B^2 = 4$.

Vietnamese Mathematical Olympiad 2003

P 71. Suppose that a, b, c, A, B, C are real numbers, $a \neq 0$ and $A \neq 0$, such that

$$|ax^2 + bx + c| \leq |Ax^2 + Bx + C|$$

for all $x \in \mathbb{R}$. Show that

$$|b^2 - 4ac| \leq |B^2 - 4AC|.$$

William Lowell Putnam Competition 2003

P 72. Do there exist polynomials $a(x), b(x), c(y), d(y)$ such that $1 + xy + x^2y^2 = a(x)c(y) + b(x)d(y)$?

William Lowell Putnam Competition 2003

P 73. Find all polynomials $f \in \mathbb{R}[x]$ such that
$$f(a-b) + f(b-c) + f(c-a) = 2f(a+b+c)$$
for all $a, b, c \in \mathbb{R}$ with $ab + bc + ca = 0$.

IMO Shortlist 2004 A4, IMO 2004 Problem 2
Proposed by Hojoo Lee (Korea)

P 74. Find all polynomials $p(x)$ for which the polynomials $p(x)p(x+1) = p(x + p(x))$.

Középiskolai Matematikai és Fizikai Lapok, April 2004, B. 3724

P 75. Let P_n be the polynomial function on $\mathbb{R}$ given by
$$P_n(x) = \sum_{j=0}^{n} \binom{n}{j}^2 x^{2j}(1-x)^{2(n-j)}.$$

(1) Show that on the interval $[0, 1]$, $P_n(x)$ is minimized at $x = \frac{1}{2}$.
(2) Show that when $P_n(x) = \sum_{k=0}^{2n} a_{n,k}\left(x - \frac{1}{2}\right)^k$, the odd coefficients $a_{n,2k+1}$ are zero and the even coefficients $a_{n,2k}$ are non-negative.

Amer. Math. Monthly 2005 Problem 11155
Proposed by Emmanuel Kowalski, Tanguy Rivoal

P 76. It is well known that $\cos(2x)$ can be expressed as a polynomial of $\cos x$:
$$\cos(2x) = 2\cos^2 x - 1.$$
Is it possible to express $\sin(2x)$ as a polynomial of $\sin x$?

Középiskolai Matematikai és Fizikai Lapok, December 2005, B. 3868

P 77. Show that the only polynomial $p(x) \in \mathbb{R}[x]$ of odd degree such that
$$p\left(x^2 - 1\right) = p(x)^2 - 1$$
is $p(x) = x$.

Polish Mathematical Olympiad 2005

P 78. Consider the polynomials
$$f(x) = \sum_{k=1}^{n} a_k x^k \quad \text{and} \quad g(x) = \sum_{k=1}^{n} \frac{a_k}{2^k - 1} x^k,$$
where $a_1, a_2, \cdots, a_n$ are real numbers and n is a positive integer. Show that if 1 and 2^{n+1} are zeros of g, then f has a positive zero less than 2^n.

USA Team Selection Test 2005

P 79. Find a nonzero polynomial $P(x,y)$ such that

$$P(\lfloor a \rfloor, \lfloor 2a \rfloor) = 0$$

for all $a \in \mathbb{R}$. Here, $\lfloor \nu \rfloor$ is the greatest integer less than or equal to ν.

William Lowell Putnam Competition 2005

P 80. Given a quadratic trinomial $f(x) = x^2 + ax + b$. Assume that the equation

$$f(f(x)) = 0$$

has four different real solutions, and that the sum of two of these solutions is -1. Prove that $b \le -\frac{1}{4}$

All-Russian Mathematical Olympiad 2006

P 81. Let $P(x) \in \mathbb{C}[x]$ be a polynomial such that $P(0) \neq 0$. Prove that there exists a multiple of $P(x)$ with real positive coefficients if and only if $P(x)$ has no real positive root.

Italy Team Selection Test 2006

P 82. Find a polynomial p of rational coefficients such that $p\left(\sqrt{2} + \sqrt{3}\right) = \sqrt{2}$.

Középiskolai Matematikai és Fizikai Lapok, September 2006, B. 3931
Proposed by Ervin Fried

P 83. Find all pairs of integers (a,b) such that there exists a polynomial $P(x) \in \mathbb{Z}[X]$ such that

$$(x^2 + ax + b)P(x) = x^n + c_{n-1}x^{n-1} + \cdots + c_1 x + c_0$$

for some $c_0, c_1, \cdots, c_{n-1} \in \{-1, 1\}$.

Polish Mathematical Olympiad 2006

P 84. Find all polynomials $P(x) \in \mathbb{R}[x]$ such that

$$P\left(x^2\right) + x(3P(x) + P(-x)) = (P(x))^2 + 2x^2$$

for all $x \in \mathbb{R}$.

Vietnamese Mathematical Olympiad 2006

P 85. Let $P(x) \in \mathbb{Z}[x]$ be a monic polynomial with even degree. Prove that, if for infinitely many integers x, the number $P(x)$ is a square of a positive integer, then there exists a polynomial $Q(x) \in \mathbb{Z}[x]$ such that $P(x) = Q(x)^2$.

Bulgarian Mathematical Olympiad 2007

P 86. Find all polynomials $P \in \mathbb{R}[x]$ such that

$$P\left(x^2\right) = P(x)P(x+2).$$

Czech-Polish-Slovak Match 2007

P 87. Does there exist a polynomial $f(x) \in \mathbb{Z}[x]$ with the degree 2007 such that, for all integers n,

$$f(n),\ f(f(n)),\ f(f(f(n))),\ \cdots$$

is coprime to each other?

Hong Kong Mathematics Olympiad 2007

P 88. Let $t \geq 3$ be a given real number. We assume that the polynomial $f(x) \in \mathbb{R}[x]$ satisfies
$$|f(k) - t^k| < 1,$$
for all $k \in \{0, 1, 2, \cdots, n\}$. Prove that the degree of the polynomial $f(x)$ is at least n

Hungary-Israel Binational Mathematical Competition 2007

P 89. Find all polynomials of degree 3 such that
$$p(x + y) \geq p(x) + p(y)$$
for all $x, y \geq 0$.

Iran Team Selection Test 2007
Proposed by Omid Hatami

P 90. A *repunit* is a positive integer whose digits in base 10 are all ones. Find all polynomials f with real coefficients such that if n is a repunit, then so is $f(n)$.

William Lowell Putnam Competition 2007

P 91. Let n be a positive integer. Find the number of pairs $P, Q \in \mathbb{R}[x]$ such that
$$P(x)^2 + Q(x)^2 = x^{2n} + 1$$
and $\deg P > \deg Q$.

William Lowell Putnam Competition 2007

P 92. Let k be a positive integer. Prove that there exist polynomials $P_0(n), P_1(n), \cdots, P_{k-1}(n)$ (which may depend on k) such that for any integer n,
$$\left\lfloor \frac{n}{k} \right\rfloor^k = P_0(n) + P_1(n) \left\lfloor \frac{n}{k} \right\rfloor + \cdots + P_{k-1}(n) \left\lfloor \frac{n}{k} \right\rfloor^{k-1}.$$
($\lfloor a \rfloor$ means the largest integer $\leq a$.)

William Lowell Putnam Competition 2007

P 93. Determine all polynomials $p(x) \in \mathbb{R}[x]$ such that $p\left((x+1)^3\right) = (p(x) + 1)^3$ and $p(0) = 0$.

Baltic Way 2008

P 94. Find all polynomials $f(x) \in \mathbb{R}[x]$ such that
$$2yf(x + y) + (x - y)(f(x) + f(y)) \geq 0.$$

Germany Team Selection Test 2008

P 95. If $p(x)$ is a polynomial with degree less than $2n$, show that
$$|p(n)| \leq 2\sqrt{n} \max\{p(0), p(1), \cdots, p(2n)\}.$$

Középiskolai Matematikai és Fizikai Lapok, September 2008, A. 460

P 96. Find all values of the positive integer m such that there exists polynomials $P(x)$, $Q(x)$, $R(x,y)$ with real coefficient satisfying the condition: For every real numbers a, b which satisfying $a^m - b^2 = 0$, we always have that $P(R(a,b)) = a$ and $Q(R(a,b)) = b$.

Vietnam Team Selection Test 2008

P 97. Let $n \geq 3$ be an integer. Let $f(x)$ and $g(x)$ be polynomials with real coefficients such that the points

$$(f(1), g(1)),\ (f(2), g(2)),\ \cdots,\ (f(n), g(n))$$

in $\mathbb{R}^2$ are the vertices of a regular n-gon in counterclockwise order. Prove that at least one of $f(x)$ and $g(x)$ has degree greater than or equal to $n-1$.

William Lowell Putnam Competition 2008

P 98. Find all complex polynomial $P(x)$ such that

$$\frac{P(a)+P(b)+P(c)}{a+b+c}$$

is an integer for all three integers a, b, c with $a+b+c \neq 0$.

China Team Selection Test 2009

P 99. Find all Polynomials $P(x,y)$ such that

$$P\left(x^2, y^2\right) = P\left(\frac{(x+y)^2}{2}, \frac{(x-y)^2}{2}\right).$$

Iran Team Selection Test 2009

P 100. Find all polynomials f with integer coefficient such that, for every prime p and positive integers u and v with the condition that $p \mid f(u)f(v) - 1$ implies that $p \mid uv - 1$.

Iran Team Selection Test 2009

P 101. For each positive integer n, let $c(n)$ be the largest real number such that

$$c(n) \leq \left|\frac{f(a)-f(b)}{a-b}\right|$$

for all triples (f, a, b) such that

- f is a polynomial of degree n taking integers to integers, and
- a, b are integers with $f(a) \neq f(b)$.

Find $c(n)$.

USA Team Selection Test 2009
Proposed by Shaunak Kishore

P 102. Find all polynomials $p(x) \in \mathbb{R}[x]$ such that

$$p(a+b-2c) + p(b+c-2a) + p(c+a-2b) = 3p(a-b) + 3p(b-c) + 3p(c-a)$$

for all $a, b, c \in \mathbb{R}$.

Benelux Mathematical Olympiad 2010

P 103. Find all polynomials $P(x) \in \mathbb{R}[x]$ such that
$$P(x)P(x+1) = P\left(x^2\right).$$

Bulgarian Mathematical Olympiad 2010

P 104. Let $P(x), Q(x) \in \mathbb{Z}[x]$ be polynomials. Let $a_n = n! + n$. Suppose that $\frac{P(a_n)}{Q(a_n)}$ is an integer for all integres n. Show that $\frac{P(n)}{Q(n)}$ is an integer for all integers n with $Q(n) \neq 0$.

Canadian Mathematical Olympiad 2010

P 105. Find the smallest number n such that there exist polynomials $f_1, f_2, \cdots, f_n$ with rational coefficients satisfying
$$x^2 + 7 = [\, f_1\,(x)\,]^2 + [\, f_2\,(x)\,]^2 + \cdots + [\, f_n\,(x)\,]^2.$$

IMO Shortlist 2010 N3
Proposed by Mariusz Skałba (Poland)

P 106. Find all two-variable polynomials $p(x, y)$ such that
$$p\left(ab, c^2+1\right) + p\left(bc, a^2+1\right) + p\left(ca, b^2+1\right) = 0$$
for all $a, b, c \in \mathbb{R}$.

Iran Team Selection Test 2010

P 107. Suppose that the polynomial $p(x) = x^{2010} \pm x^{2009} \pm \ldots \pm x \pm 1$ does not have a real root. What is the maximum number of coefficients to be -1?

Iranian Mathematical Olympiad 2010

P 108. Is there a polynomial of degree at least two that represents a one-to-one mapping of the set of rational numbers onto itself?

Középiskolai Matematikai és Fizikai Lapok, April 2010, B. 4271

P 109. Show that, for arbitrary positive integers n, k, there exists a polynomial $p(x)$, with degree at most $100\sqrt{nk}$, such that
$$p(0) > (\,|p(1)| + \cdots + |p(n)|\,) + (\,|p(-1)| + \cdots + |p(-k)|\,).$$

Középiskolai Matematikai és Fizikai Lapok, May 2010, A. 511

P 110. Find all pairs of polynomials $p(x), q(x) \in \mathbb{R}[x]$ such that
$$p(x)q(x+1) - p(x+1)q(x) = 1.$$

William Lowell Putnam Competition 2010

P 111. Let P be a polynomial over $\mathbb{R}$ given by $P(x) = x^3 + a_2x^2 + a_1x + a_0$, with $a_1 > 0$. Show that P has a least one zero between $-\frac{a0}{a1}$ and a_2.

Amer. Math. Monthly 2011 Problem 11589
Proposed by Catalin Barboianu

P 112. Let $P(x)$ and $Q(x)$ be two polynomials with integer coefficients, such that no non-constant polynomial with rational coefficients divides both $P(x)$ and $Q(x)$. Suppose that for every positive integer n the integers $P(n)$ and $Q(n)$ are positive, and $2^{Q(n)}-1$ divides $3^{P(n)}-1$. Prove that $Q(x)$ is a constant polynomial.

IMO Shortlist 2011 N6
Proposed by Oleksiy Klurman (Ukraine)

P 113. Find the polynomial p of two variables such that

$$p\,(x+y,xy)=\sum_{k=0}^{20}x^{20-k}y^k.$$

Középiskolai Matematikai és Fizikai Lapok, January 2011, B. 4330
Proposed by A. Hraskó

P 114. Find all pairs $f(x)$, $g(x)$ of polynomials of real coefficients such that

$$f(x+1)g(x-1)-g(x+1)f(x-1)=1.$$

Középiskolai Matematikai és Fizikai Lapok, February 2011, B. 4341
Proposed by P. Kutas

P 115. Prove that there exist a real number $c>0$ such that whenever each of the numbers $a_0,\,a_1,\,\cdots,\,a_n$ are 1 or -1 and the polynomial $(x-1)^k$ divides the polynomial $a_0+\cdots+a_nx^n$ then $k<c(\,\ln(n+1)\,)^2$.

Középiskolai Matematikai és Fizikai Lapok, March 2011, A. 532

P 116. The teacher in the mathematics club had an at most nth-degree polynomial in mind. The students may ask the value of the polynomial at any point on the real axis they want. What is the minimum number of points they need, in order to be able to decide in all cases whether the function is even?

Középiskolai Matematikai és Fizikai Lapok, December 2011, B. 4410

P 117. Prove that the following equality holds for $x,y\in\mathbb{C}$ and n a positive integer.

$$x^{2n}-x^ny^n+y^{2n}=\prod_{1\le k<3n,\gcd(k,6)=1}\left(x^2-2\cos\left(\frac{k\pi}{3n}\right)xy+y^2\right).$$

Mathematics Magazine 2011 Problem 1876
Proposed by Roman Wituła, Edyta Hetmaniok, Damian Słota

P 118. A Mediterranean polynomial has only real roots and it is of the form

$$P(x)=x^{10}-20x^9+135x^8+a_7x^7+a_6x^6+a_5x^5+a_4x^4+a_3x^3+a_2x^2+a_1x+a_0$$

with real coefficients $a_0\cdots,a_7$. Determine the largest real number that occurs as a root of some Mediterranean polynomial.

Mediterranean Mathematics Olympiad 2011
Proposed by Gerhard Wöginger (Austria)

P 119. Determine all positive integers n for which there exists a polynomial $f(x) \in \mathbb{R}[x]$ satisfying the following conditions.

- For each integer k, the number $f(k)$ is an integer if and only if k is not divisible by n.
- The degree of f is less than n.

Romanian Masters In Mathematics 2011
Proposed by Géza Kós (Hungary)

P 120. A polynomial $P(x)$ is called *nice* if $P(0) = 1$ and the nonzero coefficients of $P(x)$ alternate between 1 and -1 when written in order. Suppose that $P(x)$ is nice, and let m and n be two relatively prime positive integers. Show that

$$Q(x) = P(x^n) \frac{(x^{mn}-1)(x-1)}{(x^m-1)(x^n-1)}$$

is nice as well.

USA Team Selection Test 2011

P 121. Show that the polynomial $P(x, y) = x^n + xy + y^n$ is irreducible in $\mathbb{R}[x, y]$.

Vietnamese Mathematical Olympiad 2011

P 122. We are given a real number a, not equal to 0 or 1. Sacho and Deni play the following game. First is Sasho and then Deni and so on (they take turns). On each turn, a player changes one of the $\star$ symbols in the equation:

$$\star x^4 + \star x^3 + \star x^2 + \star x + \star = 0$$

with a number of the type a^n, where n is a positive integer. Sasho wins if at the end the equation has no real roots, Deni wins otherwise. Determine (in term of a) who has a winning strategy

Bulgarian Mathematical Olympiad 2012

P 123. We say that a function $f : \mathbb{R}^k \to \mathbb{R}$ is a metapolynomial if, for some positive integers m and n, it can be represented in the form

$$f(x_1, \cdots, x_k) = \max_{i=1,\cdots,m} \min_{j=1,\cdots,n} P_{i,j}(x_1, \cdots, x_k),$$

where $P_{i,j}$ are multivariate polynomials. Prove that the product of two metapolynomials is also a metapolynomial.

IMO Shortlist 2012 A7

P 124. Let $n \geq 2$ and let $p(x) = x^n + a_{n-1}x^{n-1} + \cdots + a_1 x + a_0 \in \mathbb{R}[x]$ be a polynomial. Prove that if for some positive integer k the polynomial $(x-1)^{k+1}$ divides $p(x)$ then

$$\sum_{l=0}^{n-1} |a_l| > 1 + \frac{2k^2}{n}.$$

Középiskolai Matematikai és Fizikai Lapok, November 2012, A. 574
CIIM 2012, Guanajuato, Mexico

P 125. Let $P(x)$ and $Q(x)$ be monic polynomials with real coefficients and $\deg P(x) = \deg Q(x) = 10$. Prove that if the equation

$$P(x) = Q(x)$$

has no real solutions, then the equation

$$P(x+1) = Q(x-1)$$

has a real solution.

All-Russian Mathematical Olympiad 2013

P 126. Let $m \neq 0$ be an integer. Find all polynomials $P(x) \in \mathbb{R}[x]$ such that

$$\left(x^3 - mx^2 + 1\right) P(x+1) + \left(x^3 + mx^2 + 1\right) P(x-1) = 2\left(x^3 - mx + 1\right) P(x)$$

for all $x \in \mathbb{R}$.

IMO Shortlist 2013 A6

P 127. For nonnegative integers m and n, define the sequence $a(m,n)$ of real numbers as follows. Set $a(0,0) = 2$ and for every natural number n, set $a(0,n) = 1$ and $a(n,0) = 2$. Then for $m, n \geq 1$, define

$$a(m,n) = a(m-1,n) + a(m,n-1).$$

Prove that for every natural number k, all the roots of the polynomial

$$P_k(x) = \sum_{i=0}^{k} a(i, 2k+1-2i) x^i$$

are real.

Iran Team Selection Test 2013

P 128. Let $a_0, a_1, \cdots, a_n \in \mathbb{N}$. Prove that there exist positive integers $b_0, b_1, \cdots, b_n$ such that for $a_i \leq b_i \leq 2a_i$ and the polynomial

$$P(x) = b_0 + b_1 x + \cdots + b_n x^n$$

is irreducible over $\mathbb{Q}[x]$.

Iranian Mathematical Olympiad 2013

P 129. Show that for every positive integer k and real number $\epsilon > 0$ there exists a polynomial $p(x) = x^n + a_{n-1}x^{n-1} + \cdots + a_1 x + a_0 \in \mathbb{R}[x]$ such that $p(x)$ is divisible by the polynomial $(x-1)^{k+1}$ and

$$\sum_{l=0}^{n-1} |a_l| > 1 + \frac{(2+\epsilon)\, k^2}{n}.$$

Középiskolai Matematikai és Fizikai Lapok, January 2013, A. 580

P 130. Are there two distinct non-constant polynomials p and q such that

$$\{p(1), p(2), p(3), \cdots\} = \{q(1), q(2), q(3), \cdots\}.$$

Középiskolai Matematikai és Fizikai Lapok, September 2013, B. 4561

P 131. Do there exist two real monic polynomials $P(x)$ and $Q(x)$ of degree 3 such that the roots of $P(Q(X))$ are nine pairwise distinct nonnegative integers that add up to 72?

Mediterranean Mathematics Olympiad 2013

P 132. Given an integer $n \geq 2$, determine all non-constant polynomials $f(x) \in \mathbb{C}[x]$ such that

$$1 + f\left(x^n + 1\right) = f(x)^n.$$

Romania Team Selection Test 2013

P 133. Suppose that the real numbers $a_0, a_1, \cdots, a_n$ and x, with $0 < x < 1$, satisfy

$$\frac{a_0}{1-x} + \frac{a_1}{1-x^2} + \cdots + \frac{a_n}{1-x^{n+1}} = 0.$$

Prove that there exists a real number y with $0 < y < 1$ such that

$$a_0 + a_1 y + \cdots + a_n y^n = 0.$$

William Lowell Putnam Competition 2013

P 134. Consider all polynomials $P(x) \in \mathbb{R}[x]$ that have the following property: for all $x, y \in \mathbb{R}$, one has

$$|y^2 - P(x)| \le 2|x| \quad \text{if and only if} \quad |x^2 - P(y)| \le 2|y|.$$

Determine all possible values of $P(0)$.

IMO Shortlist 2014 A5
Proposed by Belgium

P 135. Show that for each positive integer n, all the roots of the polynomial

$$\sum_{k=0}^{n} 2^{k(n-k)} x^k$$

are real numbers.

William Lowell Putnam Competition 2014

P 136. Each coefficient of a polynomial in $\mathbb{Z}[x]$ with absolute value not exceeding 2015. Prove that every positive root of this polynomial exceeds $\frac{1}{2016}$

Tournament of Towns 2015

P 137. Let $n > 1$ be an integer. Find all non-constant polynomials $P(x) \in \mathbb{R}[x]$ such that

$$P(x)P\left(x^2\right)P\left(x^3\right)\cdots P\left(x^n\right) = P\left(x^{\frac{n(n+1)}{2}}\right).$$

Baltic Way 2015

P 138. Find the least positive integer n, such that there is a polynomial

$$P(x) = a_{2n}x^{2n} + a_{2n-1}x^{2n-1} + \cdots + a_1 x + a_0$$

with real coefficients that satisfies both of the following properties.

- $2014 \le a_i \le 2015$ for $i = 0, 1, \cdots, 2n$.
- There exists a real number ξ with $P(\xi) = 0$.

Germany Team Selection Test 2015

P 139. Let n be a fixed integer with $n \ge 2$. We say that two polynomials P and Q with real coefficients are *block-similar* if for each $i \in \{1, 2, \cdots, n\}$ the sequences

$$P(2015i), P(2015i-1), \cdots, P(2015i-2014) \quad \text{and}$$
$$Q(2015i), Q(2015i-1), \cdots, Q(2015i-2014)$$

are permutations of each other.

(1) Prove that there exist distinct block-similar polynomials of degree $n+1$.
(2) Prove that there do not exist distinct block-similar polynomials of degree n.

IMO Shortlist 2015 A6
Proposed by David Arthur (Canada)

P 140. Let $p(x)$ be a polynomial of degree at most n such that
$$|p(x)| \leq \frac{1}{\sqrt{x}}$$
for all $x \in (0, 1]$. Prove that $|p(0)| \leq 2n + 1$.

Középiskolai Matematikai és Fizikai Lapok, November 2015, A. 654

P 141. Let $p(x) = a_0 + a_1x + \cdots + a_nx^n$ be a polynomial with real coefficients such that $p(x) \geq 0$ for all $x \geq 0$. Prove that for every pair of positive numbers c and d,
$$a_0 + a_1(c + d) + a_2(c + d)(c + 2d) + \cdots + a_n(c + d)(c + 2d) \cdots (c + nd) \geq 0.$$

Középiskolai Matematikai és Fizikai Lapok, December 2015, A. 656

P 142. Find all polynomials $P, Q \in \mathbb{Q}[x]$ such that
$$P(x)^3 + Q(x)^3 = x^{12} + 1.$$

Iran Team Selection Test 2015

P 143. Determine all positive integers n satisfying that $x^n + x^{n-1} + \cdots + x + 1 = g(h(x))$ for some non-constant polynomials $g(x), h(x) \in \mathbb{R}[x]$ with $\deg g(x) < n$ and $\deg h(x) < n$.

Miklós Schweitzer Competition 2015

P 144. Let $P(x) \in \mathbb{R}[x]$. Prove that if there exists an integer k such that $P(k)$ is not an integer, then here exist infinitely many integers m such that $P(m)$ is not an integer.

Polish Mathematical Olympiad 2015

P 145. Let $r(x)$ be a polynomial with real coefficients whose degree n is odd. Prove that the number of pairs of polynomials $p(x)$ and $q(x)$ with real coefficients satisfying the equation
$$p(x)^3 + q\left(x^2\right) = r(x)$$
is smaller than $2n$.

Középiskolai Matematikai és Fizikai Lapok, September 2016, A. 675
Based on the problem of the 1st International Olympiad of Metropolises

P 146. Let $M(x)$ be a real polynomial having no real root. Prove that for every real polynomial $P(x)$ there exists a real polynomial $Q(x)$ such that $P(x)^2 + Q(x)^2$ is divisible by the polynomial $M(x)$.

Középiskolai Matematikai és Fizikai Lapok, November 2016, A. 680
Based on a problem of the Miklos Schweitzer competition

P 147. Let $\{c_i\}_{i=0}^{\infty}$ be a sequence of non-negative real numbers with $c_{2017} > 0$. We consider a sequence of polynomials satisfying the recurrence

$$P_{-1}(x) = 0, \quad P_0(x) = 1, \quad P_{n+1}(x) = xP_n(x) + c_n P_{n-1}(x)$$

Prove that there doesn't exist any integer $n > 2017$ and some real number c such that

$$P_{2n}(x) = P_n\left(x^2 + c\right).$$

Iran Team Selection Test 2017
Proposed by Navid Safaei

P 148. Let $P, Q \in \mathbb{R}[x]$ be relatively prime non-constant polynomials. Show that there can be at most three real numbers λ such that $P + \lambda Q$ is the square of a polynomial.

USA Team Selection Test 2017
Proposed by Alison Miller

My mother said, "Even you, Paul, can be in only one place at one time." Maybe soon I will be relieved of this disadvantage. Maybe, once I've left, I'll be able to be in many places at the same time. Maybe then I'll be able to collaborate with Archimedes and Euclid.

– Paul Erdős

3. Sequences in $\mathbb{Z}$

Why are numbers beautiful? It's like asking why is Beethoven's Ninth Symphony beautiful. If you don't see why, someone can't tell you. I know numbers are beautiful. If they aren't beautiful, nothing is.

– Paul Erdős

D 1. Prove that for every positive integer k there exists a sequence of k consecutive positive integers none of which can be represented as the sum of two squares.

Miklós Schweitzer Competition 1950

D 2. The sequence $x_1 = 1, x_2 = 1, x_3, \cdots$ is defined by the recurrence relation

$$x_{n+2} = 14x_{n+1} - x_n - 4.$$

Prove that all terms of this sequence are perfect squares.

Bulgarian Mathematical Olympiad 1987 (Fourth Round)

D 3. Suppose $A = \{a_1, a_2, a_3, \cdots\}$ is a sequence of distinct positive integers and $B = \{b_1, b_2, b_3, \cdots\}$ is another. Suppose A contains arbitrarily long arithmetic progressions.

(1) If the inequality
$$|b_n - a_n| < K$$
holds for all n, where K is a fixed positive number, prove that B must contain arbitrarily long arithmetic progressions.

(2) If the infinite series
$$\sum_{n=1}^{\infty} \frac{1}{1 + |b_n - a_n|}$$
diverges, does the same conclusion hold?

Amer. Math. Monthly 1988 Problem 6565
Proposed by L. A. Rubel

D 4. Let S_n be the number of polygonal paths in the plane that start at $(0, 0)$, end at (n, n), contain no points above the line $y = x$, and are composed of steps taken from the set $\{(0, 1), (1, 0), (1, 1)\}$. For example, $S_0 = 1$, $S_1 = 2$, $S_2 = 6$, $S_3 = 22$, $S_4 = 90$. (The numbers in the sequence $\{S_n\}$ are often called the Schröder numbers.) Prove that S_n is divisible by 3 for every positive even integer n.

Amer. Math. Monthly 1989 Problem E 3343
Proposed by Lou Shapiro, Douglas Rogers

D 5. Define a function on the positive integers by putting $f(1) = 1$ and setting

$$f(n+1) = f(n) + 2 \quad \text{if} \quad f(f(n) - n + 1) = n$$

and

$$f(n+1) = f(n) + 1 \quad \text{otherwise}.$$

(1) Find and prove a simple explicit formula for $f(n)$ involving the greatest integer function.
(2) Show that if $f(f(n) - n + 1) \neq n$, then $f(f(n) - n + 1) = n + 1$.

Amer. Math. Monthly 1990 Problem E 3391
Proposed by David Doster

D 6. Given an initial integer $n_0 > 1$, two players, $\mathcal{A}$ and $\mathcal{B}$, choose integers n_1, n_2, n_3, $\cdots$ alternately according to the following rules :

- Knowing n_{2k}, $\mathcal{A}$ chooses any integer n_{2k+1} such that
$$n_{2k} \leq n_{2k+1} \leq n_{2k}^2.$$
- Knowing n_{2k+1}, $\mathcal{B}$ chooses any integer n_{2k+2} such that
$$\frac{n_{2k+1}}{n_{2k+2}}$$
is a prime raised to a positive integer power.

Player $\mathcal{A}$ wins the game by choosing the number 1990; player $\mathcal{B}$ wins by choosing the number 1. For which n_0 does :

(1) $\mathcal{A}$ have a winning strategy?
(2) $\mathcal{B}$ have a winning strategy?
(3) Neither player have a winning strategy?

IMO Shortlist 1990, IMO 1990 Problem 5

D 7.

(1) Let $\mathcal{A} = \{a_1, a_2, \cdots\}$ be a strictly increasing sequence of positive integers such that no sum of two or more distinct members of the sequence is equal to another member of the sequence. Let $A(x) = |\{ i \mid a_i \leq x \}|$. Show that
$$\lim_{x\to\infty} \frac{A(x)}{x} = 0.$$
(2) Show that if
$$\lim_{x\to\infty} \frac{f(x)}{x} = 0$$
(no matter how slowly), then there exists a sequence $\mathcal{A}$ for which
$$A(x) > \frac{x}{f(x)}$$
holds on an unbounded set of values of x.

Amer. Math. Monthly 1991 Problem E 3430
Proposed by Paul Erdős

D 8. Find all positive integers k such that the sequence
$$\binom{0}{0}, \binom{2}{1}, \binom{4}{2}, \binom{6}{3}, \cdots$$
is periodic modulo k from some point onward.

Amer. Math. Monthly 1991 Problem E 3457
Proposed by Herbert S. Wilf

D 9. Is there a sequence of natural numbers having the following two properties:

- the sequence is periodic modulo m for every positive integer m,
- each natural number appears in the sequence?

Amer. Math. Monthly 1992 Problem 10814
Proposed by Gerry Myerson

D 10. Let $1 < a_1 < a_2 < a_3 < \cdots$ be an increasing sequence of positive integers.

Is there such a sequence $\{a_k\}$ having the property that, for all integers n (positive, negative, or zero), $\{a_k + n\}$ contains only finitely many primes?

Is there such a sequence $\{a_k\}$ and a constant $B > 0$ having the property that $\{a_k + n\}$ contains no more than B primes for every integer n?

Amer. Math. Monthly 1992 Problem 10208
Proposed by Solomon Golomb

D 11. Choose integers a, b, c, d, and let $K = 2\left(b^2 + a^2 d - abc\right)$. Show that every member of the sequence defined by $y_0 = a_2$, $y_1 = b_2$ and

$$y_{n+1} = (c^2 - 2d)y_n - d^2 y_{n-1} + Kd_n, \quad n > 1$$

is the square of an integer.

Amer. Math. Monthly 1992 Problem 10211
Proposed by Herbert S. Wilf

D 12.

(1) Show that there exist infinitely many positive integers a such that both $a+1$ and $3a+1$ are perfect squares.
(2) Let $a_1 < a_2 < \cdots$ be the sequence of all solutions of (1). Show that $a_n a_{n+1} + 1$ is also a perfect square.

Amer. Math. Monthly 1992 Problem 10238
Proposed by David M. Bloom

D 13. Define a sequence $a_0 = 3, a_1 = 0, a_2 = 2, a_3, \cdots$ by

$$a_{n+3} = a_n + a_{n+1}, \quad n \geq 0.$$

If p is a prime, show that $p \,|\, a_n$.

Amer. Math. Monthly 1992 Problem 10268
Proposed by Ondrej Šuch

D 14. For a natural number n, let $t(n)$ be the sum of the divisors d of n in the range $1 < d < n$ with $\frac{n}{d}$ being squarefree. Is there an integer n for which the sequence

$$n,\ t(n),\ t(t(n)), \cdots$$

is unbounded?

Amer. Math. Monthly 1993 Problem 10323
Proposed by David E. Penney, Carl Pomerance

D 15. Let k be a positive integer and r_n be the remainder when $\binom{2n}{n}$ by k. Find all k for which the sequence $r_1, r_2, r_3, \cdots$ is eventually periodic.

Bulgarian Mathematical Olympiad 1994 (Fourth Round)

D 16. Let $a_1 = 1$, $a_2 = 2$, $\cdots$ be the sequence of positive integers of the form $2^\alpha 3^\beta$, where α and β are nonnegative integers. Prove that every positive integer is expressible in the form $a_{i_1} + \cdots a_{i_n}$, where no summand is a multiple of any other.

Mathematics Magazine 1994 Problem Q814
Proposed by Paul Erdős

D 17. As the Minister of Finance of a newly independent country, it is your job to design a new currency: a sequence $a_1 < a_2 < a_3 < \cdots$. of positive integers, with $a_l = 1$ to be the denominations of various coins and bills. Although you are authorized to create an infinite number of denominations, the legislature has passed some laws restricting your choices.

(1) Rule l: There must be a bound b on the number of items needed for any payment.
(2) Rule 2: the "denomination density",
$$\lim_{k\to\infty} \frac{k}{d_k}$$
must be zero.
(3) Rule 3: repeatedly choosing the largest denomination less than or equal to the amount remain- ing to be paid (the greedy algorithm) always leads to the use of the minimal number of items to pay any amount.

Can you design a currency meeting these rules?

Amer. Math. Monthly 1995 Problem 10465
Proposed by Paul K. Stockmeyer

D 18. Can the numbers $1, 2, 3, \cdots, 100$ be covered with 12 geometric progressions?

All-Russian Mathematical Olympiad 1995
Proposed by A. Golovanov

D 19. Fix a positive integer k. Let $f_k(m,n)$ be the number of m-tuples $a = (a_0, a_1, \cdots, a_{m-1})$ of integers satisfying: (a) $0 < a_i < n-1$ for all i, and (b) any k circularly consecutive entries of a (i.e., $a_i, a_{i+l}, \cdots, a_{i+k-1}$, where the subscripts are taken modulo m so that they lie between 0 and $m-1$) are all distinct. Show that the generating function
$$F_k(x,n) = \sum_{m\geq 1} f_k(m,n)x^m$$
is a quotient of two polynomials in x and n.

Amer. Math. Monthly 1995 Problem 10493
Proposed by Richard R Stanley, Christophe Reutenaue

D 20. The numbers from 1 to 100 are written in an unknown order. One may ask about any 50 numbers and find out their relative order. What is the fewest questions needed to find the order of all 100 numbers?

All-Russian Mathematical Olympiad 1996
Proposed by S. Tokarev

D 21. Consider the sequence $y_2, y_3, \cdots$ defined by the recurrence relation
$$(n+1)(n-2)y_{n+1} = n\left(n^2-n-1\right)y_n - (n-1)^3 y_{n-1}$$
and initial conditions $y_2 = y_3 = 1$. Show that y_n is an integer if and only if n is prime.

Amer. Math. Monthly 1997 Problem 10578
Proposed by Herbert S. Wilf

D 22. The sequence $a_0 = 1, a_1 = 1, a_2, \cdots$ is defined by the recursion
$$(n+1)a_{n+1} = (2n+1)a_n + 3na_{n-1}.$$
Prove that the sequence consists of integer numbers.

Középiskolai Matematikai és Fizikai Lapok, October 1997, N. 149

D 23. In an infinite arithmetic progression of distinct positive integers, we replace each term by the sum of its digits. Can it happen that the new sequence is again an arithmetic progression?

Középiskolai Matematikai és Fizikai Lapok, November 1997, Gy. 3160

D 24. The sequence $a_0 = 1, a_1, a_2, \cdots$ is defined by the recursion

$$a_{n+1} = \sin a_n$$

Prove that the sequence $\{n{a_n}^2\}_{n\geq 0}$ is bounded.

Középiskolai Matematikai és Fizikai Lapok, November 1997, F. 3196

D 25. In an infinite sequence of positive integers, each term has the same number of divisors. Prove that there is an infinite subsequence in which any two terms have the same greatest common divisor.

Középiskolai Matematikai és Fizikai Lapok, December 1997, N. 155

D 26. Suppose that $a_1 = 0, a_2, \cdots$ and $b_1 = 0, b_2, \cdots$ are integer sequences such that

$$a_n = nb_n + a_1 b_{n-1} + a_2 b_{n-2} + \cdots + a_{n-1} b_1$$

holds for all $n \geq 2$. Prove that, for any prime number p, a_p is divisible by p.

Középiskolai Matematikai és Fizikai Lapok, December 1997, N. 156

D 27. Let n be a natural number. Find the least natural number k for which there exist k sequences of 0 and 1 of length $2n+2$ with the following property: any sequence of 0 and 1 of length $2n+2$ coincides with some of these k sequences in at least $n+2$ positions.

Bulgarian Mathematical Olympiad 1998 (Fourth Round)

D 28. The sequence $a_1 = 0, a_2 = 2, a_3 = 3, a_4 \cdots$ is defined by the recursion

$$a_n = \max_{1<d<n} \{a_d a_{n-d}\}$$

for all $n \geq 4$. Find a_{1998}.

Középiskolai Matematikai és Fizikai Lapok, January 1998, Gy. 3176

D 29. Is there any number in the Fibonacci sequence whose six last digits (in decimal system) are all 9?

Középiskolai Matematikai és Fizikai Lapok, February 1998, N. 164

D 30. Is there any sequence $a_1, a_2, a_3, \cdots$ of natural numbers such that, for every natural number k, the sequence

$$a_1 + k,\ a_2 + k,\ a_3 + k,\ \cdots$$

contains only a finite number of primes?

Középiskolai Matematikai és Fizikai Lapok, November 1998, Gy. 3230

D 31. The Lucas sequence $L_0 = 2, L_1 = 1, L_2, \cdots$ is defined by the recursion

$$L_{n+2} = L_n + L_{n-1}.$$

Let, in addition, $a_1, a_2, a_3, \cdots$ be a sequence of integers such that, for all $n \in \mathbb{N}$,

$$\sum_{d|n} a_d = L_n.$$

Prove that

$$n|a_n$$

for all $n \in \mathbb{N}$.

Középiskolai Matematikai és Fizikai Lapok, December 1998, N. 191

D 32. Consider the function

$$f(n) = \begin{cases} \frac{3n}{2} & \text{if } n \text{ is even,} \\ \frac{3n+1}{2} & \text{if } n \text{ is odd.} \end{cases}$$

Let $N > 1$ be a positive integer, and define $a_n(i)$ to be the remainder when the ith iterate of f, $f^i(n)$, is divided by N. Prove that, for any $n \geq 1$, the sequence $\{a_n(i)\}_{i \geq 0}$ is not periodic.

Mathematics Magazine 1998 Problem 1547
Proposed by Homer White, Robert Bailey

D 33. Let $a_1, a_2, a_3, \cdots$ be a sequence of integers satisfying the recurrence relation

$$(n-1)a_{n+1} = (n+1)a_n - 2(n-1).$$

If $2000 \,|\, a_{1999}$, find the smallest $n \geq 2$ such that $2000 \,|\, a_n$.

Bulgarian Mathematical Olympiad 1999 (Fourth Round)
Proposed by O. Mushkarov, N. Nikolov

D 34. Initially numbers from 1 to 1000000 are all colored black. A move consists of picking one number, then change the color (black to white or white to black) of itself and all other numbers not coprime with the chosen number. Can all numbers become white after finite numbers of moves?

All-Russian Mathematical Olympiad 1999

D 35. Let $\alpha \in (1, 2)$ be a constant. Define the sequence $a_0, a_1, a_2, \cdots$ by

$$a_n = \lfloor \alpha^n \rfloor$$

for all $n \in \mathbb{N}_0$. Is it true that all but a finite number of positive integers are expressible as the sum of the elements of some finite subsequence of $\{a_n\}_{n \geq 0}$?

Középiskolai Matematikai és Fizikai Lapok, January 1999, N. 197

D 36. Count those n-term sequences of positive integers in which the number of terms greater than i is at most $n - i$, for each $i \in \{1, \cdots, n\}$.

Középiskolai Matematikai és Fizikai Lapok, May 1999, N. 211

D 37. Prove that the set of positive integers cannot be partitioned into three nonempty subsets such that, for any two integers x, y taken from two different subsets, the number $x^2 - xy + y^2$ belongs to the third subset.

IMO Shortlist 1999 A4

D 38. Prove that one can partition the set of natural numbers into 100 nonempty subsets such that among any three natural numbers a, b, c satisfying $a + 99b = c$, there are two that belong to the same subset.

All-Russian Mathematical Olympiad 2000

D 39. A sequence of numbers is called of Fibonacci-type if each term, after the first two, is the sum of the previous two. Prove that the set of positive integers can be partitioned into the disjoint union of infinite Fibonacci-type sequences.

Középiskolai Matematikai és Fizikai Lapok, September 2000, A. 244
Proposed by E. Fried

D 40. Find all positive integers $a_0 = 1, a_1, a_2, \cdots, a_n$ satisfying the following conditions.

- $(a_{k+1} - 1)a_{k-1} \geq {a_k}^2(a_k - 1)$ for all $k \in \{1, 2, \cdots, n-1\}$.
- $\frac{99}{100} = \frac{a_0}{a_1} + \frac{a_1}{a_2} + \cdots + \frac{a_{n-1}}{a_n}$.

IMO Shortlist 2001 A5

D 41. Find all finite sequences $(x_0, x_1, \cdots, x_n)$ such that for every j, $0 \leq j \leq n$, x_j equals the number of times j appears in the sequence.

IMO Shortlist 2001 C5

D 42. In a sequence $a_1, a_2, \cdots$ of positive integers, let f_n denote the number of times the positive integer n occurs in the sequence, provided that this number is finite. If f_n is finite for every n, then the sequence $f_1, f_2, \cdots$ is called the frequency sequence of $a_1, a_2, \cdots$. If the sequence $f_1, f_2, \cdots$ also admits a frequency sequence, it is then called the second frequency sequence of the sequence $a_1, a_2, \cdots$. Frequency sequences of higher order are defined similarly. Is there any sequence which admits a k-th frequency sequence for every positive integer k?

Középiskolai Matematikai és Fizikai Lapok, March 2001, B. 3444

D 43. The sequence $a_1 = 1, a_2 = 1, a_3 = 1, a_4, \cdots$ is defined by the recursion

$$a_{n+1} = \frac{{a_n}^2 + {a_{n-1}}^2}{a_{n-2}}$$

for all $n \geq 3$. Prove that each term of the sequence is an integer.

Középiskolai Matematikai és Fizikai Lapok, April 2001, A. 265

D 44. Let A be a non-empty set of positive integers. Suppose that there are positive integers $b_1, \cdots b_n$ and $c_1, \cdots, c_n$ such that

- for each i the set $b_iA + c_i = \{b_ia + c_i \colon a \in A\}$ is a subset of A, and
- the sets $b_iA + c_i$ and $b_jA + c_j$ are disjoint whenever $i \neq j$

Prove that

$$\frac{1}{b_1} + \cdots + \frac{1}{b_n} \leq 1.$$

IMO Shortlist 2002 A6

D 45. Each member of the sequence $a_1, a_2, \cdots, a_{2n}, a_{2n+1} = a_1$ is either 2, 5 or 9. No two consecutive members are equal. Prove that

$$a_1a_2 - a_2a_3 + a_3a_4 - a_4a_5 + \cdots - a_{2n}a_{2n+1} = 0.$$

Középiskolai Matematikai és Fizikai Lapok, May 2002, B. 3552

D 46. A random sequence is produced using the digits 0, 1, 2. How long should the sequence be so that the probability that all three digits occur in the sequence is at least 0.61?

Középiskolai Matematikai és Fizikai Lapok, October 2002, B. 3576
Proposed by E. Fried

D 47. The Lucas sequence $L_0 = 2, L_1 = 1, L_2, \cdots$ is defined by the recursion

$$L_{n+2} = L_n + L_{n-1}.$$

Prove that, for all prime $p > 3$, the integer $\left[L_{\frac{p+1}{2}}\right]^p - 1$ is divisible by p.

Középiskolai Matematikai és Fizikai Lapok, November 2003, A. 330

D 48. Does there exist an increasing arithmetic progression $a_1, \cdots, a_{100}$ of positive integers such that

$$\gcd(a_i, a_j) = 1$$

for all $i, j \in \{1, \cdots 100\}$ with $i \neq j$?

Tournament of Towns 2003

D 49. Let $f(0,0) = 5^{2003}, f(0,n) = 0$ for every integer $n \neq 0$ and

$$f(m,n) = f(m-1,n) - 2 \cdot \left\lfloor \frac{f(m-1,n)}{2} \right\rfloor + \left\lfloor \frac{f(m-1,n-1)}{2} \right\rfloor + \left\lfloor \frac{f(m-1,n+1)}{2} \right\rfloor$$

for every natural number $m > 0$ and for every integer n. Prove that there exists a positive integer M such that $f(M,n) = 1$ for all integers n such that $|n| \leq \frac{(5^{2003}-1)}{2}$ and $f(M,n) = 0$ for all integers n such that $|n| > \frac{5^{2003}-1}{2}$.

Vietnam Team Selection Test 2003

D 50. Consider the sequence $a_0 = 1, a_1, a_2, \cdots$ such that $a_{2n+1} = a_n$ and $a_{2n+2} = a_n + a_{n+1}$ for all $n \in \mathbb{N}_0$. Prove that every positive rational number occurs among the elements of the set

$$\left\{ \frac{a_1}{a_0}, \frac{a_2}{a_1}, \frac{a_3}{a_2}, \cdots \right\} = \left\{ \frac{1}{1}, \frac{1}{2}, \frac{2}{1}, \frac{1}{3}, \frac{3}{2}, \cdots \right\}.$$

Középiskolai Matematikai és Fizikai Lapok, April 2004, B. 3728

D 51. Prove that the sequences $\left\{2^{2^n} + 1 \mid n \in \mathbb{N}\right\}$ and $\left\{6^{2^n} + 1 \mid n \in \mathbb{N}\right\}$ together contain infinitely many composite numbers.

Középiskolai Matematikai és Fizikai Lapok, October 2004, A. 353

D 52. Does there exist a sequence of 2005 consecutive positive integers that contains exactly 25 prime numbers?

Hungary-Israel Binational Mathematical Competition 2005

D 53. Let $a_1, a_2, \cdots$ be a sequence of integers with infinitely many positive and negative terms. Suppose that for every positive integer n the numbers $a_1, a_2, \cdots, a_n$ leave n different remainders upon division by n. Prove that every integer occurs exactly once in the sequence $a_1, a_2, \cdots$.

IMO Shortlist 2005 N2, IMO 2005 Problem 2

D 54. Prove that if $n > 1$ is an integer then there exists a sequence of integers $a_1 = n < a_2 < a_3 < \cdots$ such that ${a_1}^2 + \cdots + {a_k}^2$ is divisible by $a_1 + \cdots + a_k$ for all $k \in \mathbb{N}$.

Középiskolai Matematikai és Fizikai Lapok, May 2007, B. 4002

D 55. A set of balls contains n balls which are labeled with numbers $1, 2, 3, \cdots, n$. We are given $k > 1$ such sets. We want to color the balls with two colors, black and white in such a way, that

- the balls labeled with the same number are of the same color,
- any subset of $k+1$ balls with (not necessarily different) labels $a_1, a_2, \cdots, a_{k+1}$ satisfying the condition $a_1 + a_2 + \cdots + a_k = a_{k+1}$, contains at least one ball of each color.

Find, depending on k, the greatest possible number n which admits such a coloring.

Middle European Mathematical Olympiad 2007

D 56. Prove Green-Tao Theorem that there are arbitrarily long arithmetic progressions of primes.

Ben Green, Terence Tao, Ann. of Math. (2) 167 (2008), no. 2, 481–547.

D 57. The sequence $a_0 = 1$, $a_1 = 1$, $a_2, \cdots$ satisfies the recurrence relation

$$a_{n+1} = \frac{1 + {a_n}^2}{a_{n-1}}.$$

for all $n \in \mathbb{N}$. Prove that all the terms of the sequence are integers.

Hungary-Israel Binational Mathematical Competition 2008

D 58. The sequence $a_1, a_2, \cdots$ of integers has infinitely many positive terms and infinitely many negative terms. For every n, the remainders of $a_1, a_2, \cdots, a_n$ divided by n are pairwise different. How many times does the number 2008 occur in the sequence?

Középiskolai Matematikai és Fizikai Lapok, January 2008, B. 4053

D 59. Let $t_3(a_1, a_2, \cdots, a_k)$ denote the number of three term arithmetic progressions that can be selected from the terms of a sequence $a_1 < a_2 < \cdots < a_k$. Prove that $t_3(a_1, a_2, \cdots, a_k) \le t_3(1, 2, \cdots, k)$.

Középiskolai Matematikai és Fizikai Lapok, December 2008, B. 4134

D 60. Let $a_1 < a_2 < \cdots$ be an increasing sequence of integers. For all quadruple (i, j, k, l) of indices such that $1 \le i < j \le k < l$ and $i + l + j + k$ we have

$$a_i + a_l > a_j + a_k.$$

Determine the least possible value of a_{2008}.

Middle European Mathematical Olympiad 2008

D 61.

(1) Do there exist 2009 distinct positive integers such that their sum is divisible by each of the given numbers?
(2) Do there exist 2009 distinct positive integers such that their sum is divisible by the sum of any two of the given numbers?

Hungary-Israel Binational Mathematical Competition 2009

D 62. Suppose that $s_1, s_2, s_3, \cdots$ is a strictly increasing sequence of positive integers such that the sub-sequences

$$s_{s_1},\, s_{s_2},\, s_{s_3},\, \cdots \qquad \text{and} \qquad s_{s_1+1},\, s_{s_2+1},\, s_{s_3+1},\, \cdots$$

are both arithmetic progressions. Prove that the sequence $s_1, s_2, s_3, \cdots$ is itself an arithmetic progression.

IMO Shortlist 2009 A6, IMO 2009 Problem 3
Proposed by Gabriel Carroll (USA)

D 63. Prove that the sequence $\left\{\lfloor n\sqrt{2}\rfloor\right\}_{n\geq 1}$ contains infinitely many integer powers of 3.

Középiskolai Matematikai és Fizikai Lapok, April 2009, B. 4180

D 64. The terms of a sequence are positive integers. The first two terms are 1 and 2. No term of the sequence is equal to the sum of two different terms. Prove that for any natural number k, the number of terms less than k is at most $2+\frac{k}{3}$.

Középiskolai Matematikai és Fizikai Lapok, September 2010, B. 4285

D 65. There are given a positive integer m and an infinite sequence $a_1 < a_2 < \cdots$ of positive integers such that $a_k \leq mk$ holds for infinitely many indices k. Prove that there exist positive integers $b_1, \cdots, b_m$ with the property that every integer can be written as $a_i - a_j + b_k$ where i, j are positive integers and $1 \leq k \leq m$.

Középiskolai Matematikai és Fizikai Lapok, December 2010, A. 521

D 66. Let a be an arbitrary integer. Consider the recursive sequence of integers defined by $u_0 = 4$, $u_1 = 0$, $u_2 = 2$, $u_3 = 3$, and

$$u_{n+4} = u_{n+2} + u_{n+1} + a\, u_n$$

for every integer $n \geq 0$. Prove that p divides u_p for every prime p.

Mathematics Magazine 2010 Problem 1851
Proposed by Éric Pité

D 67. Let $f(n)$ be the number of binary words $a_1 \cdots a_n$ of length n that have the same number of pairs $a_i a_{i+1}$ equal to 00 as 01. Show that

$$\sum_{n=0}^{\infty} f(n)t^n = \frac{1}{2}\left(\frac{1}{1-t} + \frac{1+t}{\sqrt{(1-t)(1-2t)\,(1+t+2t^2)}}\right).$$

Amer. Math. Monthly 2011 Problem 11610
Proposed by Richard Stanley

D 68. Determine all sequences $(x_1, x_2, \cdots, x_{2011})$ of positive integers, such that for every positive integer n, there exists an integer a with

$$\sum_{j=1}^{2011} j\, {x_j}^n = a^{n+1} + 1$$

IMO Shortlist 2011 A2
Proposed by Warut Suksompong (Thailand)

D 69. The elements of the set $\mathcal{H}$ are finite sequences whose elements are from $\{1, 2, 3\}$. Suppose that no element of $\mathcal{H}$ is a subsequence of any other element of $\mathcal{H}$. Show that $\mathcal{H}$ is finite.

Középiskolai Matematikai és Fizikai Lapok, September 2011, A. 541
Proposed by András Pongrácz

D 70. Given a positive integer number k, define the function $f : \mathbb{N} \to \mathbb{N}$ by

$$f(n) = \begin{cases} 1, & \text{if } n \leq k+1, \\ f(f(n-1)) + f(n - f(n-1)), & \text{if } n > k+1. \end{cases}$$

Show that the preimage of every positive integer number under f is a finite non-empty set of consecutive positive integers.

Romania Team Selection Test 2011

D 71. Let $f : \mathbb{N} \to \mathbb{N}$ be a function. Suppose that for every $n \in \mathbb{N}$ there exists a $k \in \mathbb{N}$ such that $f^{2k}(n) = n + k$, and let k_n be the smallest such k. Prove that the sequence $k_1, k_2, \cdots$ is unbounded.

IMO Shortlist 2012 A6

D 72. The sequence $a_1 = 1, a_2 = 1, a_3 = 2, a_4 \cdots$ is defined by the recursion

$$a_{n+3} = \frac{a_{n+2}a_{n+1} + n!}{a_n}.$$

Prove that the sequence consists of integer numbers.

Középiskolai Matematikai és Fizikai Lapok, December 2012, B. 4497

D 73. Let $a_1, a_2, \cdots$ be a strictly increasing sequence of positive integers such that

$$\sum_{k=2}^{\infty} \frac{1}{a_k \log a_k}$$

diverges. Prove that

$$\operatorname{lcm}(a_1, \cdots, a_k) = \operatorname{lcm}(a_1, \cdots, a_k, a_{k+1})$$

for infinitely many $k \in \mathbb{N}$.

Amer. Math. Monthly 2013 Problem 11699
Proposed by Bakir Farhi

D 74. Let n be a positive integer, and consider a sequence $a_1, a_2, \cdots, a_n$ of positive integers. Extend it periodically to an infinite sequence $a_1, a_2, \cdots$ by defining $a_{n+i} = a_i$ for all $i \geq 1$. If

$$a_1 \leq a_2 \leq \cdots \leq a_n \leq a_1 + n$$

and

$$a_{a_i} \leq n + i - 1 \quad \text{for} \quad i = 1, 2, \cdots, n,$$

prove that

$$a_1 + \cdots + a_n \leq n^2.$$

IMO Shortlist 2013 A4

D 75. Find all positive integers n for which there exists a sequence $a_1 < a_2 < \cdots < a_n$ of positive integers with the property that the sums $a_i + a_j$ with $1 \leq i < j \leq n$ are distinct and among them each modulo 4 residue occurs the same number of times.

Középiskolai Matematikai és Fizikai Lapok, April 2013, A. 587, Proposed by Ilya Bogdanov

D 76. Let $a_0 < a_1 < a_2 \cdots$ be an infinite sequence of positive integers. Prove that there exists a unique integer $n \geq 1$ such that
$$a_n < \frac{a_0 + a_1 + a_2 + \cdots + a_n}{n} \leq a_{n+1}.$$

IMO Shortlist 2014 A1
Proposed by Gerhard Wöginger (Austria)

D 77. Is it true that for every infinite sequence $x_1, x_2, \cdots$ of integers satisfying $|x_{k+1}x_k| = 1$ for every positive integer k, there exists a sequence $k_1 < k_2 < \cdots < k_{2014}$ of positive integers such that as well the indices $k_1, k_2, \cdots, k_{2014}$ as the numbers $x_{k_1}, x_{k_2}, \cdots, x_{k_2014}$ (in this order) form arithmetic progressions?

Középiskolai Matematikai és Fizikai Lapok, November 2014, A. 628
Proposed by E. Csóka, Ben Green

D 78. Let n, m be integers greater than 1, and let $a_1, a_2, \ldots, a_m$ be positive integers not greater than n^m. Prove that there exist positive integers $b_1, b_2, \ldots, b_m$ not greater than n, such that
$$\gcd(a_1 + b_1, a_2 + b_2, \ldots, a_m + b_m) < n,$$
where $\gcd(x_1, x_2, \ldots, x_m)$ denotes the greatest common divisor of $x_1, x_2, \ldots, x_m$

European Girls' Mathematical Olympiad 2015

D 79. Determine whether there exists an infinite sequence $a_1, a_2, a_3, \ldots$ of positive integers such that
$$a_{n+2} = a_{n+1} + \sqrt{a_{n+1} + a_n}$$
for all $n \in \mathbb{N}$.[/quote]

European Girls' Mathematical Olympiad 2015

D 80. The sequence $a_1, a_2, \cdots$ of integers satisfies the conditions:
- $1 \leq a_j \leq 2015$ for all $j \geq 1$,
- $k + a_k \neq \ell + a_\ell$ for all $1 \leq k < \ell$.

Prove that there exist two positive integers b and N for which
$$\left| \sum_{j=m+1}^{n} (a_j - b) \right| \leq 1007^2$$
for all integers m and n such that $n > m \geq N$.

IMO Shortlist 2015 C5, IMO 2015 Problem 6
Proposed by Ivan Guo and Ross Atkins (Australia)

D 81. Let $f : \mathbb{N} \to \mathbb{N}$ be a function. For any $m, n \in \mathbb{N}$, we write
$$f^n(m) = \underbrace{f(f(\cdots f}_{n}(m)\cdots)).$$
Suppose that f has the following two properties.
- if $m, n \in \mathbb{N}$, then $\frac{f^n(m) - m}{n} \in \mathbb{N}$.
- The set $\mathbb{N} \setminus \{f(n) \mid n \in \mathbb{N}\}$ is finite.

Prove that the sequence $f(1) - 1, f(2) - 2, f(3) - 3, \cdots$ is periodic.

IMO Shortlist 2015 N6
Proposed by Ang Jie Jun (Singapore)

D 82. Let $f : \mathbb{N}_0 \to \mathbb{N}_0$ be a function satisfying the following conditions.

- $f(0) = 0$.
- $f(2n + 1) = 2f(n)$ for all $n \in \mathbb{N}_0$.
- $f(2n) = 2f(n) + 1$for all $n \in \mathbb{N}_0$.

If $g(n) = f(f(n))$, prove that $g(n - g(n)) = 0$ for all $n \geq 0$.

Indian National Mathematical Olympiad Program 2015

D 83. The sequence $a_1, a_2, \cdots$ satisfies the following conditions.

- $a_1 = a_2 = 1$, $a_3 = 2$.
- $a_{n+3} = \frac{a_{n+2}a_{n+1}+n!}{a_n}$ for all $n \in \mathbb{N}_0$.

Prove that all the terms in the sequence are integers.

Indian National Mathematical Olympiad Program 2015

D 84. Let m, n be positive integers. Let $S(n, m)$ be the number of sequences of length n and consisting of 0 and 1 in which there exists a 0 in any consecutive m digits. Prove that

$$S(2015n, n)S(2015m, m) \geq S(2015n, m)S(2015m, n)$$

Turkey Team Selection Test 2015

D 85. Let n be an odd positive integer, and let $x_1, x_2, \cdots, x_n, x_{n+1} = x_1$ be non-negative real numbers. Show that

$$\min_{i=1,\ldots,n}(x_i^2 + x_{i+1}^2) \leq \max_{j=1,\ldots,n}(2x_jx_{j+1}).$$

European Girls' Mathematical Olympiad 2016

D 86. Determine whether the set of rational numbers can be ordered to in a sequence $q_1, q_2, \cdots$ in such a way that there is no sequence of indices $1 \leq i_1 < i_2 < \cdots < i_6$ such that $q_{i_1}, q_{i_2}, \cdots, q_{i_6}$ form an arithmetic progression.

Középiskolai Matematikai és Fizikai Lapok, April 2016, A. 669
Proposed by Gyula Károlyi, Péter Komjath

D 87. Let $a_1, a_2, a_3, \cdots$ be a sequence of nonnegative integers such that

$$\sum_{i=1}^{2n} a_{id} \leq n$$

holds for all $n, d \in \mathbb{N}$. Prove that for every positive integer K, there are some positive integers N and D such that

$$\sum_{i=1}^{2N} a_{iD} = N - K.$$

Középiskolai Matematikai és Fizikai Lapok, April 2016, A. 670
Chinese problem

D 88. Let $a_1, a_2, \cdots, a_n \in \{2, 3, 4, \cdots\}$ be integers satisfying the inequality

$$\sum_{i=1}^{n} \frac{1}{1+a_i} \leq \frac{7}{12}.$$

Every year, the government of Optimistica publishes its Annual Report with n economic indicators. For each $i \in \{1, \cdots, n\}$, the possible values of the i-th indicator are $1, 2, \cdots, a_i$. The Annual Report is said to be optimistic if at least $n-1$ indicators have higher values than in the previous report. Prove that the government can publish optimistic Annual Reports in an infinitely long sequence.

Középiskolai Matematikai és Fizikai Lapok, September 2016, A. 674
Based on the problem of the 1st International Olympiad of Metropolises

D 89. Let $n \geq 4$ be an integer. Find all functions $W : \{1, \cdots, n\} \times \{1, \cdots, n\} \to \mathbb{R}$ such that for every partition $\{1, \cdots, n\} = A \cup B \cup C$ into disjoint sets,

$$\sum_{a \in A} \sum_{b \in B} \sum_{c \in C} W(a, b) W(b, c) = |A||B||C|.$$

USA Team Selection Test 2016

D 90. Prove that each term of the sequence $a_1 = 1, a_2, a_3, \cdots$ defined by the recurrence relation

$$a_n = \frac{4a_{n-1} + \sqrt{7{a_{n-1}}^2 - 3}}{3}$$

is rational.

Középiskolai Matematikai és Fizikai Lapok, January 2017, B. 4845
Proposed by J. Szoldatics

D 91. Let $c \geq 3$ be an integer, and define the sequence $a_1 = c^2 - 1, a_2, \cdots$ by the recurrence

$$a_{n+1} = {a_n}^3 - 3{a_n}^2 + 3.$$

Show that for every integer $n \geq 2$, the number a_n has a prime divisor that does not divide any of $a_1, \cdots, a_{n1}$.

Középiskolai Matematikai és Fizikai Lapok, February 2017, A. 691

D 92. The sequence $a_1, a_2, a_3, \cdots$ of positive integers satisfies $a_n a_{n+1} = a_{n+2} a_{n+3}$ for all $n \in \mathbb{N}$. Show that the sequence is eventually periodic.

Középiskolai Matematikai és Fizikai Lapok, May 2017, B. 4880
Proposed by M. E. Gáspár

There'll be plenty of time to rest in the grave.

– Paul Erdős (1913 – 1996)

4. Sequences in $\mathbb{R}$

Katherine: **Euler's Method.**
Paul Stafford: **That's ancient.**
Katherine: **Yes. But it works. It works numerically!**

Hidden Figures (2016)

S 1. Let $0 < k < 1$ be a given real number and let $a_1, a_2, \cdots$ be an infinite sequence of real numbers which satisfies

$$a_{n+1} \leq \left(1 + \frac{k}{n}\right) a_n - 1.$$

Prove that there is an integer t such that $a_t < 0$.

Bulgarian Mathematical Olympiad 1986 (Fourth Round)

S 2. The sequence $a_0 = 2, a_1, a_2, \cdots$ is defined by the recurrence relation

$$a_{n+1} = a_n + \frac{1}{a_n}.$$

Prove that $62 < a_{1974} < 77$.

Bulgarian Mathematical Olympiad 1974
Proposed by I. Tonov

S 3. Let $r_1, r_2, r_3, \cdots$ be any enumeration of the rationals in the interval $(0, 1)$. For each $j > 1$, let

$$r_j = .\, a_{j,1} a_{j,2} a_{j,3} \cdots$$

be a decimal representation of r_j.

(1) Prove that the main diagonal $.\, a_{1,1} a_{2,2} a_{3,3} \cdots$ is irrational.
(2) For arbitrary positive integers j and k, prove that the diagonal $.\, a_{j,k} a_{j+1,k+1} a_{j+2,k+2} \cdots$ is irrational.

Mathematics Magazine 1986 Problem 1239
Proposed by Bruce Hanson

D 93. For each positive integer n, let $f(n)$ be the smallest integer r for which there is an increasing sequence of integers $n = a_1 < a_2 < \cdots < a_k = r$ such that the product $a_1 a_2 \cdots a_k$ is a perfect square. For example, $f(2) = 6$, $f(3) = 8$, $f(4) = 4$, $f(8) = 15$. Prove that f is a one-to-one function.

Mathematics Magazine 1986 Problem 1242
Proposed by Ronald Graham

S 4. Suppose $x_1, x_2, x_3, \cdots$ is a sequence of numbers in $[0, 1)$ such that at least one of its sequential limit points is irrational. For given n, consider the 2^n numbers

$$\epsilon_1 x_1 + \epsilon_2 x_2 + \cdots + \epsilon_n x_n$$

where each ϵ_i takes the two values ± 1. If $0 \leq a < b \leq 1$, let $N_n(a, b)$ be the number of n-tuples $(\epsilon_1, \epsilon_2, \cdots, \epsilon_n)$ such that the fractional part of $\epsilon_1 x_1 + \epsilon_2 x_2 + \cdots + \epsilon_n x_n$ is in $[a, b)$. Prove that for every a, b, we have

$$\lim_{n \to \infty} \frac{N_n(a, b)}{2^n} = b - a.$$

Amer. Math. Monthly 1987 Problem 6542
Proposed by Andrew M. Odlyzko

S 5. Assume that $a_1,\, a_2,\, a_3,\, \cdots$ are real numbers satisfying the inequality

$$|\, a_{m+n} - a_m - a_n \,| \leq C$$

for all $m, n \in \mathbb{N}$. Prove that there exists a constant k such that

$$|\, a_n - k\, n \,| \leq C$$

for all $n \in \mathbb{N}$.

Crux Mathematicorum 1988 Problem 1320
Proposed by Themistocles M. Rassias

S 6. An infinite sequence $x_0, x_1, x_2, \cdots$ of real numbers is said to be *bounded* if there is a constant C such that $|x_i| \leq C$ for every $i \geq 0$. Given any real number $a > 1$, construct a bounded infinite sequence $x_0, x_1, x_2, \cdots$ such that

$$|x_i - x_j|\, |i-j|^a \geq 1$$

for every pair of distinct nonnegative integers i, j.

IMO Shortlist 1991, IMO 1991 Problem 6

S 7. The sequence $a_1 = 1, a_2, a_3, \cdots$ is defined by the recurrence relation

$$a_{n+1} = \frac{a_n}{n} + \frac{n}{a_n}.$$

Prove that $\lfloor {a_n}^2 \rfloor = n$ for all $n \geq 4$.

Bulgarian Mathematical Olympiad 1996 (Fourth Round)

S 8. The sequence $x_1 \in (0,1), x_2, x_3, \cdots$ is defined by the recursion

$$x_{n+1} = x_n - {x_n}^2$$

for all $n \in \mathbb{N}$. Prove that

$${x_1}^2 + {x_2}^2 + \cdots + {x_n}^2 < 1$$

no matter how large n is.

Középiskolai Matematikai és Fizikai Lapok, October 1998, F. 3246

S 9. Let the sequence $\{a_n\}_{n\geq 1}$ be defined by $a_1 = t$ and $a_{n+1} = 4a_n(1-a_n)$ for $n \geq 1$. How many possible values of t are there, if $a_{1998} = 0$?

Turkey Team Selection Test 1998

S 10. Let $a_0, a_1, a_2, \cdots$ be an arbitrary infinite sequence of positive numbers. Show that the inequality

$$1 + a_n > 2^{\frac{1}{n}}\, a_{n-1}$$

holds for infinitely many positive integers n.

IMO Shortlist 2001 A2

S 11. Let $x_1, x_2, \cdots, x_n$ be arbitrary real numbers. Prove the inequality

$$\frac{x_1}{1+x_1^2} + \frac{x_2}{1+x_1^2+x_2^2} + \cdots + \frac{x_n}{1+x_1^2+\cdots+x_n^2} < \sqrt{n}.$$

IMO Shortlist 2001 A3

S 12. Let $a_1, a_2, \cdots$ be an infinite sequence of real numbers, for which there exists a real number c with $0 \leq a_i \leq c$ for all i, such that

$$|a_i - a_j| \geq \frac{1}{i+j} \quad \text{for all } i,\ j \text{ with } i \neq j.$$

Prove that $c \geq 1$.

IMO Shortlist 2002 A2

S 13. Consider two monotonically decreasing sequences (a_k) and (b_k), where $k \geq 1$, and a_k and b_k are positive real numbers for all $k \in \mathbb{N}$. Now, define the sequences

$$c_k = \min(a_k, b_k), \quad A_k = a_1 + a_2 + \cdots + a_k, \quad B_k = b_1 + b_2 + \cdots + b_k, \quad C_k = c_1 + c_2 + \cdots + c_k, k \in \mathbb{N}.$$

(1) Do there exist two monotonically decreasing sequences (a_k) and (b_k) of positive real numbers such that the sequences (A_k) and (B_k) are not bounded, while the sequence (C_k) is bounded?

(2) Does the answer to problem (1) change if we stipulate that the sequence (b_k) must be $b_k = \dfrac{1}{k}$ for all $k \in \mathbb{N}$?

IMO Shortlist 2003 A3

S 14. Let $n \in \mathbb{N}$ and let $x_1, \cdots, x_n, y_1, \cdots, y_n > 0$. Suppose $(z_2, \cdots, z_{2n})$ is a sequence of positive real numbers such that ${z_{i+j}}^2 \geq x_i y_j$ for all $1 \leq i, j \leq n$. Let $M = \max\{z_2, \cdots, z_{2n}\}$. Prove that

$$\left(\frac{M + z_2 + \cdots + z_{2n}}{2n}\right)^2 \geq \left(\frac{x_1 + \cdots + x_n}{n}\right)\left(\frac{y_1 + \cdots + y_n}{n}\right).$$

IMO Shortlist 2003 A6
Proposed by Reid Barton (USA)

S 15. Let a_{ij} (with the indices i and j from the set $\{1,\ 2,\ 3\}$) be real numbers such that

- $a_{ij} > 0$ for $i = j$;
- $a_{ij} < 0$ for $i \neq j$.

Prove the existence of positive real numbers c_1, c_2, c_3 such that the numbers

$$a_{11}c_1 + a_{12}c_2 + a_{13}c_3, \quad a_{21}c_1 + a_{22}c_2 + a_{23}c_3, \quad a_{31}c_1 + a_{32}c_2 + a_{33}c_3$$

are either all negative, or all zero, or all positive.

IMO Shortlist 2003 A1
Proposed by Kiran Kedlaya (USA)

S 16. Assume that a periodic sequence $a_1, a_2, 2_3, \cdots$ is defined by the recursion

$$a_{n+1} = \frac{1}{1 - a_n} - \frac{1}{1 + a_n}$$

for all $n \in \mathbb{N}$. Find the first term of this sequence.

Középiskolai Matematikai és Fizikai Lapok, February 2003, B. 3620
Proposed by P. Zsíros

S 17. The sequence $a_1 = 1, a_2, a_3, \cdots$ is defined by the recursion

$$a_{n+1} = a_n + \frac{1}{a_1 + \cdots + a_n}$$

for all $n \in \mathbb{N}$. Is the sequence bounded?

Középiskolai Matematikai és Fizikai Lapok, April 2003, B. 3640

S 18. Let $n \geq 3$ be an integer. Let t_1, t_2, ..., t_n be positive real numbers such that

$$n^2 + 1 > (t_1 + t_2 + \cdots + t_n)\left(\frac{1}{t_1} + \frac{1}{t_2} + \cdots + \frac{1}{t_n}\right).$$

Show that t_i, t_j, t_k are side lengths of a triangle for all i, j, k with $1 \leq i < j < k \leq n$.

IMO Shortlist 2004 A1, IMO 2004 Problem 4
Proposed by Hojoo Lee (Korea)

S 19. Let a_0, a_1, a_2, ... be an infinite sequence of real numbers satisfying the equation

$$a_n = |a_{n+1} - a_{n+2}|$$

for all $n \geq 0$, where a_0 and a_1 are two different positive reals. Can this sequence a_0, a_1, a_2, ... be bounded?

IMO Shortlist 2004 A2
Proposed by Mihai Bălună (Romania)

S 20. Let $a_1, a_2, \cdots, a_n$ be positive real numbers, $n > 1$. Denote by $g_n = (a_1 a_2 \cdots a_n)^{\frac{1}{n}}$ their geometric mean, and by $A_1, A_2, \cdots, A_n$ the sequence of arithmetic means defined by

$$A_k = \frac{a_1 + a_2 + \cdots + a_k}{k}, \qquad k = 1, 2, \cdots, n.$$

Let G_n be the geometric mean of $A_1, A_2, \cdots, A_n$. Prove the inequality

$$n\left(\frac{G_n}{A_n}\right)^{\frac{1}{n}} + \frac{g_n}{G_n} \leq n + 1.$$

and establish the cases of equality.

IMO Shortlist 2004 A7
Proposed by Finbarr Holland (Ireland)

S 21. A sequence of real numbers $a_0,\ a_1,\ a_2, \cdots$ is defined by the formula

$$a_{i+1} = \lfloor a_i \rfloor \langle a_i \rangle$$

for all $i \in \mathbb{N}_0$. Here, a_0 is an arbitrary real number, $\lfloor a_i \rfloor$ denotes the greatest integer not exceeding a_i, and $\langle a_i \rangle = a_i - \lfloor a_i \rfloor$. Prove that $a_i = a_{i+2}$ for all sufficiently large integers i.

IMO Shortlist 2006 A1
Proposed by Harmel Nestra (Estionia)

S 22. We define a sequence $(a_1, a_2, a_3, \cdots)$ by

$$a_n = \frac{1}{n}\left(\left\lfloor\frac{n}{1}\right\rfloor + \left\lfloor\frac{n}{2}\right\rfloor + \cdots + \left\lfloor\frac{n}{n}\right\rfloor\right),$$

where $\lfloor x \rfloor$ denotes the integer part of x.

(1) Prove that $a_{n+1} > a_n$ infinitely often.
(2) Prove that $a_{n+1} < a_n$ infinitely often.

IMO Shortlist 2006 N3

S 23. Let

$$a_n = ((n+1)!)^{\frac{1}{n}} - (n!)^{\frac{1}{n}}.$$

Show that $a_n \geq 1$ for all $n \in \mathbb{N}$. Show that the sequence is not bounded.

Középiskolai Matematikai és Fizikai Lapok, January 2006, A. 391

S 24. Construct a sequence $a_1, a_2, \cdots, a_N$ of positive real numbers such that
$$n_1 a_{n_0} + n_2 a_{n_1} + \cdots + n_k a_{n_{k-1}} > 2.7\,(a_1 + a_2 + \cdots + a_N)$$
for arbitrary integers $1 = n_0 < n_1 < \cdots < n_k = N$.

Középiskolai Matematikai és Fizikai Lapok, January 2006, A. 391

S 25. Let $a_1, a_2, \cdots, a_N$ be nonnegative real numbers, not all 0. Prove that there exists a sequence $1 = n_0 < n_1 < \cdots < n_k = N + 1$ of integers such that
$$n_1 a_{n_0} + n_2 a_{n_1} + \cdots + n_k a_{n_{k-1}} < 3\,(a_1 + a_2 + \cdots + a_N)\,.$$

Középiskolai Matematikai és Fizikai Lapok, February 2006, A. 394

S 26. Real numbers $a_1,\, a_2,\, \cdots,\, a_n$ are given. For each i, $(1 \le i \le n)$, define
$$d_i = \max\{a_j \mid 1 \le j \le i\} - \min\{a_j \mid i \le j \le n\}$$
and let
$$d = \max\{d_i \mid 1 \le i \le n\}.$$

(1) Prove that
$$\max\{\,|x_i - a_i| \mid 1 \le i \le n\} \ge \frac{d}{2}$$
for all $x_1, x_2, \cdots, x_n \in \mathbb{R}$ with $x_1 \le x_2 \le \cdots \le x_n$,

(2) Show that there exist $x_1, x_2, \cdots, x_n \in \mathbb{R}$ such that $x_1 \le x_2 \le \cdots \le x_n$ and
$$\max\{|x_i - a_i| \mid 1 \le i \le n\} = \frac{d}{2}.$$

IMO Shortlist 2007 A1
Proposed by Michael Albert (New Zealand)

S 27. Let $a_1, a_2, \cdots$ be a sequence of nonnegative real numbers satisfying the following conditions

- $a_{m+n} \le 2a_m + 2a_n$ for all $n \in \mathbb{N}$.
- There exists a constant $c > 2$ such that $a_{2^k} \le \frac{1}{(k+1)^c}$ for all $k \in \mathbb{N}_0$.

Prove that the sequence $a_1, a_2, \cdots$ is bounded.

IMO Shortlist 2007 A5

S 28. Let $a_1, a_2, \cdots, a_{100}$ be nonnegative real numbers such that
$${a_1}^2 + {a_2}^2 + \cdots + {a_{100}}^2 = 1.$$
Prove that
$${a_1}^2 a_2 + {a_2}^2 a_3 + \cdots + {a_{100}}^2 a_1 < \frac{12}{25}.$$

IMO Shortlist 2007 A6
Proposed by Marcin Kuczma (Poland)

S 29. In a sequence of real numbers, the sum every 5 consecutive terms is positive, and the sum of every 7 consecutive terms is negative. How long may be the sequence?

Középiskolai Matematikai és Fizikai Lapok, April 2007, B. 3993

S 30. Let $a_0, a_1, a_2, \cdots$ be an infinite sequence of real numbers satisfying the following conditions.

- $a_0 \in \left(0, \frac{1}{2}\right)$ is an irrational number.
- $a_n = \min\left\{ 2a_{n-1},\, 1 - 2a_{n-1} \right\}$ for all $n \in \mathbb{N}$.

(1) Prove that $a_n < \frac{3}{16}$ for some $n \in \mathbb{N}$.
(2) Can it happen that $a_n > \frac{7}{40}$ for all $n \in \mathbb{N}$?

Tournament of Towns 2007

S 31. Let $a_1 = 1, a_2, \cdots$ be an infinite sequence of real numbers. Each a_n, $n > 1$, is obtained from a_{n-1} as follows:

- if the greatest odd divisor of n has residue 1 modulo 4, then $a_n = a_{n-1} + 1$, and
- if this residue equals 3, then $a_n = a_{n-1} - 1$.

Prove that, in this sequence, each positive integer occurs infinitely many times.

Tournament of Towns 2008

S 32. Let $a_1 = 3, a_2, \cdots$ be a sequence of positive numbers defined by

$$a_n = \frac{1}{2}\left({a_{n-1}}^2 + 1\right)$$

for all $n \in \mathbb{N}$. Show that

$$\left[\sum_{k=1}^{n} \frac{a_k}{1+a_k} \sum_{k=1}^{n} \frac{1}{a_k\left(1+a_k\right)}\right]^{\frac{1}{2}} \le \frac{1}{4}\frac{a_1 + a_n}{\sqrt{a_1 a_n}}.$$

Amer. Math. Monthly 2009 Problem 11442
Proposed by José Luis Díaz-Barrero, José Gibergans-Báguena

S 33. A sequence $x_1 = 1, x_2, \cdots$ satisfies

$$x_{2k} = -x_k, \quad x_{2k-1} = (-1)^{k+1} x_k$$

for all $k \ge 1$. Prove that

$$x_1 + x_2 + \cdots + x_n \ge 0.$$

for all $n \ge 1$.

IMO Shortlist 2010 A4
Proposed by Gerhard Wöginger (Austria)

S 34. Let $a_1, a_2, a_3, \cdots$ be a sequence of positive real numbers, and s be a positive integer, such that

$$a_n = \max\{a_k + a_{n-k} \mid 1 \le k \le n-1\} \text{ for all } n > s.$$

Prove there exist positive integers $\ell \le s$ and N, such that

$$a_n = a_\ell + a_{n-\ell} \text{ for all } n \ge N.$$

IMO Shortlist 2010 A7, IMO 2010 Problem 6
Proposed by Morteza Saghafiyan (Iran)

S 35. Let n be a positive integer and let $a_1, \cdots, a_{n-1}$ be arbitrary real numbers. Define the sequences $u_0, \cdots, u_n$ and $v_0, \cdots, v_n$ inductively by $u_0 = u_1 = v_0 = v_1 = 1$, and $u_{k+1} = u_k + a_k u_{k-1}$, $v_{k+1} = v_k + a_{n-k} v_{k-1}$ for $k = 1, \cdots, n-1$.

IMO Shortlist 2013 A1

S 36. Define the function $f : (0,1) \to (0,1)$ by

$$f(x) = \begin{cases} x + \frac{1}{2} & \text{if } x < \frac{1}{2} \\ x^2 & \text{if } x \geq \frac{1}{2} \end{cases}$$

Let a and b be two real numbers such that $0 < a < b < 1$. We define the sequences a_n and b_n by $a_0 = a, b_0 = b$, and $a_n = f(a_{n-1})$, $b_n = f(b_{n-1})$ for $n > 0$. Show that there exists a positive integer n such that

$$(a_n - a_{n-1})(b_n - b_{n-1}) < 0.$$

IMO Shortlist 2014 A2

S 37. For a sequence $x_1, x_2, \cdots, x_n$ of real numbers, we define its *price* as

$$\max_{1 \leq i \leq n} |x_1 + \cdots + x_i|.$$

Given n real numbers, Dave and George want to arrange them into a sequence with a low price. Diligent Dave checks all possible ways and finds the minimum possible price D. Greedy George, on the other hand, chooses x_1 such that $|x_1|$ is as small as possible; among the remaining numbers, he chooses x_2 such that $|x_1 + x_2|$ is as small as possible, and so on. Thus, in the i-th step he chooses x_i among the remaining numbers so as to minimise the value of $|x_1 + x_2 + \cdots x_i|$. In each step, if several numbers provide the same value, George chooses one at random. Finally he gets a sequence with price G. Find the least possible constant c such that for every positive integer n, for every collection of n real numbers, and for every possible sequence that George might obtain, the resulting values satisfy the inequality $G \leq cD$.

IMO Shortlist 2014 A3
Proposed by Georgia

S 38.

(1) Prove that for every infinite sequence $x_1, x_2, \cdots \in [0,1]$, there exists some $C > 0$ such that for every positive integer r, there are positive integers n, m satisfying $|n - m| \geq r$ and $|x_n x_m| < \frac{C}{|n-m|}$.

(2) Show that for every $C > 0$ there exists an infinite sequence $x_1, x_2, \cdots \in [0,1]$ and a positive integer r such that $|x_n x_m| > \frac{C}{|n-m|}$ holds true for every pair n, m of positive integers with $|n - m| \geq r$.

Középiskolai Matematikai és Fizikai Lapok, October 2014, A. 624
CIIM6, Costa Rica

S 39. Let $a_1, \cdots, a_n$ and $b_1, \cdots, b_n$ be real numbers. Prove that there exists an integer k with $1 \leq k \leq n$ such that

$$\sum_{i=1}^{n} |a_i - a_k| \leq \sum_{i=1}^{n} |b_i - a_k|$$

Mediterranean Mathematics Olympiad 2014
Proposed by Gerhard Woeginger (Austria)

S 40. Suppose that a sequence $a_1, a_2, \cdots$ of positive real numbers satisfies

$$a_{k+1} \geq \frac{k a_k}{a_k^2 + (k-1)}$$

for every positive integer k. Prove that $a_1 + a_2 + \cdots + a_n \geq n$ for every $n \geq 2$.

IMO Shortlist 2015 A1

S 41. Let K be a fixed positive integer. Let $a_0 = -1, a_1, a_2, \cdots$ be the sequence such that

$$\sum_{i_0,\cdots,i_K \geq 0,\, i_0+\cdots+i_K=n} \frac{a_{i_1} \cdots a_{i_K}}{1+i_0} = 0$$

for all $n \in \mathbb{N}$. Show that $a_n > 0$ for all $n \in \mathbb{N}$.

Középiskolai Matematikai és Fizikai Lapok, January 2016, A. 661

The views of space and time which I wish to lay before you have sprung from the soil of experimental physics, and therein lies their strength. They are radical. Henceforth, space by itself, and time by itself, are doomed to fade away into mere shadows, and only a kind of union of the two will preserve an independent reality.

– Hermann Minkowski
(Address to the 80th Assembly of German Natural Scientists and Physicians)

5. Functional equations over $\mathbb{Z}$

Hamlet: O God, I could be bounded in a nutshell and count myself a king of infinite space.

– William Shakespeare, Hamlet, Act II, Scene 2

N 1. Let $f : \mathbb{N} \to \mathbb{N}$ be a function such that

$$f(n+1) > f(f(n))$$

for all $n \in \mathbb{N}$. Show that $f(n) = n$ for all $n \in \mathbb{N}$.

IMO 1977 Problem 6

N 2. Let $f : \mathbb{N} \to \mathbb{N}_0$ be a function satisfying the following conditions.

- $f(2) = 0,\ f(3) > 0,\ f(9999) = 3333.$
- $f(m+n) - f(m) - f(n) \in \{0, 1\}$ for all $m, n \in \mathbb{N}$.

Determine the value of $f(1982)$.

IMO 1982 Problem 1

N 3. Find all surjective functions $f : \mathbb{N} \to \mathbb{N}$ such that

$$f(n) \geq n + (-1)^n$$

for all $n \in \mathbb{N}$.

Romanian Mathematical Olympiad 1986

N 4. Prove that there exist no function $f : \mathbb{N}_0 \to \mathbb{N}_0$ such that

$$f(f(n)) = n + 1987$$

for all $n \in \mathbb{N}_0$.

IMO 1987 Problem 4
Proposed by Vietnam

N 5. A function f is defined on the positive integers by

$$\left\{\begin{array}{rcl} f(1) & = & 1, \\ f(3) & = & 3, \\ f(2n) & = & f(n), \\ f(4n+1) & = & 2f(2n+1) - f(n), \\ f(4n+3) & = & 3f(2n+1) - 2f(n), \end{array}\right.$$

for all positive integers n. Determine the number of positive integers n, less than or equal to 1988, for which $f(n) = n$.

IMO 1988 Problem 3

N 6. Find all functions $f : \mathbb{N} \to \mathbb{N}$ such that

$$f(f(m) + f(n)) = m + n$$

for all $m, n \in \mathbb{N}$.

IMO Short List 1988

N 7. Does there exists a function $f : \mathbb{N} \mapsto \mathbb{N}$ such that

$$f^{1989}(n) = 2n$$

for all $n \in \mathbb{N}$?

China Team Selection Test 1989

N 8. Let $g : \mathbb{N} \to \mathbb{N}$ be a bijection, and $a \in \mathbb{N}$ be an odd number.

(1) Prove that there is no function f such that $f(f(n)) = g(n) + a$ for all $n \in \mathbb{N}$.
(2) What if a is even?

Mathematics Magazine 1988 Problem 1308
Proposed by Nicholas A. Martin

N 9. Find all functions $f : \mathbb{N}_0 \to \mathbb{N}_0$ such that

$$f(f(n)) + f(n) = 2n + 6$$

for all $n \in \mathbb{N}_0$.

Austria 1989

N 10. Find all functions $f : \mathbb{N} \times \mathbb{N} \to \mathbb{N}$ satisfying the following conditions.

- $f(m, m) = m$ for all $m \in \mathbb{N}$.
- $f(m, k) = f(k, m)$ for all $m, k \in \mathbb{N}$.
- $f(m, m + k) = f(m, k)$ for all $m, k \in \mathbb{N}$.

Turkey Team Selection Test 1989

N 11. Suppose that $f : \mathbb{N}_0 \to \mathbb{N}_0$ is a function such that $f(1) > 0$ and

$$f\left(m^2 + n^2\right) = f(m)^2 + f(n)^2$$

for all $m, n \in \mathbb{N}_0$. Show that f is the identity function.

Amer. Math. Monthly 1991 Problem E 3458
Proposed by Umberto Zannier

N 12. Find all functions $f : \mathbb{Z} - \{0\} \to \mathbb{Q}$ such that

$$f\left(\frac{x + y}{3}\right) = \frac{f(x) + f(y)}{2}$$

for all $x, y \in \mathbb{Z} - \{0\}$.

Austrian Mathematical Olympiad (Final Round) 1991

N 13. Prove that there is no bijective function $f : \mathbb{N} \to \mathbb{N}_0$ such that

$$f(mn) = f(m) + f(n) + 3f(m)f(n)$$

for all $m, n \in \mathbb{N}$.

Balkan Mathematical Olympiad 1991

N 14. Let f be a function $f : \mathbb{N}_0 \mapsto \mathbb{N}$, and satisfies the following conditions:

- $f(0) = 0$,
- $f(1) = 1$,
- $f(n+2) = 23f(n+1) + f(n)$ for all $n \in \mathbb{N}_0$.

Prove that for any $m \in \mathbb{N}$, there exist $d \in \mathbb{N}$ such that

$$m \mid f(f(n)) \quad \text{if and only if} \quad d \mid n.$$

China Team Selection Test 1991

N 15. Let $f, g : \mathbb{Z} \to \mathbb{Z}$ be functions satisfying the following conditions.

- $f(m + f(f(n))) = -f(f(m+1) - n$ for all $M, n \in \mathbb{Z}$.
- g is a polynomial function with integer coefficients and $g(n) = g(f(n))$ for all $n \in \mathbb{Z}$.

IMO Shortlist 1991
Proposed by India

N 16. If P is a polynomial with coefficients in $\mathbb{Z}$, it is immediate that $P(b+c) - P(b)$ is an integer multiple of c for all integers b and c. Are there any non-polynomial functions $f : \mathbb{Z} \to \mathbb{Z}$ with this property?

Amer. Math. Monthly 1992 Problem 10185
Proposed by William C. Waterhouse

N 17. Prove that $f(n) = 1 - n$ is the only function $f : \mathbb{Z} \to \mathbb{Z}$ that satisfies the following conditions.

(i) $f(f(n)) = n$, for all integers n.
(ii) $f(f(n+2) + 2) = n$ for all integers n.
(iii) $f(0) = 1$.

William Lowell Putnam Competition 1992

N 18. Let $\mathcal{F}$ be the class of integer-valued functions defined on $\mathbb{N}$ that satisfy the following constraints. For $j \in \mathcal{F}$, $j(1) = 1$ and

$$j(m) + j(n) - 1 \leq j(m+n) \leq j(m) + j(n).$$

for all $m, n \in \mathbb{N}$. Show that the number of distinct initial sequences of length N generated by the $j \in \mathcal{F}$ is

$$\sum_{n=1}^{N} \phi(n)$$

where $\phi(n)$ is Euler's totient function.

Amer. Math. Monthly 1993 Problem 10302
Proposed by Jeffrey A. Barnet

N 19. Find all functions $f : \mathbb{Z} \to \mathbb{Z}$ satisfying the following conditions.

- $f(-1) = f(1)$.
- $f(x) + f(y) = f(x + 2xy) + f(y - 2xy)$ for all $m, n \in \mathbb{N}$.

Czechoslovak Mathematical Olympiad 1993

N 20. Does there exist a function $f : \mathbb{N} \to \mathbb{N}$ satisfying the following conditions?

- $f(1) = 2$.

- $f(f(n)) = f(n) + n$ for all $n \in \mathbb{N}$.
- $f(n) < f(n+1)$ for all $n \in \mathbb{N}$.

IMO 1993 Problem 5

N 21. Determine all real valued functions f on the integer lattice $\mathbb{Z}^2$ such that
$$f(\mathbf{u} + \mathbf{v}) = f(\mathbf{u}) + f(\mathbf{v})$$
for every pair of orthogonal vectors $\mathbf{u}, \mathbf{v} \in \mathbb{Z}^2$.

Amer. Math. Monthly 1994 Problem 10381
Proposed by Marcin E. Kuczma

N 22. Does there exist a sequence $F(1), F(2), F(3), \cdots$ of non-negative integers that simultaneously satisfies the following three conditions?

- Each of the integers $0, 1, 2, \cdots$ occurs in the sequence.
- Each positive integer occurs in the sequence infinitely often.
- $F(F(n^{163})) = F(F(n)) + F(F(361))$ for all $n \geq 2$.

IMO Shortlist 1995

N 23. For positive integers n, the numbers $f(n)$ are defined inductively as follows: $f(1) = 1$, and for every positive integer n, $f(n+1)$ is the greatest integer m such that there is an arithmetic progression of positive integers $a_1 < a_2 < \cdots < a_m = n$ for which
$$f(a_1) = f(a_2) = \cdots = f(a_m).$$
Prove that there are positive integers a and b such that $f(an + b) = n + 2$ for every positive integer n.

IMO Shortlist 1995

N 24. Find all functions $f : \mathbb{Z} - \{0\} \to \mathbb{Q}$ such that
$$f\left(\frac{x+y}{3}\right) = \frac{f(x) + f(y)}{2}$$
for all $x, y \in \mathbb{Z} - \{0\}$.

Iranian Mathematical Olympiad 1995

N 25. The sequence $a_1, a_2, \cdots$ of positive integers satisfies
$$\gcd(a_i, a_j) = \gcd(i, j)$$
for all $i, j \in \mathbb{N}$ with $i \neq j$. Prove that $a_i = i$ for all $i \in \mathbb{N}$.

All-Russian Mathematical Olympiad 1995

N 26. Find all surjective functions $f : \mathbb{N} \to \mathbb{N}$ such that
$$f(m) \mid f(n) \quad \text{if and only if} \quad m \mid n$$
for all $m, n \in \mathbb{N}$.

Turkish Mathematical Olympiad 1995

N 27. Does there exist a function $f : \mathbb{N} \to \mathbb{N}$ satisfying the following three conditions?

- $f(1996) = 1$.
- For all primes p, every prime occurs in the sequence $f(p), f(2p), f(3p), \cdots, f(kp)$ infinitely often.
- $f(f(n)) = 1$ for all $n \in \mathbb{N}$.

Crux Mathematicorum 1996 Problem 2111
Proposed by Hoe Teck Wee

N 28. Show that there exists a bijective function $f : \mathbb{N}_0 \to \mathbb{N}_0$ such that

$$f(3mn + m + n) = 4f(m)f(n) + f(m) + f(n)$$

for all $m, n \in \mathbb{N}_0$.

IMO ShortList 1996

N 29. Find all functions $f : \mathbb{N}_0 \to \mathbb{N}_0$ such that

$$f(m + f(n)) = f(f(m)) + f(n)$$

for all $m, n \in \mathbb{N}_0$.

IMO 1996 Problem 3

N 30. Determine all functions $f : \mathbb{N}_0 \to \mathbb{N}_0 - \{1\}$ such that

$$f(n+1) + f(n+3) = f(n+5)f(n+7) - 1375$$

for all $n \in \mathbb{N}$.

Iranian Mathematical Olympiad 1996

N 31. Find all functions $f : \mathbb{N}_0 \to \mathbb{N}_0$ satisfying the following conditions.

- $2f\left(m^2 + n^2\right) = f(m)^2 + f(n)^2$ for all $m, n \in \mathbb{N}_0$.
- $f\left(m^2\right) \geq f\left(n^2\right)$ for all $m, n \in \mathbb{N}_0$ with $m \geq n$.

Korean Mathematical Olympiad (Final Round) 1996

N 32. Find all functions $f : \mathbb{N} \to \mathbb{N}$ such that

$$f(f(n)) = n^2$$

for all $n \in \mathbb{N}$.

Singapore 1996

N 33. Find all functions $f : \mathbb{N} \to \mathbb{N}$ such that

$$f(n) + f(n+1) = f(n+2)f(n+3) - 1996$$

for all $n \in \mathbb{N}$.

Vietnamese Mathematical Olympiad 1996

N 34. Find all functions $f : \mathbb{N} \to \mathbb{N}$ such that

$$f^{(19)}(n) + 97f(n) = 98n + 232$$

for all $n \in \mathbb{N}$.

IMO unused 1997

N 35. Suppose the function $f : \mathbb{N} \to \mathbb{Z}$ satisfies the property that $m+n$ divides $f(m)+f(n)$ for all $m, n \in \mathbb{N}$ (for example, $f(x) = x^3$). Show that $m-n$ divides $f(m)-f(n)$ for $m, n \in \mathbb{N}$.

Mathematics Magazine 1997 Problem Q 861
Proposed by David Callan

N 36. Find all functions $f : \mathbb{Z} \to \mathbb{Z}$ such that

$$f(f(m)) = m+1$$

for all $m \in \mathbb{Z}$.

Slovenia 1997

N 37. Find all functions $f : \mathbb{Z} \to \mathbb{Z}$ such that

$$f(m+f(n)) = f(m)-n$$

for all $m, n \in \mathbb{Z}$.

Austrian-Polish Mathematics Competition 1997

N 38. Find all functions $f : \mathbb{Z} \to \mathbb{Z}$ such that

$$f(m+f(n)) = f(m)+n$$

for all $m, n \in \mathbb{Z}$.

South Africa 1997

N 39. Find the number of functions $f : \mathbb{N} \to \mathbb{N}$ satisfying the following conditions.

- $f(1) = 1$.
- $f(n)f(n+2) = f(n+1)^2 + 1997$ for all $n \in \mathbb{N}$.

Vietnamese Mathematical Olympiad 1997

N 40. Consider all functions $f : \mathbb{N} \to \mathbb{N}$ such that

$$f(t^2 f(s)) = s(f(t))^2$$

for all $s, t \in \mathbb{N}$. Determine the least possible value of $f(1998)$.

IMO Shortlist 1998, IMO 1998 Problem 6

N 41. For any two nonnegative integers n and k satisfying $n \geq k$, we define the number $c(n, k)$ as follows:

- $c(n, 0) = c(n, n) = 1$ for all $n \geq 0$;
- $c(n+1, k) = 2^k c(n, k) + c(n, k-1)$ for $n \geq k \geq 1$.

Prove that $c(n, k) = c(n, n-k)$ for all $n \geq k \geq 0$.

IMO Shortlist 1998

N 42. A sequence of integers $a_1, a_2, a_3, \cdots$ is defined as follows: $a_1 = 1$ and for $n \geq 1$, a_{n+1} is the smallest integer greater than a_n such that

$$a_i + a_j \neq 3a_k$$

for all $i, j, k \in \{1, 2, 3, \cdots, n+1\}$, not necessarily distinct. Determine a_{1998}.

IMO Shortlist 1998

N 43. Let $a_0, a_1, a_2, \cdots$ be an increasing sequence of nonnegative integers such that every nonnegative integer can be expressed uniquely in the form

$$a_i + 2a_j + 4a_k,$$

where i, j and k are not necessarily distinct. Determine a_{1998}.

IMO Shortlist 1998

N 44. Find all strictly increasing functions $f : \mathbb{N} \to \mathbb{N}$ such that

$$f(n + f(n)) = 2f(n)$$

for all $n \in \mathbb{N}$.

Spanish Mathematical Olympiad 1998

N 45. Find all functions $h : \mathbb{Z} \to \mathbb{Z}$ such that

$$h(x + y) + h(xy) = h(x)h(y) + 1$$

for all $x, y \in \mathbb{Z}$.

Belarus 1999

N 46. Let $f, g : \mathbb{N} \to \mathbb{N}$ be functions satisfying the following conditions.

- The function f is surjective.
- The function g is injective.
- $f(n) \geq g(n)$ for all $n \geq \mathbb{N}$.

Prove that $f = g$.

Crux Mathematicorum 1999 Problem A227
Proposed by Mohammed Aassila

N 47. Find all functions $f : \mathbb{N}_0 \to \{0, 1, \cdots, 1999\}$ satisfying the following conditions.

- $f(s) = s$ for all $s \in \{0, 1, \cdots, 1999\}$.
- $f(m + n) = f(f(m) + f(n))$ for all $m, n \in \mathbb{N}_0$.

Vietnamese Mathematical Olympiad 1999

N 48. Let $q = \frac{1+\sqrt{5}}{2}$. Let $f : \mathbb{N} \to \mathbb{N}$ be a function such that

$$|f(n) - qn| < \frac{1}{q}$$

for all $n \in \mathbb{N}$. Prove that

$$f(f(n)) = f(n) + n$$

for all $n \in \mathbb{N}$.

Középiskolai Matematikai és Fizikai Lapok, Decober 2000, B. 3429

N 49. Suppose that $p \in \mathbb{N}_0$. Show that the following two conditions are equivalent.

- $3 \mid p$.
- There exists a function $f : \mathbb{N}_0 \to \mathbb{N}_0$ such that

$$f(f(n)) + f(n) = 2n + p$$

for all $n \in \mathbb{N}_0$.

Crux Mathematicorum 2001 Problem 2646
Proposed by Hojoo Lee

N 50. Let $\mathcal{T}$ denote the set of all ordered triples (p,q,r) of nonnegative integers. Find all functions $f:\mathcal{T}\to\mathbb{R}$ satisfying

$$f(p,q,r)=\begin{cases}0 & \text{if } pqr=0,\\ 1+\frac{1}{6}(f(p+1,q-1,r)+f(p-1,q+1,r) & \\ +f(p-1,q,r+1)+f(p+1,q,r-1) & \text{otherwise,}\\ +f(p,q+1,r-1)+f(p,q-1,r+1)) & \end{cases}$$

for all nonnegative integers p, q, r.

IMO Shortlist 2001 A1

N 51. Find all functions $f:\mathbb{N}\to\mathbb{N}$ such that

$$f(f(f(n)))+6f(n)=3f(f(n))+4n+2001.$$

for all $n\in\mathbb{N}$.

USAMO Summer Program 2001

N 52. Determine all functions $f:\mathbb{N}\to\mathbb{N}$ satisfying

$$2n+2001\le f(f(n))+f(n)\le 2n+2002.$$

for all $n\in\mathbb{N}$.

Balkan Mathematical Olympiad 2002

N 53. Find all functions $f:\mathbb{N}_0\to\mathbb{N}_0$ such that

$$mf(n)+nf(m)=(m+n)f\left(m^2+n^2\right).$$

for all $m,n\in\mathbb{N}_0$.

Canada 2002

N 54. Let P denote the set of all ordered pairs (p,q) of nonnegative integers. Find all functions $f:P\to\mathbb{R}$ satisfying

$$f(p,q)=\begin{cases}0 & \text{if } pq=0,\\ 1+\frac{1}{2}f(p+1,q-1)+\frac{1}{2}f(p-1,q+1) & \text{otherwise}\end{cases}$$

Germany Team Selection Test 2002

N 55. Let $n\ge 2$ be a positive integer. Find the number of functions $f:\{1,2,\cdots,n\}\to\{1,2,3,4,5\}$ such that

$$|f(k+1)-f(k)|\ge 3$$

for all $k\in\{1,2,\cdots,n-1\}$.

Romania Team Selection Test 2000
Proposed by Vasile Pop

N 56. Find all functions $f : \mathbb{N} \to \mathbb{N}$ such that

$$f(m+n)\,f(m-n) = f\left(m^2\right)$$

for $m, n \in \mathbb{N}$.

USA Team Selection Test 2003

N 57. Find all functions $f : \mathbb{N} \to \mathbb{N}$ such that

$$f^2\,(m) + f\,(n) \mid \left(m^2 + n\right)^2$$

for all $m, n \in \mathbb{N}$.

IMO Shortlist 2004 N3

N 58. Find all functions $f : \mathbb{N} \to \mathbb{N}$ such that

$$(2^m + 1)\,f(n)f\,(2^m n) = 2^m f(n)^2 + f(2^m n)^2 + (2^m - 1)^{2n}$$

for all $m, n \in \mathbb{N}$.

Italy Team Selection Test 2004

N 59. Find all one-to-one mappings $f : \mathbb{N} \to \mathbb{N}$ such that

$$f(f(n)) \le \frac{n + f(n)}{2}$$

for all $n \in \mathbb{N}$.

Romania Team Selection Test 2004

N 60. Find all strictly increasing functions $f : \{1, 2, \cdots, 10\} \to \{1, 2, \cdots, 100\}$ such that

$$x + y \mid xf(x) + yf(y)$$

for all $x, y \in \{1, 2, \cdots, 10\}$.

Romanian Mathematical Olympiad 2004
Proposed by Cristinel Mortici

N 61. Find all functions $f : \mathbb{Z} \to \mathbb{Z}$ such that

$$f(x + f(y)) = f(x) - y$$

for all $x, y \in \mathbb{Z}$.

Spanish Mathematical Olympiad 2004

N 62. Let $f : \mathbb{N}_0 \to \mathbb{Q}$ be a function satisfying the following conditions.

- $f(0) = 0$.
- $f(3n + k) = -\frac{3f(n)}{2} + k$ for all $n \in \mathbb{Z}$ and $k = 0, 1, 2$.

Show that f is one-to-one and determine the range of f.

USA Team Selection Test 2004

N 63. Let α be given positive real number, find all functions $f : \mathbb{N} \to R$ such that

$$f(k + m) = f(k) + f(m)$$

holds for all $k, m \in \mathbb{N}$ with $\alpha m \le k \le (\alpha + 1)m$.

China Team Selection Test 2005

N 64. Suppose that $f : \{1, 2, \cdots, 1600\} \to \{1, 2, \cdots, 1600\}$ satisfies $f(1) = 1$ and

$$f^{2005}(x) = x \quad \text{for } x = 1, 2, \cdots, 1600.$$

- Prove that f has a fixed point different from 1.
- Find all $n > 1600$ such that any function $f : \{1, \cdots, n\} \to \{1, \cdots, n\}$ satisfying the above conditions has at least two fixed points.

Italy Team Selection Test 2005

N 65. Let $f, g : \mathbb{N} \to \mathbb{N}$ be functions satisfying the following conditions.

- g is surjective.
- $2f(n)^2 = n^2 + g(n)^2$ for all $n \in \mathbb{N}$.
- $|f(n) - n| \le 2005\sqrt{n}$ for all $n \in \mathbb{N}$.

Prove that there exist infinitely many $n \in \mathbb{N}$ such that $f(n) = n$.

Moldova Team Selection Test 2005

N 66. Find all functions $f : \mathbb{N} \to \mathbb{N}$ such that

$$f(m)^2 + f(n) \mid \left(m^2 + n\right)^2$$

for all $m, n \in \mathbb{N}$.

Romania Team Selection Test 2005

N 67.

(1) Prove that there are no injective functions $f : \mathbb{N} \to \mathbb{N}_0$ such that

$$f(mn) = f(m) + f(n)$$

for all $m, n \in \mathbb{N}$.

(2) Prove that, for all positive integers k, there exist one-to-one functions $f : \{1, 2, \cdots, k\} \to \mathbb{N}_0$ such that

$$f(mn) = f(m) + f(n)$$

for all $m, n \in \{1, 2, \cdots, k\}$ with $mn \le k$.

Romanian Mathematical Olympiad 2005
Proposed by Mihai Baluna

N 68. Find all functions $f : \mathbb{Z} \to \mathbb{Z}$ such that

$$f\left(x^3 + y^3 + z^3\right) = f(x)^3 + f(y)^3 + f(z)^3$$

for all $x, y, z \in \mathbb{Z}$.

Vietnam Team Selection Test 2005

N 69. Find all functions $f : \mathbb{Z} \to \mathbb{Z}$ such that

$$f(m - n + f(n)) = f(m) + f(n)$$

for all $m, n \in \mathbb{Z}$.

Italy Team Selection Test 2006

N 70. Consider those functions $f : \mathbb{N} \to \mathbb{N}$ such that

$$f(m+n) \geq f(m) + f(f(n)) - 1$$

for all $m, n \in \mathbb{N}$. Find all possible values of $f(2007)$.

IMO Shortlist 2007 A2, Germany Team Selection Test 2008
Proposed by Nikolai Nikolov (Bulgaria)

N 71. Find all surjective functions $f : \mathbb{N} \to \mathbb{N}$ such that, for every $m, n \in \mathbb{N}$ and every prime p, the number $f(m+n)$ is divisible by p if and only if $f(m) + f(n)$ is divisible by p.

IMO Shortlist 2007 N5
Proposed by Mohsen Jamaali and Nima Ahmadi Pour Anari (Iran)

N 72. For an integer m, denote by $t(m)$ the unique number in $\{1, 2, 3\}$ such that $m + t(m)$ is a multiple of 3. A function $f : \mathbb{Z} \to \mathbb{Z}$ satisfies the following conditions.

- $f(-1) = 0$, $f(0) = 1$, $f(1) = -1$.
- $f(2^n + m) = f(2^n - t(m)) - f(m)$ for all integers $m, n \in \mathbb{Z}$ with $n \geq 0$ and $2^n > m$.

Prove that $f(3p) \geq 0$ for all integers $p \geq 0$.

IMO Shortlist 2008 A4
Proposed by Gerhard Wöginger (Austria)

N 73. For every $n \in \mathbb{N}$, let $d(n)$ denote the number of (positive) divisors of n. Find all functions $f : \mathbb{N} \to \mathbb{N}$ satisfying the following conditions.

- $d(f(x)) = x$ for all $x \in \mathbb{N}$.
- $f(xy) \mid (x-1)y^{xy-1}f(x)$ for all x, $y \in \mathbb{N}$.

IMO Shortlist 2008 N5, Germany Team Selection Test 2009
Proposed by Bruno Le Floch (France)

N 74. Let k be a given positive integer. Find all functions $f : \mathbb{N} \to \mathbb{N}$ such that

$$f(m) + f(n) \mid (m+n)^k$$

for all $m, n \in \mathbb{N}$.

Iran Team Selection Test 2008

N 75. Prove that every bijective function $f : \mathbb{Z} \to \mathbb{Z}$ can be written in the way $f = u + v$ where $u, v : \mathbb{Z} \to \mathbb{Z}$ are bijective functions.

Romanian Masters In Mathematics 2008

N 76. Find all functions $f : \mathbb{N} \to \mathbb{N}$ satisfying the following conditions.

- $f(n)$ is a perfect square for all $n \in \mathbb{N}$.
- $f(m+n) = f(m) + f(n) + 2mn$ for all $m, n \in \mathbb{N}$.

Benelux Mathematical Olympiad 2009

N 77. Determine all functions $f : \mathbb{N} \to \mathbb{N}$ such that

$$f\left(f(m)^2 + 2f(n)^2\right) = m^2 + 2n^2$$

for all $m, n \in \mathbb{N}$.

Balkan Mathematical Olympiad 2009

N 78. Do there exist strictly monotonic functions $f, g : \mathbb{N} \to \mathbb{N}$ such that

$$f(g(g(n))) < g(f(n))$$

for all $n \in \mathbb{N}$?

Hungary-Israel Binational Mathematical Competition 2009

N 79. Determine all functions $f : \mathbb{N} \to \mathbb{N}$ such that, for all positive integers a and b, there exists a non-degenerate triangle with sides of lengths a, $f(b)$ and $f(b+f(a)-1)$. (A triangle is non-degenerate if its vertices are not collinear.)

IMO Shortlist 2009 A3, IMO 2009 Problem 5
Proposed by Bruno Le Floch (France)

N 80. Let $n \geq 2$ be an integer. The numbers 0, 1, $\cdots$, n are written on a blackboard. In each step we erase an integer which is the arithmetic mean of two different numbers which are still left on the blackboard. We make such steps until no further integer can be erased. Find the smallest possible number $g(n)$ of integers left on the blackboard at the end.

Middle European Mathematical Olympiad 2009

N 81. Determine all sequences of non-negative integers $a_1, \cdots, a_{2016}$ all less than or equal to 2016 such that

$$i + j \mid i\, a_i \; + \; j\, a_j$$

for all $i, j \in \{1, 2, \cdots, 2016\}$.

Nordic Mathematics Contest 2009

N 82. Let $f : \mathbb{N} \to \mathbb{N}$ be a function satisfying the following conditions.

- $f(1) = 1$.
- $f(n) = n - f(f(n-1))$ for all $n \geq 2$.

Prove that

$$f(n + f(n)) = n$$

for all $n \in \mathbb{N}$.

Bulgarian Mathematical Olympiad 2010

N 83. Find all functions $f : \mathbb{Z} \to \mathbb{Z}$ satisfying the following conditions.

- $f(1) = 1$.
- $f(m+n)(f(m) - f(n)) = f(m-n)(f(m) + f(n))$ for all $m, n \in \mathbb{Z}$.

Indian National Mathematical Olympiad Program 2010

N 84. Does there exist a function $f : \mathbb{N} \to \mathbb{N}$ satisfying the following condition?

$$f(f(n-1)) = f(n+1) - f(n)$$

for all $n\{2, 3, \cdots\}$.

Indian National Mathematical Olympiad Program 2010

N 85. Determine all functions $g : \mathbb{N} \to \mathbb{N}$ such that

$$(g(m) + n)\,(g(n) + m)$$

is a perfect square for all $m, n \in \mathbb{N}$.

IMO 2010 Problem 3
Proposed by Gabriel Carroll (USA)

N 86. Suppose that f and g are two functions defined on the set of positive integers and taking positive integer values. Suppose also that the equations $f(g(n)) = f(n) + 1$ and $g(f(n)) = g(n) + 1$ hold for all positive integers. Prove that $f(n) = g(n)$ for all positive integer n.

IMO Shortlist 2010 A6
Proposed by Alex Schreiber (Germany)

N 87. Let $a_1 = 1, a_2 = 2, a_3, \cdots$ denote a sequence of positive real numbers satisfying the following conditions.

- $a_{mn} = a_m a_n$ for all $m, n = 1, 2, 3, \cdots$.
- There exists a constant $C > 1$ such that $a_{m+n} \leq C(a_m + a_n)$ for all $m, n = 1, 2, 3, \cdots$.

Prove that $a_n = n$ for $n = 1, 2, 3, \cdots$.

Polish Mathematical Olympiad 2010

N 88. Consider the set $\mathcal{F}$ of functions $f : \mathbb{N}_0 \to \mathbb{N}_0$ such that

$$f\left(a^2 - b^2\right) = f(a)^2 - f(b)^2$$

for all $a, b \in \mathbb{N}$ with $a \geq b$.

(1) Determine the set $\{f(1) \mid f \in \mathcal{F}\}$.
(2) Prove that the function $\mathcal{F}$ has exactly two elements.

Romanian Mathematical Olympiad 2010
Proposed by Nelu Chichirim

N 89. Let $f : \mathbb{Z} \to \mathbb{Z}$ be a function such that

$$f(f(x) - y) = f(y) - f(f(x))$$

for all $x, y \in \mathbb{Z}$. Show that the function f is bounded.

Baltic Way 2011

N 90. Determine all functions $f, g : \mathbb{N} \to \mathbb{N}$ such that

$$f^{g(n)+1}(n) + g^{f(n)}(n) = f(n+1) - g(n+1) + 1$$

for every positive integer n. Here, $f^k(n)$ means $\underbrace{f(f(\cdots f)}_{k}(n)\cdots))$.

IMO Shortlist 2011 A4
Proposed by Bojan Bašić (Serbia)

N 91. Let $n \geq 1$ be an odd integer. Determine all functions $f : \mathbb{Z} \to \mathbb{Z}$ such that

$$f(x) - f(y) \mid x^n - y^n$$

for all $x, y \in \mathbb{Z}$.

IMO Shortlist 2011 N3
Proposed by Mihai Baluna (Romania)

N 92. Let $f : \mathbb{Z} \to \mathbb{N}$ be a function such that

$$f(m) - f(n) \mid f(m - n)$$

for all $m, n \in \mathbb{Z}$. Prove that, for all integers m and n with $f(m) \leq f(n)$, the number $f(n)$ is divisible by $f(m)$.

IMO Shortlist 2011 N5, IMO 2011 Problem 5
Proposed by Mahyar Sefidgaran (Iran)

N 93. Let $f : \mathbb{N} \to \mathbb{N}$ be a function such that the positive integer

$$af(a) + bf(b) + 2ab$$

is a perfect square for all $a, b \in \mathbb{N}$. Prove that $f(a) = a$ for all $a \in \mathbb{N}$.

Iran Team Selection Test 2011

N 94. Let p be a prime, n be a positive integer, and let $\mathbb{Z}_{p^n}$ denote the set of congruence classes modulo p^n. Determine the number of functions $f : \mathbb{Z}_{p^n} \to \mathbb{Z}_{p^n}$ satisfying the condition

$$f(a) + f(b) \equiv f(a + b + pab) \pmod{p^n}$$

for all $a, b \in \mathbb{Z}_{p^n}$.

Turkey Team Selection Test 2011

N 95. Does exists a function $f : \mathbb{N} \to \mathbb{N}$ such that

$$f(f(n) + 2011) = f(n) + f(f(n))$$

for all $n \in \mathbb{N}$?

Uzbekistan Mathematical Olympiad 2011

N 96. Determine all functions $f : \mathbb{N} \to \mathbb{N}$ satisfying that

- $f(n!) = f(n)!$ for every positive integer n,
- $m - n$ divides $f(m) - f(n)$ whenever m and n are different positive integers.

Balkan Mathematical Olympiad 2012

N 97. Given an integer $n \geq 2$, a function $f : \mathbb{Z} \to \{1, 2, \cdots, n\}$ is called *good*, if for any integer $k, 1 \leq k \leq n - 1$ there exists an integer $j(k)$ such that for every integer m we have

$$f(m + j(k)) \equiv f(m + k) - f(m) \pmod{n + 1}.$$

Find the number of *good* functions.

China Team Selection Test 2012

N 98. Find all functions $f : \mathbb{Z} \to \mathbb{Z}$ such that

$$f(a)^2 + f(b)^2 + f(c)^2 = 2f(a)f(b) + 2f(b)f(c) + 2f(c)f(a)$$

for all integers $a, b, c \in \mathbb{Z}$ with $a + b + c = 0$.

IMO Shortlist 2012 A1, IMO 2012 Problem 4
Proposed by Liam Baker (South Africa)

N 99. Let $f : \mathbb{N} \to \mathbb{N}$ be a function. Suppose that for every $n \in \mathbb{N}$ there exists a $k \in \mathbb{N}$ such that $f^{2k}(n) = n + k$, and let k_n be the smallest such k. Prove that the sequence $k_1, k_2, \cdots$ is unbounded.

IMO Shortlist 2012 A6

N 100. Each positive integer is coloued red or blue. A function $f : \mathbb{N} \to \mathbb{N}$ has the following two properties.

- if $x \le y$, then $f(x) \le f(y)$.
- if x, y and z are (not necessarily distinct) positive integers of the same color and $x + y = z$, then $f(x) + f(y) = f(z)$.

Prove that there exists a positive number a such that $f(x) \le ax$ for all positive integers x.

Romanian Masters In Mathematics 2012
Proposed by Ben Elliott (United Kingdom)

N 101. Is there a bijection $f : \mathbb{N} \to \mathbb{N} \times \mathbb{N}$ such that, for all distinct $a, b, c \in \mathbb{N}$ with $\gcd(a, b, c) > 1$, the points $f(a)$, $f(b)$, $f(c)$ are not collinear?

Serbian Mathematical Olympiad 2012

N 102. Find all functions $f : \mathbb{N} \to \mathbb{N}$ satisfying the following conditions.

- $f(n!) = f(n)!$ for all positive integers n.
- $m - n$ divides $f(m) - f(n)$ for all distinct positive integers m, n.

United States of America Mathematical Olympiad 2012

N 103. Find all functions $f : \mathbb{N}_0 \to \mathbb{N}_0$ such that

$$f(f(f(n))) = f(n + 1) + 1$$

for all $n \in \mathbb{N}_0$.

IMO Shortlist 2013 A5

N 104. Find all functions $f : \mathbb{N} \to \mathbb{N}$ such that

$$m^2 + f(n) \mid mf(m) + n$$

for all $m, n \in \mathbb{N}$.

IMO Shortlist 2013 N1, Germany Team Selection Test 2014

N 105. Let $f : \mathbb{Z} \to \mathbb{Z}$ be a function satisfying the following conditions.

- $f(m) + f(n) + f\left(f\left(m^2 + n^2\right)\right) = 1$ for all $m, n \in \mathbb{Z}$.
- There exist integers a and b such that $f(a) - f(b) = 3$.

Prove that there exist integers c and d can be found such that $f(c) - f(d) = 1$.

Iran Team Selection Test 2013
Proposed by Amirhossein Gorzi

N 106. Find all functions $f : \mathbb{Z} \to \mathbb{R}$ such that

$$f(m) + f(n) = f(mn) + f(m + n + mn)$$

for all $m, n \in \mathbb{Z}$.

Japanese Mathematical Olympiad 2013

N 107. Determine all functions $f : \mathbb{Z} \to \mathbb{Z}$ such that

$$f\big(f(m) + n\big) + f(m) = f(n) + f(3m) + 2014$$

for all $m, n \in \mathbb{Z}$.

IMO Shortlist 2014 A4
Proposed by Netherlands

N 108. Find all functions $f : \mathbb{Z} \to \mathbb{Z}$ such that

$$n^2 + 4f(n) = f(f(n))^2$$

for all $n \in \mathbb{Z}$.

IMO Shortlist 2014 A6
Proposed by Sahl Khan (United Kingdom)

N 109. Find all $f : \mathbb{N} \to \mathbb{N}$ such that

$$f(a) + f(b) \mid 2(a + b - 1)$$

for all $a, b \in \mathbb{N}$.

Middle European Mathematical Olympiad 2014

N 110. Let $f : \mathbb{N} \to \mathbb{N}$ be a function defined by $f(1) = 1$, $f(2n) = f(n)$ and $f(2n + 1) = f(n) + f(n + 1)$ for all $n \in \mathbb{N}$. Show that, for any positive integer n, the number of positive odd integers m such that $f(m) = n$ is equal to the number of positive integers less or equal to n and coprime to n.

Romania Team Selection Test 2014

N 111. There is an even number of cards on a table. A positive integer is written on each card. Let a_k be the number of cards having k written on them. It is known that

$$a_n - a_{n-1} + a_{n-2} - \cdots \geq 0$$

for each positive integer n. Prove that the cards can be partitioned into pairs so that the numbers in each pair differ by 1.

Tuymaada Olympiad 2014
Proposed by A. Golovanov

N 112. Let $S = \{2, 3, 4, \cdots\}$. Does there exist a function $f : S \to S$ such that

$$f(a)f(b) = f\left(a^2b^2\right)$$

for all $a, b \in S$ with $a \neq b$?

Asian Pacific Mathematics Olympiad 2015
Proposed by Angelo Di Pasquale (Australia)

N 113. Find all functions $f : \mathbb{N} \to \mathbb{N}$ such that

$$(n-1)^2 < f(n)f(f(n)) < n^2 + n$$

for all $n \in \mathbb{N}$.

Canadian Mathematical Olympiad 2015

N 114. Determine all functions $f : \mathbb{Z} \to \mathbb{Z}$ such that

$$f(x - f(y)) = f(f(x)) - f(y) - 1$$

for all $x, y \in \mathbb{Z}$.

IMO Shortlist 2015 A2, Greece Team Selection Test 2016

N 115. Let $2\mathbb{Z}+1$ denote the set of odd integers. Find all functions $f : \mathbb{Z} \to 2\mathbb{Z}+1$ such that

$$f\left(x + f(x) + y\right) + f\left(x - f(x) - y\right) = f(x + y) + f(x - y)$$

for all $x, y \in \mathbb{Z}$.

IMO Shortlist 2015 A5

N 116. Suppose that $a_0, a_1, \cdots$ and $b_0, b_1, \cdots$ are two sequences of positive integers satisfying the following conditions.

- $a_0, b_0 \geq 2$.
- $a_{n+1} = \gcd\left(a_n, b_n\right) + 1$ for all $n \in \mathbb{N}$.
- $b_{n+1} = \operatorname{lcm}\left(a_n, b_n\right) - 1$ for all $n \in \mathbb{N}$.

Show that the sequence a_n is eventually periodic; in other words, there exist integers $N \geq 0$ and $t > 0$ such that

$$a_{n+t} = a_n$$

for all $n \in \mathbb{N}$ with $n \geq N$.

IMO Shortlist 2015 N4

N 117. For any positive integer k, a function $f : \mathbb{N} \to \mathbb{N}$ is called *k-good* if

$$\gcd(f(m) + n, f(n) + m) \leq k$$

for all $m, n \in \mathbb{N}$ with $m \neq n$. Find all k such that there exists a k-good function.

IMO Shortlist 2015 N7, Iran Team Selection Test 2016
James Rickards (Canada)

N 118. Find all functions $f : \mathbb{N} \to \mathbb{N}$ such that

$$f\left(mf(n)\right) = n + f(2015m)$$

for all $m, n \in \mathbb{N}$.

Moldova Team Selection Test 2015

N 119. Find all functions $f : \mathbb{N} \to \mathbb{N}$ satisfying the following conditions.

- The integer $f(m) + f(n) - mn$ is nonzero for all $m, n \in \mathbb{N}$.
- $f(m) + f(n) - mn \mid mf(m) + nf(n)$ for all $m, n \in \mathbb{N}$.

IMO Shortlist 2016 N6
Proposed by Dorlir Ahmeti (Albania)

N 120. Find all functions $f : \mathbb{N} \to \mathbb{N}$ satisfying the following conditions.

- $f(mn) = f(m)f(n)$ for all $m, n \in \mathbb{N}$.
- $m + n \mid f(m) + f(n)$ for all $m, n \in \mathbb{N}$.

Turkey Team Selection Test 2016
Proposed by Melih Üçer

N 121. Let $S = \{1, \cdots, n\}$. Given a bijection $f : S \to S$ an *orbit* of f is a set of the form $\{x, f(x), f(f(x)), \cdots\}$ for some $x \in S$. We denote by $c(f)$ the number of distinct orbits of f. For example, if $n = 3$ and $f(1) = 2$, $f(2) = 1$, $f(3) = 3$, the two orbits are $\{1, 2\}$ and $\{3\}$, hence $c(f) = 2$. Given k bijections $f_1, \cdots, f_k$ from S to itself, prove that

$$c(f_1) + \cdots + c(f_k) \leq n(k-1) + c(f),$$

where $f : S \to S$ is the composed function $f_1 \circ \cdots \circ f_k$.

USA Team Selection Test 2016
Proposed by Maria Monks Gillespie

N 122. Find all functions $f : \mathbb{Z} \to \mathbb{Z}$ such that

$$xf(2f(y) - x) + y^2 f(2x - f(y)) = \frac{f(x)^2}{x} + f(yf(y))$$

for all $x, y \in \mathbb{Z}$ with $x \neq 0$.

United States of America Mathematical Olympiad 2016

N 123. Find all functions $f : \mathbb{N} \to \mathbb{N}$ such that

$$n + f(m) \mid f(n) + nf(m)$$

for all $m, n \in \mathbb{N}$.

Balkan Mathematical Olympiad 2017
Proposed by Dorlir Ahmeti (Albania)

N 124. Find the smallest positive integer k for which there exists a coloring of the positive integers with k colors and a function $f : \mathbb{N} \to \mathbb{N}$ with the following two properties.

- For all positive integers m, n of the same color, $f(m + n) = f(m) + f(n)$.
- There are positive integers m, n such that $f(m + n) \neq f(m) + f(n)$.

In a coloring of $\mathbb{N}$ with k colors, every integer is colored in exactly one of the k colors. In both (i) and (ii) the positive integers m, n are not necessarily distinct.

European Girls' Mathematical Olympiad 2017

Don't just read it; fight it! Ask your own questions, look for your own examples, discover your own proofs. Is the hypothesis necessary? Is the converse true? What happens in the classical special case? What about the degenerate cases? Where does the proof use the hypothesis?

– Paul Halmos

6. Functional equations over $\mathbb{Q}$

It gives me the same pleasure when someone else proves a good theorem as when I do it myself.

– Edmund Landau

Q 1. Find all functions $f : \mathbb{Q} \to \mathbb{Q}$ satisfying the following conditions.

- $f(1) = 2$.
- $f(xy) = f(x)f(y) - f(x+y) + 1$ for all $x, y \in \mathbb{Q}$.

IMO Shortlist 1980

Q 2. Find all functions $f : \mathbb{Q} \to \mathbb{R}$ such that

$$f(xy) = f(x)f(y) - f(x+y) + 1$$

for all $x, y \in \mathbb{Q}$.

Austrian Polish Mathematical Competition 1984

Q 3. Find all functions $f : \mathbb{Q} \to \mathbb{C}$ satisfying the following conditions.

- $\overline{f(x_1 + x_2 + \cdots + x_{1988})} = f(x_1)f(x_2)\cdots f(x_{1988})$ for all $x_1, x_2, \cdots, x_{1988} \in \mathbb{Q}$.
- $\overline{f(1988)}f(x) = f(1988)\overline{f(x)}$ for all $x \in \mathbb{Q}$.

China Team Selection Test 1988

Q 4. Construct a function $f : \mathbb{Q}^+ \to \mathbb{Q}^+$ such that

$$f(xf(y)) = \frac{f(x)}{y}$$

for all $x, y \in \mathbb{Q}^+$.

IMO 1990 Problem 4

Q 5. Let $f : \mathbb{Q} \to \mathbb{Z}$ be a function. Prove that there exist two distinct rational numbers a and b such that

$$\frac{f(a) + f(b)}{2} \leq f\left(\frac{a+b}{2}\right).$$

Russian Mathematical Olympiad 1990
Proposed by S. Berlov

Q 6. Find all functions $f : \mathbb{Q}^+ \to \mathbb{Q}^+$, where $\mathbb{Q}^+$ satisfying the following conditions.

- $f(x+1) = f(x) + 1$ for all $x \in \mathbb{Q}^+$.
- $f\left(x^3\right) = f(x)^3$ for all $x \in \mathbb{Q}^+$.

Polish Mathematical Olympiad 1992

Q 7. Determine all functions $f : \mathbb{Q}^+ \to \mathbb{Q}^+$ such that

$$f\left(x + \frac{y}{x}\right) = f(x) + f\left(\frac{y}{x}\right) + 2y$$

for all $x, y \in \mathbb{Q}^+$.

Turkey Team Selection Test 1993

Q 8. Find all functions $f : \mathbb{Q}^+ \to \mathbb{Q}^+$ satisfying the following conditions.

- $f(x+1) = f(x) + 1$ for all $x \in \mathbb{Q}^+$.
- $f\left(x^2\right) = f(x)^2$ for all $x \in \mathbb{Q}^+$.

Ukrine 1997

Q 9. Find all functions $f : \mathbb{Q} \to \mathbb{Q}$ such that

$$f(x+y) + f(x-y) = 2\left(f(x) + f(y)\right).$$

for all $x, y \in \mathbb{Q}$.

Nordic Mathematics Contest 1998

Q 10. Determine all functions $f : \mathbb{Q} \to \mathbb{R}$ such that

$$f(x+y) = f(x)f(y) - f(xy) + 1$$

for all $x, y \in \mathbb{Q}$.

Hungary-Israel Binational Mathematical Competition 1999

Q 11. Suppose that the functions $f, g : \mathbb{Q} \to \mathbb{Q}$ are strictly monotone increasing functions which attain every rational value. Is it necessarily true that the range of values of the function $f + g$ is also the whole set $\mathbb{Q}$?

Középiskolai Matematikai és Fizikai Lapok, January 2000, A. 228
Proposed by E. Fried

Q 12. Determine all functions $f : \mathbb{Q} \to \mathbb{Q}$ satisfying the following conditions.

- $f(x) + f\left(\frac{1}{x}\right) = 1$ for all $x \in \mathbb{Q} - \{0\}$.
- $2f(f(x)) = f(2x)$ for all $x \in \mathbb{Q}$.

Iranian Mathematical Olympiad 2001

Q 13. Determine all functions $f : \mathbb{Q}^+ \to \mathbb{Q}^+$ satisfying the following conditions.

- $f\left(\frac{1}{x}\right) = f(x)$ for all $x \in \mathbb{Q}^+$.
- $\left(1 + \frac{1}{x}\right) f(x) = f(x+1)$ for all $x \in \mathbb{Q}^+$.

Baltic Way 2003

Q 14. Does there exist a function $s\colon \mathbb{Q} \to \{-1, 1\}$ such that if x and y are distinct rational numbers satisfying $xy = 1$ or $x + y \in \{0, 1\}$, then $s(x)s(y) = -1$?

IMO Shortlist 2004 A3
Proposed by Dan Brown (Canada)

Q 15. Find all functions $f : \mathbb{Q} \to \mathbb{R}$ satisfying the following conditions.
- $f(xy) = f(x)f(y)$ for all $x, y \in \mathbb{Q}$.
- $f(x + y) \leq \max(f(x), f(y))$ for all $x, y \in \mathbb{Q}$.

Középiskolai Matematikai és Fizikai Lapok, April 2004, A. 346

Q 16. Find all functions $f : \mathbb{Q} \to \mathbb{R}$ satisfying the following conditions.
- $f(1) + 1 > 0$.
- $f(x + y) - xf(y) - yf(x) = f(x)f(y) - x - y + xy$ for all $x, y \in \mathbb{Q}$.
- $f(x) = 2f(x + 1) + x + 2$ for all $x \in \mathbb{Q}$.

Balkan Mathematical Olympiad 2005

Q 17. Determine all functions $f : \mathbb{Q} \to \mathbb{Q}$ such that
$$f(2f(x) + f(y)) = 2x + y,$$
for all $x, y \in \mathbb{Q}$.

Canadian Mathematical Olympiad 2005

Q 18. Find all functions $f : \mathbb{Q}^+ \to \mathbb{R}^+$ such that
$$f(x) + f(y) + 2xyf(xy) = \frac{f(xy)}{f(x + y)}$$
for all $x, y \in \mathbb{Q}^+$.

Bundeswettbewerb Mathematik 2006

Q 19. Let r and s be two rational numbers. Find all functions $f : \mathbb{Q} \to \mathbb{Q}$ such that
$$f(x + f(y)) = f(x + r) + y + s$$
for all $x, y \in \mathbb{Q}$.

Romania Team Selection Test 2006

Q 20. Find all functions $f : \mathbb{Q}^+ \mapsto \mathbb{Q}^+$ such that
$$f(x) + f(y) + 2xyf(xy) = \frac{f(xy)}{f(x + y)}.$$
for all $x, y \in \mathbb{Q}^+$.

China Team Selection Test 2007

Q 21. Let $f : \mathbb{Q} \to \mathbb{R}$ be a function such that
$$|f(x) - f(y)| \leq (x - y)^2$$
for all $x, y \in \mathbb{Q}$. Prove that the function f is constant.

Romania Team Selection Test 2007

Q 22. Let $S \subseteq \mathbb{R}$ be a set of real numbers. We say that a pair (f, g) of functions from S into S is a *Spanish Couple* on S, if they satisfy the following conditions.

(1) Both functions are strictly increasing, i.e. $f(x) < f(y)$ and $g(x) < g(y)$ for all $x, y \in S$ with $x < y$.
(2) The inequality $f(g(g(x))) < g(f(x))$ holds for all $x \in S$.

Decide whether there exists a Spanish Couple

- on the set $S = \mathbb{N}$ of positive integers,
- on the set $S = \{a - \frac{1}{b} \mid a, b \in \mathbb{N}\}$.

IMO Shortlist 2008 A3, Germany Team Selection Test 2009
Proposed by Hans Zantema (Netherlands)

Q 23. Find all $f : \mathbb{Q}^+ \to \mathbb{Z}$ functions satisfying the following conditions.

- $f\left(\frac{1}{x}\right) = f(x)$ for all $x \in \mathbb{Q}^+$.
- $(x+1)\, f(x-1) = xf(x)$ for all $x \in \mathbb{Q}^+$ with $x > 1$.

Turkey Team Selection Test 2009

Q 24. Determine all functions $f : \mathbb{Q}^+ \mapsto \mathbb{Q}^+$ such that

$$f\left(f(x)^2 y\right) = x^3 f(xy).$$

for all $x, y \in \mathbb{Q}^+$.

IMO Shortlist 2010 A5
Proposed by Thomas Huber (Switzerland)

Q 25. Determine all functions $f : \mathbb{Q}^+ \to \mathbb{Q}^+$ satisfying the conditions.

- $f\left(\frac{x}{x+1}\right) = \frac{f(x)}{x+1}$ for all $x \in \mathbb{Q}^+$.
- $f\left(\frac{1}{x}\right) = \frac{f(x)}{x^3}$ for all $x \in \mathbb{Q}^+$.

Turkey Team Selection Test 2011

Q 26. Let $f : \mathbb{Q}^+ \to \mathbb{R}$ be a function satisfying the following conditions.

- $f(x)f(y) \geq f(xy)$ for all $x, y \in \mathbb{Q}^+$.
- $f(x+y) \geq f(x) + f(y)$ for all $x, y \in \mathbb{Q}^+$.
- There exists a rational number $a > 1$ such that $f(a) = a$.

Prove that $f(x) = x$ for all $x \in \mathbb{Q}^+$.

IMO Shortlist 2013 A3, IMO 2013 Problem 5
Proposed by Bulgaria

Q 27. Determine all functions $f : \mathbb{Q} \to \mathbb{Z}$ such that

$$f\left(\frac{f(x)+a}{b}\right) = f\left(\frac{x+a}{b}\right)$$

for all $x \in \mathbb{Q}$, $a \in \mathbb{Z}$, and $b \in \mathbb{N}$.

IMO Shortlist 2013 N6

Q 28. Find all functions $f : \mathbb{Q} \mapsto \mathbb{Q}$ such that

$$f(x+y) + f(y+z) + f(z+t) + f(t+x) + f(x+z) + f(y+t) \geq 6f(x - 3y + 5z + 7t)$$

for all $x, y \in \mathbb{Q}$.

Uzbekistan Mathematical Olympiad 2013

Q 29. Determine all functions $f : \mathbb{Q}^+ \to \mathbb{R}^+$ such that

$$f(xy) = f(x+y)(f(x) + f(y))$$

for all $x, y \in \mathbb{Q}^+$.

Bulgarian Mathematical Olympiad 2014
Proposed by Nikolay Nikolov

Q 30. Find all functions $f : \mathbb{Q}^+ \to \mathbb{Q}^+$ such that

$$\underbrace{f(f(f(\cdots f(f}_{n}(q))\cdots))) = f(nq)$$

for all integers $n \geq 1$ and rational numbers $q > 0$.

Polish Mathematical Olympiad 2014

Q 31. Let $f : \mathbb{Q} \to \mathbb{Q}$ be a function such that for any $x, y \in \mathbb{Q}$, the number

$$f(x+y) - f(x) - f(y)$$

is an integer. Decide whether it follows that there exists a constant c such that

$$f(x) - cx$$

is an integer for all $x \in \mathbb{Q}$.

USA Team Selection Test 2015
Proposed by Victor Wang

Q 32. Let $\alpha \in \mathbb{Q}^+$ be a constant. Find all functions $f : \mathbb{Q}^+ \to \mathbb{Q}^+$ such that

$$f\left(\frac{x}{y} + y\right) = \frac{f(x)}{f(y)} + f(y) + \alpha x$$

for all $x, y \in \mathbb{Q}^+$.

Austrian Mathematical Olympiad 2016
Proposed by Walther Janous

Q 33. Find all functions $f : \mathbb{Q}^+ \to \mathbb{N}$ such that

$$f(xy)\ \gcd\left(f(x)f(y), f\left(\frac{1}{x}\right) f\left(\frac{1}{y}\right)\right) = xyf\left(\frac{1}{x}\right) f\left(\frac{1}{y}\right)$$

for all $x, y \in \mathbb{Q}^+$.

Benelux Mathematical Olympiad 2017

Q 34. Do there exist bijective functions $f, g : \mathbb{Q} \to \mathbb{Q}$ such that the function $f(g(x))$ is strictly increasing, but the function $g(f(x))$ is strictly decreasing?

Középiskolai Matematikai és Fizikai Lapok, March 2017, A. 692

The art of doing mathematics consists in finding that special case which contains all the germs of generality.

– David Hilbert

7. Functional equations over $\mathbb{R}$

Mathematicians do not study objects, but relations between objects. – Henri Poincaré

R 1. Find all functions $f : \mathbb{R} \to \mathbb{R}$ such that
$$xf(y) + yf(x) = (x + y)f(x)f(y)$$
for all $x, y \in \mathbb{R}$. Prove that exactly two of them are continuous.

Bulgarian Mathematical Olympiad 1968 (Fourth Round)
Proposed by I. Dimovski

R 2. Let f be a real-valued function defined for all real numbers, such that for some $a > 0$ we have
$$f(x + a) = \frac{1}{2} + \sqrt{f(x) - f(x)^2}$$
for all $x \in \mathbb{R}$.

- Prove that the function f is periodic.
- For $a = 1$, give an example of a non-constant function with the required properties.

IMO 1968 Problem 5

R 3. Find all continuous functions $f : \mathbb{R}^+ \to \mathbb{R}$ such that
$$f(x + y) + g(xy) = h(x) + h(y)$$
for all $x, y > 0$.

Miklós Schweitzer Competition 1969

R 4. Let $f, g : \mathbb{R} \to \mathbb{R}$ be functions such that
$$f(x + y) + f(x - y) = 2f(x)g(y).$$
for all $x, y \in \mathbb{R}$. Prove that if $f(x)$ is not identically zero, and if $|f(x)| \leq 1$ for all $x \in \mathbb{R}$, then $|g(y)| \leq 1$ for all $y \in \mathbb{R}$.

IMO 1972 Problem 5

R 5. Let G be a set of non-constant functions of the real variable x of the form
$$f(x) = ax + b \quad \text{(a and b are real numbers)}.$$
The set G has the following properties:

- If f and g are in G, then $g \circ f$ is in G; here $(g \circ f)(x) = g(f(x))$.
- If f and g are in G, its inverse $f^{-1}$is in G; here $f^{-1}(x) = \frac{x-b}{a}$.
- For every f in G, there exists a real number x_f such that $f(x_f) = x_f$.

Prove that there exists a constant $k \in \mathbb{R}$ such that $f(k) = k$ for all functions f in G.

IMO 1973 Problem 5

R 6. Let $f : \mathbb{R}^+ \to \mathbb{R}$ be a function satisfying the following conditions.

- $f(xy) = xf(y) + yf(x)$ for all $x, y > 0$.
- $f(x + 1) \leq f(x)$ for all $x > 0$.

- $f\left(\frac{1}{2}\right) = \frac{1}{2}$.

Show that
$$f(x) + f(1-x) \geq -x\log_2 x - (1-x)\log_2(1-x)$$
for all $x \in (0,1)$.

Miklós Schweitzer Competition 1977
Proposed by Z. Daroczy, Gy. Maksa

R 7. Let $f : \mathbb{R} \to \mathbb{R}$ be a function. Show that the following two conditions are equivalent.

(1) $f(x+y) = f(x) + f(y)$ for all $x, y \in \mathbb{R}$.
(2) $f(x+y+xy) = f(x) + f(y) + f(xy)$ for all $x, y \in \mathbb{R}$.

IMO Shortlist 1979

R 8. Find all functions $f : \mathbb{R}^+ \to \mathbb{R}^+$ satisfying the following conditions.

- $f(xf(y)) = yf(x)$ for all $x, y > 0$.
- $f(x) \to 0$ as $x \to \infty$.

IMO 1983 Problem 1

R 9. Let $f : \mathbb{R} \to \mathbb{R}$ be a function such that
$$f(f(x)) = -x$$
for all $x \in \mathbb{R}$. Prove that f has infinitely many points of discontinuity.

Vietnam Team Selection Test 1985

R 10.

(1) Show that on any real interval $[a,b]$, the exponential function e^x is characterized, up to multiplication by an arbitrary positive constant, by the inequality
$$f(x) < \frac{f(y) - f(x)}{y - x} < f(y),$$
where $x, y \in [a,b]$ with $x < y$.
(2) Are the exponentials the only functions on an interval $\mathbf{I}$ that satisfy the inequality
$$\min\{f(x), f(y)\} \leq \frac{f(y) - f(x)}{y - x} \leq \max\{f(x), f(y)\},$$
for all $x, y \in \mathbf{I}$ with $x \neq y$?

Amer. Math. Monthly 1986 Problem E 3127
Proposed by David Shelupsky

R 11. Let f be a continuous real-valued function defined on the interval $[0,L]$. If the is an equilateral triangle ABC of side length L such that
$$f(AP) + f(BP) + f(CP) = 0$$
for all points P inside the triangle, prove that f is identically zero.

Amer. Math. Monthly 1986 Problem 6526
Proposed by Grzegorz Rzadkowski

R 12. Find all functions $f : [0,\infty) \to [0,\infty)$ satisfying the following conditions.

- $f(xf(y))f(y) = f(x+y)$ for all $x, y \geq 0$.
- $f(2) = 0$,
- $f(x) \neq 0$ for $0 \leq x < 2$.

IMO 1986 Problem 5

R 13. Let $a \in \mathbb{R}$ be a constant. Let $f : \mathbb{R} \to \mathbb{R}$ be a function satisfying the following conditions.
- $f(0) = \frac{1}{2}$.
- $f(x+y) = f(x)f(a-y) + f(y)f(a-x)$ for all $x, y \in \mathbb{R}$.

Prove that the function f is constant.

Balkan Mathematical Olympiad 1987

R 14. Let $f : [0,1] \to \mathbb{R}$ be a function such that
$$f(x_1 - x_2) < |x_1 - x_2|$$
for all $x_1, x_2 \in [0,1]$ with $x_1 \neq x_2$. Prove that
$$f(x_1 - x_2) < \frac{1}{2}$$
for all $x_1, x_2 \in [0,1]$.

Hungarian Mathematical Olympiad 1987

R 15. Let $f : \mathbb{R} \to \mathbb{R}$ be a function satisfying the following conditions.
- If $x > y$ and $f(y) - y \geq v \geq f(x) - x$, then $f(z) = v + z$, for some number z between x and y.
- The equation $f(x) = 0$ has at least one solution, and among the solutions of this equation, there is one that is not smaller than all the other solutions.
- $f(0) = 1$.
- $f(1987) \leq 1988$.
- $f(x)f(y) = f(xf(y) + yf(x) - xy)$.

Find the value of $f(1987)$.

IMO Shortlist 1987
Proposed by Australia

R 16. Prove that there exists a *unique* function $f : \mathbb{R}^+ \to \mathbb{R}^+$ satisfying the following conditions.
- $f(f(x)) = 6x - f(x)$ for all $x > 0$.
- $f(x) > 0$ for all $x > 0$.

William Lowell Putnam Competition 1988

R 17. Determine all functions $f : \mathbb{R} \to \mathbb{R}$ satisfying the following conditions.
- $f(x)$ is strictly increasing.
- $f(x) + f^{-1}(x) = 2x$ for all $x \in \mathbb{R}$.

Asian Pacific Mathematics Olympiad 1989

R 18. Find all functions $f, g, h : \mathbb{R} \mapsto \mathbb{R}$ such that

$$f(x) - g(y) = (x - y)h(x + y)$$

for all $x, y \in \mathbb{R}$.

China Team Selection Test 1990

R 19. Find all functions $f : [-1, 1] \to [-1, 1]$ such that

$$|xf(y) - yf(x)| \geq |x - y|$$

for all $x, y \in [-1, 1]$.

Crux Mathematicorum 1990 Problem 1550
Proposed by Mihály Bencze

R 20. Does there exist a continuous real-valued function on $\mathbb{R}$ whose values are rational on the irrationals and irrational on the rationals?

Mathematics Magazine 1990 Problem Q 757
Proposed by Jozsef Mala, Sandor Roka

R 21. Find all functions $f : \mathbb{R} \to \mathbb{R}$ satisfying the following conditions.

- The function f is (not necessarily strict) monotonic.
- $f(xy) = f\left(\frac{f(y)}{x}\right)$ for all $x, y \in \mathbb{R}$.

Swedish Mathematics Olympiad (Final Round) 1990

R 22. Prove that there exists no function $f : \mathbb{R} \to \mathbb{R}$ such that

$$f(f(x)) = x^2 - 2$$

for all $x \in \mathbb{R}$.

Vietnam Team Selection Test 1990

R 23. Let S be a non-empty interval on the real line. Let $f : S \to S$ be a continuous function having the property that for each $x \in S$ there exists a positive integer $n = n(x)$ with $f^{n(x)} = x$, where f^n denotes the n-th iterate of f. For given S, characterize all such functions.

Amer. Math. Monthly 1991 Problem E 3428
Proposed by Artin B. Boghossian

R 24. Find all functions $f : \mathbb{R} \to \mathbb{R}$ such that

$$\frac{f(xy) + f(xz)}{2} - f(x)f(yz) \geq \frac{1}{4}$$

for all $x, y, z \in \mathbb{R}$.

Vietnamese Mathematical Olympiad 1991

R 25. Consider the functional equation

$$f(a-b)f(a-c)f(a-d)f(a-e) - f(b)f(c)f(d)f(e) = q^b f(a)f(a-b-c)f(ab-d)f(a-b-e)$$

with parameter q, , where the variables a, b, c, d and e are related by

$$b + c + d + e = 2a.$$

(1) When $q=1$, show that, for any k, $f(\alpha)=\sin(k\alpha)$ is a solution.
(2) When $0<q<!1$, show that $\left(\Gamma_q(\alpha)\Gamma_q(1-\alpha)\right)^{-1}$ is a solution.

Amer. Math. Monthly 1992 Problem 10226
Proposed by Chu Wenchang

R 26. Let $n\geq 3$ be an integer. Determine all continuous functions $f:[0,1]\to\mathbb{R}$ such that

$$f(x_1)+\cdots+f(x_n)=1$$

for all $x_1,\cdots,x_n\in[0,1]$ satisfying $x_1+\cdots+x_n=1$.

Crux Mathematicorum 1992 Problem 1723
Proposed by Walther Janous

R 27. Find all functions $f:\mathbb{R}\to\mathbb{R}$ such that

$$f\left(x^2+f(y)\right)=y+(f(x))^2$$

for all $x,y\in\mathbb{R}$.

IMO 1992 Problem 2

R 28. Let $f,g:\mathbb{R}\to\mathbb{R}-\{0,1\}$ be functions such that

$$f(x+1)=\frac{f(x)}{g(x)}\quad\text{and}\quad g(x+1)=\frac{g(x)-1}{f(x)-1}$$

for all $x\in\mathbb{R}$. Prove that f and g are periodic.

Mathematics Magazine 1992 Problem 1398
Proposed by Florin S. Ptîrvănescu

R 29. Determine all pairs (a,b) of nonnegative real numbers such that the functional equation

$$f(f(x))+f(x)=ax+b$$

has a unique continuous solution $f:\mathbb{R}\to\mathbb{R}$.

Crux Mathematicorum 1993 Problem 1821
Proposed by Gerd Baron and Walther Janous

R 30. Assume that $f:\mathbb{R}\to\mathbb{R}$ has the property that $f(x+\lambda)-f(x)$ is continuous for all $\lambda\in\mathbb{R}$. Must f be continuous?

Mathematics Magazine 1993 Problem Q 810
Proposed by Daniel Goffinet, Patrick Goffine

R 31. Consider functions $f:[0,1]\to\mathbb{R}$ satisfying the following conditions.
(1) $f(x)\geq 0$ for all x in $[0,1]$,
(2) $f(1)=1$,
(3) $f(x)+f(y)\leq f(x+y)$ whenever x, y, and $x+y$ are all in $[0,1]$.
Find, with proof, the smallest constant c such that

$$f(x)\leq cx$$

for all functions f satisfying the three conditions (1), (2), (3).

United States of America Mathematical Olympiad 1993

R 32. Do there exist nonlinear C^1 functions $f : \mathbb{R} \to \mathbb{R}$ such that for any rational x, $f(x)$ is also rational and for any irrational x, $f(x)$ is also irrational?

Amer. Math. Monthly 1994 Problem 10361
Proposed by Emil A. Cornea, Florin N. Diacu

R 33. Suppose that $f_1, \cdots, f_n$ are continuous real periodic functions, and that $f_1 + \cdots + f_n$ is a constant function, while no sum of fewer than n of $f_1, \cdots, f_n$ is a constant function. Show that the $f_1, \cdots, f_n$ have a common period.

Amer. Math. Monthly 1994 Problem 10380
Proposed by Michael Slater

R 34. Find all functions $f : \mathbb{R} \to \mathbb{R}$ such that

$$xf(x)yf(y) = (xy)f(x+y)$$

for all $x, y \in \mathbb{R}$.

Bulgarian Mathematical Olympiad 1994 (Fourth Round)
Proposed by I. Dimovski

R 35. Suppose that $f : \mathbb{R} \to \mathbb{R}$ is a continuous even function satisfying the following conditions.
- $f(0) = 0$.
- $f(x + y) \le f(x) + f(y)$ for all $x, y \in \mathbb{R}$.

Must f be monotonic on $\mathbb{R}^+$?

Crux Mathematicorum 1994 Problem 1901
Proposed by Marcin E. Kuczma

R 36. Find all functions $f : \mathbb{R} \to \mathbb{R}$ satisfying the following conditions.
- $f(x) + f(y) + 1 \ge f(x + y) \ge f(x) + f(y)$ for all $x, y \in \mathbb{R}$.
- $f(0) \ge f(x)$ for all $x \in [0, 1)$.
- $-f(-1) = f(1) = 1$.

Asian Pacific Mathematics Olympiad 1994

R 37. Find all functions $f : [1, \infty) \to [1, \infty)$ satisfying the following conditions
- $f(x) \le 2(x + 1)$ for all $x \ge 1$.
- $f(x + 1) = \frac{1}{x}\left(-1 + f(x)^2\right)$ for all $x, y \ge 1$.

Chinese Mathematical Olympiad 1994

R 38. Find all functions $f : (-1, \infty) \to S$ satisfying the following conditions.
- $f(x + f(y) + xf(y)) = y + f(x) + yf(x)$ for all $x, y > -1$.
- The function $\frac{f(x)}{x}$ is strictly increasing on each of the two intervals $(-1, 0)$ and $(0, \infty)$.

IMO 1994 Problem 5

R 39. Let $\alpha, \beta \in \mathbb{R}$ be constants, not necessarily distinct. Find all functions $f : \mathbb{R}^+ \to \mathbb{R}$ such that
$$f(x)f(y) = y^\alpha f\left(\frac{x}{2}\right) + x^\beta f\left(\frac{y}{2}\right)$$
for all $x, y > 0$.

IMO Shortlist 1994

R 40. Find all functions $f : \mathbb{R} \to \mathbb{R}$ satisfying the following three conditions.

- $f(-x) = -f(x)$ for all $x \in \mathbb{R}$.
- $f(x+1) = f(x) + 1$ for all $x \in \mathbb{R}$.
- $f\left(\frac{1}{x}\right) = \frac{f(x)}{x^2}$ for all $x \in \mathbb{R} - \{0\}$.

Polish Mathematical Olympiad 1994

R 41. Determine all functions $f : \mathbb{R} \to \mathbb{R}$ such that
$$f\left(\sqrt{2}x\right) + f\left(4 + 3\sqrt{2}x\right) = 2f\left(\left(2+\sqrt{2}\right)x\right)$$
for all $x \in \mathbb{R}$.

Vietnam Team Selection Test 1994

R 42. Let f be a continuous function from the unit disc $\mathbb{D}$ in $\mathbb{R}^2$ to itself such that

- $f \circ f$ is the identity of $\mathbb{D}$; and
- f is the identity on the unit circle $\partial\mathbb{D}$.

Show that f is the identity on $\mathbb{D}$.

Amer. Math. Monthly 1995 Problem 10442
Proposed by Roger Bielawski

R 43. Does there exist a function $f : \mathbb{R} \to \mathbb{R}$ which simultaneously satisfies the following three conditions?

- There is a constant M such that $-M \le f(x) \le M$ for all $x \in \mathbb{R}$.
- $f(1) = 1$.
- If $x \neq 0$, then
$$f\left(x + \frac{1}{x^2}\right) = f(x) + \left[f\left(\frac{1}{x}\right)\right]^2.$$

IMO Shortlist 1995

R 44. Let $\alpha \in \mathbb{R}$ be a constant. Find all functions $f : \mathbb{R}^+ \to \mathbb{R}^+$ such that
$$\alpha x^2 f\left(\frac{1}{x}\right) + f(x) = \frac{x}{x+1}$$
for all $x > 0$.

Israeli Mathematical Olympiad 1995

R 45. Let $f : \mathbb{R} \to \mathbb{R}$ be a function satisfying the following conditions.

- $f(x+24) \le f(x) + 24$ for all $x \in \mathbb{R}$.
- $f(x+77) \ge f(x) + 77$ for all $x \in \mathbb{R}$.

Prove that $f(x+1) = f(x) + 1$ for all $x \in \mathbb{R}$.

Italy Team Selection Test 1995

R 46. Let f be a function from the set of real numbers $\mathbb{R}$ into itself such for all $x \in \mathbb{R}$, we have $|f(x)| \le 1$ and

$$f\left(x+\frac{13}{42}\right)+f(x)=f\left(x+\frac{1}{6}\right)+f\left(x+\frac{1}{7}\right).$$

Prove that f is a periodic function (that is, there exists a constant $c \neq 0$ such $f(x+c)=f(x)$ for all $x \in \mathbb{R}$).

IMO Shortlist 1996

R 47. Find all functions $f : \mathbb{R} \to \mathbb{R}$ satisfying the following conditions.

- $f(x+f(y)+yf(x)) = y+f(x)+xf(y)$ for all $x, y \in \mathbb{R}$.
- The set $\{\frac{f(x)}{x} \mid x \in \mathbb{R} - \{0\}\}$ is finite.

Mathematics Magazine 1996 Problem 1491
Proposed by Wu Wei Chao

R 48. Let $f : \mathbb{R}^2 \to \mathbb{R}$ be a function such that, for every regular n-gon $A_1A_2\cdots A_n$,

$$f(A_1)+f(A_2)+\cdots+f(A_n)=0.$$

Prove that $f(x)=0$ for all $x \in \mathbb{R}$.

Romania Team Selection Test 1996

R 49. Show that there is no function $f : \mathbb{R}^+ \to \mathbb{R}^+$ such that

$$f(x+y) > f(x)(1+yf(x))$$

for all $x, y > 0$.

Turkish Mathematical Olympiad 1996

R 50. Find all continuous functions $f : \mathbb{R} \to \mathbb{R}$ such that

$$f(x)=f\left(x^2+\frac{1}{4}\right)$$

for all $x \in \mathbb{R}$.

Bulgarian Mathematical Olympiad 1997

R 51. Let $c > 0$ be a constant. Give a complete description, with proof, of the set of all continuous functions $f : \mathbb{R} \to \mathbb{R}$ such that $f(x) = f\left(x^2+c\right)$ for all $x \in \mathbb{R}$.

William Lowell Putnam Competition 1996

R 52. Let $f : \mathbb{R} \to \mathbb{R}$ be a function such that

$$f\left(x^n-y^n\right)=(x-y)\left\{[f(x)]^{n-1}+[f(x)]^{n-2}f(y)+\cdots+f(x)[f(y)]^{n-2}+[f(y)]^{n-1}\right\}.$$

Prove that $f(rx)=rf(x)$ for all $r \in \mathbb{Q}$ and $x \in \mathbb{R}$.

Mathematics Magazine 1997 Problem 1521
Proposed by Wu Wei Chao

R 53. Let $n \geq 4$ be a positive integer and let M be a set of n points in the plane, where no three points are collinear and not all of the n points being concyclic. Find all real functions $f : M \to \mathbb{R}$ such that for any circle $\mathcal{C}$ containing at least three points from M, the following equality holds:

$$\sum_{P \in \mathcal{C} \cap M} f(P) = 0.$$

Romania Team Selection Test 1997

R 54. Find all functions $f : \mathbb{R} \to [0; +\infty)$ such that

$$f\left(x^2 + y^2\right) = f\left(x^2 - y^2\right) + f\left(2xy\right)$$

for all $x, y \in \mathbb{R}$.

Romania Team Selection Test 1997
Proposed by Laurentiu Panaitopol

R 55. Does there exists a continuous function $f : \mathbb{R} \to \mathbb{R}$ such that

$$f(f(x)) = e^x - e^{e^x}$$

for all $x \in \mathbb{R}$?

Bernoulli Trials 1998

R 56. Find all functions $f : \mathbb{R}^+ \to \mathbb{R}^+$ such that

$$f(x)^2 \geq f(x+y)\left(f(x) + y\right)$$

for all $x, y > 0$.

Bulgarian Mathematical Olympiad 1998

R 57. Find all functions $f : \mathbb{R} \to \mathbb{R}$ such that

$$f(f(x) + y) = f(x^2 - y) + 4f(x)y$$

for all $x, y \in \mathbb{R}$.

Iranian Mathematical Olympiad 1998

R 58. Find all functions $f : (0, 1) \to \mathbb{R}$ such that

$$f\left(xy\right) = xf(x) + yf(y)$$

for all $x, y \in (0, 1)$.

Középiskolai Matematikai és Fizikai Lapok, November 1998, F. 3250

R 59. Find all functions $f : \mathbb{R} \to \mathbb{R}$ such that

$$f(x + yf(x)) = f(x) + xf(y)$$

for all $x, y \in \mathbb{R}$.

Mathematics Magazine 1998 Problem 1552
Proposed by Wu Wei Chao

R 60. Find all monotonic functions $u : \mathbb{R} \to \mathbb{R}$ which have the property that there exists a strictly monotonic function $f : \mathbb{R} \to \mathbb{R}$ such that
$$f(x+y) = f(x)u(x) + f(y)$$
for all $x, y \in \mathbb{R}$.

Romania Team Selection Test 1998
Proposed by Vasile Pop

R 61. Find an example of three continuous functions $f, g, h : \mathbb{R} \to \mathbb{R}$ with the property that exactly five of the six composite functions $f(g(h(x)),\ f(h(g(x))),\ g(f(h(x))),\ g(h(f(x))),\ h(f(g(x)))$ and $h(g(f(x)))$ are the same function and the sixth function is different.

Crux Mathematicorum 1999 Problem C87
Proposed by Mark Krusemeyer

R 62. Let $f : \mathbb{R}^+ \to \mathbb{R}^+$ be a strictly decreasing continuous function such that
$$f(x+y) + f(f(x) + f(y)) = f\left(f(x + f(y)) + f(y + f(x))\right)$$
for all $x, y > 0$. Obviously, $f(x) = \frac{c}{x}$ is a solution for $c > 0$. Determine all other solutions.

Crux Mathematicorum 1999 Problem 2474
Proposed by Mohammed Aassila

R 63. Find all functions $f : \mathbb{R} \to \mathbb{R}$ such that
$$f(x + yf(x)) = f(x) + xf(y)$$
for all $x, y \in \mathbb{R}$.

Hong Kong Mathematics Olympiad 1999

R 64. Determine all functions $f : \mathbb{R} \to \mathbb{R}$ such that
$$f(x - f(y)) = f(f(y)) + xf(y) + f(x) - 1$$
for all $x, y \in \mathbb{R}$.

IMO Shortlist 1999 A5, IMO 1999 Problem 6

R 65.

(1) Find all strictly monotone functions $f : \mathbb{R} \to \mathbb{R}$ such that
$$f(x + f(y)) = f(x) + y$$
for all real x, y.

(2) If $n > 1$ is an integer, prove that there is no strictly monotone function $f : \mathbb{R} \to \mathbb{R}$ such that
$$f(x + f(y)) = f(x) + y^n$$
for all real x, y.

Italy Team Selection Test 1999

R 66. Determine all functions $f : \mathbb{R} \to \mathbb{R}$ satisfying the following conditions.

- $f(1999) = 1000$.
- $f(xy) = f(x)f(y) + f(-x)f(-y)$ for all $x, y \in \mathbb{R}$.
- $f(x+y) \leq f(x) + f(y)$ for all $x, y \in \mathbb{R}$.

Középiskolai Matematikai és Fizikai Lapok, March 1999, N. 206

R 67. A continuous function $f : [0, 1] \to [0, 1]$ has the property that, for any real number $x \in [0, 1]$, the sequence

$$f(x),\ f^2(x) = f(f(x)),\ f^3(x) = f(f(f(x))),\ \cdots,$$

contains 0. Does it follow that, for n large enough, the composite function f^n is identically 0?

Középiskolai Matematikai és Fizikai Lapok, April 1999, N. 207

R 68. Find all continuous functions $f : \mathbb{R}^+ \to \mathbb{R}^+$ such that

$$f\left(\frac{1}{f(x)f(y)}\right) = f(x)f(y)$$

for all $x, y > 0$.

Középiskolai Matematikai és Fizikai Lapok, October 1999, A. 218

R 69. Find all functions $f : \mathbb{R} \to \mathbb{R}$ such that

$$f(x - f(y)) = f\left(x + y^{1999}\right) + f\left(f(y) + y^{1999}\right) + 1$$

for all $x, y \in \mathbb{R}$.

Középiskolai Matematikai és Fizikai Lapok, November 1999, A. 223

R 70. Determine all functions $f : \mathbb{R} \to \mathbb{R}$ satisfying the following conditions.

- The set $\left\{\frac{f(x)}{x} : x \neq 0 \text{ and } x \in \mathbb{R}\right\}$ is finite.
- $f(x - 1 - f(x)) = f(x) - x - 1$ for all $x \in \mathbb{R}$,

Turkey Team Selection Test 1999

R 71. Find all functions $f : \mathbb{R} \to \mathbb{R}$ such that

$$f(xf(x) + f(y)) = f^2(x) + y$$

for all $x, y \in \mathbb{R}$.

Balkan Mathematical Olympiad 2000

R 72. Let $f : \mathbb{R} \to \mathbb{R}$ be a continuous and periodic function such that

$$\frac{|f(1)|}{1} + \frac{|f(2)|}{2} + \cdots + \frac{|f(n)|}{n} \leq 1$$

for all $n \in \mathbb{N}$. Prove that there exists $c \in \mathbb{R}$ such that $f(c) = 0$ and $f(c + 1) = 0$.

Crux Mathematicorum 2000 Problem 2536
Proposed by Cristinel Mortici

R 73. Find all pairs of functions $f, g : \mathbb{R} \to \mathbb{R}$ such that

$$f(x + g(y)) = xf(y) - yf(x) + g(x)$$

for all $x, y \in \mathbb{R}$.

IMO Shortlist 2000

R 74. Find all functions $f : \mathbb{R}^+ \to \mathbb{R}^+$ such that

$$f(x)f\left(yf(x)\right) = f(x+y)$$

for all $x, y > 0$.

Középiskolai Matematikai és Fizikai Lapok, December 2000, A. 253

R 75. Call a real-valued function f *very convex* if

$$\frac{f(x)+f(y)}{2} \geq f\left(\frac{x+y}{2}\right) + |x-y|$$

for all $x, y \in \mathbb{R}$. Prove that no very convex function exists.

United States of America Mathematical Olympiad 2000

R 76. Let $f : \mathbb{R} \to \mathbb{R}$ be a function such that

$$|f(x+y) - f(x) - f(y)| \leq 1$$

for all $x, y \in \mathbb{R}$. Prove that there exists a function $g : \mathbb{R} \to \mathbb{R}$ satisfying the following conditions.

- $|f(x) - g(x)| \leq 1$ for all $x \in \mathbb{R}$.
- $g(x+y) = g(x) + g(y)$ for all $x, y \in \mathbb{R}$.

Turkey Team Selection Test 2000

R 77. Let $a > 1$ and $r > 1$ be constants.

- Let $f : \mathbb{R}^+ \to \mathbb{R}^+$ be a function satisfying the following conditions.
  (1) $f(x)^2 \leq ax^r f(\frac{x}{a})$ for all $x > 0$
  (2) $f(x) < 2^{2000}$ for all $x < \frac{1}{2^{2000}}$.
  Prove the inequality $f(x) \leq x^r a^{1-r}$ for all $x > 0$.
- Construct a function $f : \mathbb{R}^+ \to \mathbb{R}^+$ satisfying the following conditions.
  (1) $f(x)^2 \leq ax^r f(\frac{x}{a})$ for all $x > 0$
  (2) $f(x) > x^r a^{1-r}$ for all $x > 0$.

Vietnam Team Selection Test 2000

R 78. Let $f : \mathbb{R} \to \mathbb{R}$ be a continuous function such that

$$f\left(2x^2 - 1\right) = 2xf(x)$$

for all $x \in \mathbb{R}$. Show that $f(x) = 0$ for $-1 \leq x \leq 1$.

William Lowell Putnam Competition 2000

R 79. Find all functions $f : \mathbb{R} \to \mathbb{R}$ that is continuous at zero and satisfies

$$f\left(x + 2f(y)\right) = f(x) + y + f(y)$$

for all $x, y \in \mathbb{R}$.

Amer. Math. Monthly 2001 Problem 10854
Proposed by Wu Wei Chao

R 80. Find all functions $f : \mathbb{R} \to \mathbb{R}$ such that

$$f\left(x^2 + y + f(y)\right) = 2y + [\,f(x)\,]^2$$

for all $x, y \in \mathbb{R}$.

Amer. Math. Monthly 2001 Problem 10908
Proposed by Wu Wei Chao

R 81. Find all functions $f : \mathbb{R} \to \mathbb{R}$ such that

$$f\left(x^2 + y\right) + f(f(x) - y) = 2f(f(x)) + 2y^2$$

for all $x, y \in \mathbb{R}$.

Czech-Polish-Slovak Match 2001

R 82. For a given natural number $k > 1$, find all functions $f : \mathbb{R} \to \mathbb{R}$ such that

$$f\left(x^k + f(y)\right) = y + f(x)^k$$

for all $x, y \in \mathbb{R}$.

China Team Selection Test 2001

R 83. Find all continuous functions $f : \mathbb{R} \to \mathbb{R}$ such that

$$f(f(x)) = f(x) + x$$

for all $x \in \mathbb{R}$.

Hungary-Israel Binational Mathematical Competition 2001

R 84. Find all functions $f : \mathbb{R} \to \mathbb{R}$ such that

$$f(xy)(f(x) - f(y)) = (x - y)f(x)f(y)$$

for all $x, y \in \mathbb{R}$.

IMO Shortlist 2001 A4

R 85. Find all continuous functions $f : \mathbb{R} \to \mathbb{R}$ such that

$$f(f(x)) = f(x) + x$$

for all $x \in \mathbb{R}$.

Középiskolai Matematikai és Fizikai Lapok, May 2001, B. 3469, Gillis-Turán Mathematics Competition 2001

R 86. Let $f : \mathbb{R}^+ \to \mathbb{R}$ be the function such that

$$f\left(\frac{x+y}{2}\right) + f\left(\frac{2xy}{x+y}\right) = f(x) + f(y)$$

for all $x, y > 0$. Prove that

$$2f(\sqrt{xy}) = f(x) + f(y)$$

for all $x, y > 0$.

Miklós Schweitzer Competition 2001

R 87. Let $\mathbf{I} \subset \mathbb{R}$ be a non-empty open interval, $\varepsilon \geq 0$ and $f : \mathbf{I} \to \mathbb{R}$ a function such that
$$f(tx + (1-t)y) \leq t f(x) + (1-t) f(y) + \varepsilon\, t(1-t)|x-y|$$
for all $x, y \in \mathbf{I}$ and $t \in [0,1]$. Prove that there exists a convex function $g : \mathbf{I} \to \mathbb{R}$ such that the function $l := f - g$ has the ε-Lipschitz property, that is,
$$|l(x) - l(y)| \leq \varepsilon\, |x - y|$$
for all $x, y \in \mathbf{I}$.

Miklós Schweitzer Competition 2001

R 88. Prove that there exists no function $f : (0,\infty) \to (0,\infty)$ such that
$$f(x+y) \geq f(x) + y f(f(x))$$
for all $x, y > 0$.

Romania Team Selection Test 2001

R 89. Show that there exists no continuous function $f : \mathbb{R} \to \mathbb{R}$ such that
$$f(x - f(x)) = \frac{x}{2}$$
for all $x \in \mathbb{R}$.

Turkey Team Selection Test 2001

R 90. Let a and b be real numbers in the interval $\left(0, \frac{1}{2}\right)$, and let $g : \mathbb{R} \to \mathbb{R}$ be a continuous function such that
$$g(g(x)) = a g(x) + b x$$
for all $x \in \mathbb{R}$. Prove that there exists a constant c such that
$$g(x) = cx$$
for all $x \in \mathbb{R}$.

William Lowell Putnam Competition 2001

R 91. Find all functions $f : \mathbb{R} \to \mathbb{R}$ satisfying the following conditions.
- There are only finitely many s in $\mathbb{R}$ such that $f(s) = 0$.
- $f(x^4 + y) = x^3 f(x) + f(f(y))$ for all x, y in $\mathbb{R}$.

Asian Pacific Mathematics Olympiad 2002
Proposed by Hojoo Lee

R 92. Find all functions $f : \mathbb{R} \to \mathbb{R}$ such that
$$f(x) f\left(y f(x) - 1 \right) = x^2 f(y) - f(x)$$
for all $x, y \in \mathbb{R}$.

Crux Mathematicorum 2002 Problem 2772
Proposed by Wu Wei Chao

R 93. Find all functions $f : \mathbb{R} \to \mathbb{R}$ such that
$$f(x f(y)) = (1-x) f(xy) + x^2 y^2 f(y)$$
for all $x, y \in \mathbb{R}$.

Crux Mathematicorum 2002 Problem 2781
Proposed by Wu Wei Chao

R 94. Find all functions $f : \mathbb{R} \to \mathbb{R}$ such that

$$(f(x) + f(z))\,(f(y) + f(t)) = f(xy - zt) + f(xt + yz)$$

for all $x, y, z, t \in \mathbb{R}$.

IMO Shortlist 2002 A4, IMO 2002 Problem 5

R 95. Find all the continuous functions $f : \mathbb{R} \to \mathbb{R}$ such that

$$f(x) + f(y) + f(xy) = f(x + y + xy)$$

for all $x, y \in \mathbb{R}$.

Iranian Mathematical Olympiad 2002

R 96. Find all functions $f : \mathbb{R}^+ \to \mathbb{R}^+$ satisfying the following conditions.

- $f(x + f(y)) = f(x)f(y)$ for all $x, y > 0$.
- There are at most finitely many $x > 0$ with $f(x) = 1$.

Italy Team Selection Test 2002

R 97. Find all functions $f : \mathbb{R} \to \mathbb{R}$ such that

$$f\left(f(x) + y\right) = 2x + f\left(f(y) - x\right)$$

for all $x, y \in \mathbb{R}$.

IMO Shortlist 2002 A1, Italy Team Selection Test 2003, Germany Team Selection Test 2003

R 98. Find all functions $f : \mathbb{R} \to \mathbb{R}$ such that

$$f\left(\frac{x - y}{f(y)}\right) = \frac{f(x)}{f(y)}$$

for all $x, y \in \mathbb{R}^+$ with $x > y$.

Korean Mathematical Olympiads Winter Program Test 2002

R 99. Find all continuous functions $f : \mathbb{R} \to \mathbb{R}$ such that

$$f\left(\frac{x + y}{1 + xy}\right) = \frac{f(x)f(y)}{|\,1 + xy\,|}$$

for all $x, y \in \mathbb{R}$.

Középiskolai Matematikai és Fizikai Lapok, February 2002, A. 286
Based on the idea of Z. Győrfi and G. Ligeti

R 100. Find all functions $f : \mathbb{R}^+ \to \mathbb{R}^+$ such that

$$f(x + y) + f(x)f(y) = f(xy) + f(x) + f(y)$$

for all $x, y > 0$.

Középiskolai Matematikai és Fizikai Lapok, November 2002, A. 304

R 101. Determine all functions $f : \mathbb{R} \to \mathbb{R}$ such that

$$f\left(x^2 - y^2\right) = xf(x) - yf(y)$$

for all $x, y \in \mathbb{R}$.

United States of America Mathematical Olympiad 2002

R 102. If a function $f : \mathbb{R} \to \mathbb{R}$ has at least two centers of symmetry, show that this function can be written as sum of a linear function and a periodic function. (For every real number x, if there is a real number a such that $f(a-x) + f(a+x) = 2f(a)$, the point $(a, f(a))$ is called a center of symmetry of the function f.)

Turkey Team Selection Test 2002

R 103. Show that for $d < -1$ there are exactly two real-valued functions f such that

$$f(x+y) - f(x)f(y) = d \sin x \sin y$$

for all $x, y \in \mathbb{R}$.

Amer. Math. Monthly 2003 Problem 11030
Proposed by Steven Butler

R 104. Find all functions $f : \mathbb{R} \to \mathbb{R}$ such that

$$f(x + f(x)f(y)) = f(x) + xf(y)$$

for all $x, y \in \mathbb{R}$.

Amer. Math. Monthly 2003 Problem 11053 (a)
Proposed by Wu Wei Chao

R 105. Find all functions $f : \mathbb{R} \to \mathbb{R}$ such that

$$f(x + f(xy)) = f(x) + xf(y)$$

for all $x, y \in \mathbb{R}$.

Amer. Math. Monthly 2003 Problem 11053 (b)
Proposed by Wu Wei Chao

R 106. Find all functions $f : \mathbb{R} \to \mathbb{R}$ such that

$$f(f(x) + y) = 2x + f(f(y) - x)$$

for all $x, y \in \mathbb{R}$.

Czech-Polish-Slovak Match 2003

R 107. Determine all injective functions $f : \mathbb{R} \to \mathbb{R}$ such that

$$f\left(x^3 + x\right) \le f(x) \le [f(x)]^3 + f(x)$$

for all $x \in \mathbb{R}$.

Crux Mathematicorum 2003 Problem 2811
Proposed by Mihály Bencze

R 108. Determine all injective functions $f : \mathbb{R} \to \mathbb{R}$ such that

$$(2a+b)f(ax+b) \geq a\left[f\left(\frac{1}{x}\right)\right]^2 + bf\left(\frac{1}{x}\right) + a$$

for all $x \in \mathbb{R}$, where $a, b \in \mathbb{R}$, $a > 0$, $a^2 = 4b > 0$, $2a + b > 0$.

Crux Mathematicorum 2003 Problem 2812
Proposed by Mihály Bencze

R 109. Find all nondecreasing functions $f : \mathbb{R} \to \mathbb{R}$ satisfying the following conditions.

- $f(0) = 0$.
- $f(1) = 1$.
- $f(a) + f(b) = f(a)f(b) + f(a + b - ab)$ for all real numbers a, b such that $a < 1 < b$.

IMO Shortlist 2003 A2
Proposed by A. Di Pisquale and D. Matthews (Australia)

R 110. Find all functions $f : \mathbb{R}^+ \to \mathbb{R}^+$ satisfying the following conditions.

- $f(xyz) + f(x) + f(y) + f(z) = f(\sqrt{xy})f(\sqrt{yz})f(\sqrt{zx})$ for all $x, y, z \in \mathbb{R}^+$.
- $f(x) < f(y)$ for $1 \leq x < y$.

IMO Shortlist 2003 A5
Proposed by Hojoo Lee (Korea)

R 111. Find all functions $f : \mathbb{R}^+ \to \mathbb{R}^+$ such that

$$f(f(x) + y) = xf\left(1 + xy\right)$$

for all $x, y > 0$.

Középiskolai Matematikai és Fizikai Lapok, October 2003, A. 328

R 112. Find all continuous functions $f : \mathbb{R}^+ \to \mathbb{R}^+$ such that

$$f\left(a\right) + f\left(b\right) + f\left(c\right) = f\left(a\right) f\left(b\right) f\left(c\right)$$

for all $a, b, c > 0$ with $ab + bc + ca = 1$.

Mathematics Magazine 2003 Problem 1685
Proposed by Michel Bataille

$f_k(x) = \tan\left(\frac{k+2}{6}\pi - k\ \arctan x\right)$ where $k \in \left[-\frac{1}{2}, 1\right]$.
$f_0(x) = \sqrt{3}, \quad f_1(x) = \frac{1}{x}, \quad f_{-\frac{1}{2}}(x) = x + \sqrt{1 + x^2}$.

R 113. Let $\mathcal{F}$ be the set of all functions $f : (0, \infty) \to (0, \infty)$ such that

$$f(3x) \geq f(f(2x)) + x$$

for all $x > 0$. Find the largest A such that $f(x) \geq Ax$ for all $f \in \mathcal{F}$ and all $x > 0$.

Vietnamese Mathematical Olympiad 2003

R 114. Let f and g be nonconstant, continuous, periodic functions mapping $\mathbb{R}$ into $\mathbb{R}$. Is it possible that the function h on $\mathbb{R}$ given by $h(x) = f(xg(x))$ is periodic?

Amer. Math. Monthly 2004 Problem 11111
Proposed by Pál Péter Dalyay

R 115. Find all functions $f : \mathbb{R} \to \mathbb{R}$ such that

$$f(x^2 + y^2 + 2f(xy)) = (f(x + y))^2.$$

for all $x, y \in \mathbb{R}$.

IMO Shortlist 2004 A6

R 116. Let $f : \mathbb{R} \to \mathbb{R}$ be a monotone function. Suppose that there exist constants $c_1, c_2 \in \mathbb{R}$ such that

$$f(x) + f(y) - c_1 \le f(x + y) \le f(x) + f(y) + c_2$$

for all $x, y \in \mathbb{R}$. Prove that there exists a constant $k \in \mathbb{R}$ such that the function $f(x) - kx$ is bounded.

Középiskolai Matematikai és Fizikai Lapok, December 2004, A. 359

R 117. Prove that there exist functions $f : \mathbb{R} \to \mathbb{R}$ such that

$$f\,(\,x\; -\; f(y)\,) = f(x) + y$$

for all $x, y \in \mathbb{R}$, and show how such functions can be constructed.

Mathematics Magazine 2004 Problem 1690
Proposed by Costas Efthimiou, Peter Hilton

Hint. Construct an additive function f such that $f(f(t)) = -t$ for all $t \in \mathbb{R}$.

R 118. Let $f : \mathbb{R} \to \mathbb{R}$ be a function satisfying the following conditions.

- $|f(x) - f(y)| \le |x - y|$, for all $x, y \in \mathbb{R}$.
- For any $x \in \mathbb{R}$, the sequence $x, f(x), f(f(x)), \cdots$ is an arithmetic progression.

Show that there exists a constant $a \in \mathbb{R}$ such that $f(x) = x + a$ for all $x \in \mathbb{R}$.

Romanian Mathematical Olympiad 2004

R 119. Find all real values of α, for which there exists one and only one function $f : \mathbb{R} \to \mathbb{R}$ such that

$$f(x^2 + y + f(y)) = (f(x))^2 + \alpha\, y$$

for all $x, y \in \mathbb{R}$.

Vietnam Team Selection Test 2004

R 120. Find all monotonically increasing or monotonically decreasing functions $f : \mathbb{R}^+ \to \mathbb{R}^+$ satisfying

$$f\,(xy)\,f\left(\frac{f\,(y)}{x}\right) = 1$$

for all $x, y > 0$.

Germany Team Selection Test 2005

R 121. Find all functions $f : \mathbb{R}^+ \to \mathbb{R}^+$ such that

$$f(x)f(y) = 2f(x + yf(x))$$

for all $x, y > 0$.

IMO Shortlist 2005 A2, Germany Team Selection Test 2006
Proposed by Nikolai Nikolov (Bulgaria)

R 122. Find all functions $f : \mathbb{R} \to \mathbb{R}$ such that
$$f(x+y) + f(x)f(y) = f(xy) + 2xy + 1$$
for all $x, y \in \mathbb{R}$.

IMO Shortlist 2005 A4, Germany Team Selection Test 2006
Proposed by B. J. Venkatachala (India)

R 123. Find all functions $f : \mathbb{R} \to \mathbb{R}$ satisfying the following two conditions:
- $f(x) \geq 0$ for all $x \in \mathbb{R}$.
- $f(x+y) + f(x-y) - 2f(x) - 2y^2 = 0$ for all $x \in \mathbb{R}$.

Korean Mathematical Olympiad 2005

R 124. Find all continuous functions $f : \mathbb{R} \to \mathbb{R}$ such that
$$f\left(x + y + f(xy)\right) = xy + f(x+y)$$
for all $x, y \in \mathbb{R}$.

Középiskolai Matematikai és Fizikai Lapok, May 2005, A. 375

R 125. Find a function $f : [0,1] \to [0,1]$ such that for each nontrivial interval $\mathbf{I} \subset [0,1]$, we have
$$f\left(\mathbf{I}\right) = [0,1].$$
for all $x, y \in \mathbb{R}$.

Mathematics Magazine 2005 Problem 1795
Proposed by Jeff Groah

R 126. It is well known that if $f : [0,1] \to [0,1]$ is a continuous function, then f must have at least one fixed point. However, the example $f(x) = x^2$, which only has 0 and 1 as fixed points shows that the set of fixed points need not be an interval. Let $f : [0,1] \to [0,1]$ be a function with
$$|f(x) - f(y)| \leq |x - y|$$
for all $x, y \in [0,1]$. Prove that the set of all fixed points of f is either a single point or an interval.

Mathematics Magazine 2005 Problem Q 982
Proposed by Michael W. Botsko

R 127. Find all functions $f : \mathbb{R} \longrightarrow \mathbb{R}$ that satisfy
$$(x-y)f(x+y) - (x+y)f(x-y) = 4xy(x^2 - y^2)$$
for all $x, y \in \mathbb{R}$.

Polish Mathematical Olympiad 2005

R 128. Let $a \in \mathbb{R} - \{0\}$. Find all functions $f : \mathbb{R} \to \mathbb{R}$ such that $f(a+x) = f(x) - x$ for all $x \in \mathbb{R}$.

Romania Team Selection Test 2005
Proposed by Dan Schwartz

R 129. Find all functions $f : \mathbb{R} \to \mathbb{R}$ $f : \mathbb{R} \to \mathbb{R}$ satisfying the following conditions.

- $x(f(x+1) - f(x)) = f(x)$ for all $x \in \mathbb{R}$.
- $|f(x) - f(y)| \leq |x - y|$ for all $x, y \in \mathbb{R}$.

Romanian Mathematical Olympiad 2005

R 130. Find all functions $f : [0, \infty) \to [0, \infty)$ such that $4f(x) \geq 3x$ and $f(4f(x) - 3x) = x$ for all $x \geq 0$.

Turkey Team Selection Test 2005

R 131. Find all function $f : \mathbb{R} \to \mathbb{R}$ such that

$$f(f(x-y)) = f(x)f(y) - f(x) + f(y) - xy$$

for all $x, y \in \mathbb{R}$.

Vietnamese Mathematical Olympiad 2005

R 132. Let f be a continuous mapping from $\mathbb{R}$ into $\mathbb{R}$ that is bounded below. Show that there exists a real number x_0 such that

$$f(x) - f(x_0) < |x - x_0|$$

holds for all x other than x_0.

Amer. Math. Monthly 2006 Problem 11232
Proposed by Michael W. Botsko

R 133. Find all functions $f : \mathbb{R}^+ \to \mathbb{R}^+$ such that

$$f(x+y) - f(x-y) = 4\sqrt{f(x)f(y)}$$

for all $x, y \in \mathbb{R}$ with $x > y > 0$.

Bulgarian Mathematical Olympiad 2006
Proposed by Nikolai Nikolov, Oleg Mushkarov

R 134. Find all functions $f : \mathbb{R} \to \mathbb{R}$ such that

$$f(x)^2 + 2yf(x) + f(y) = f(y + f(x))$$

for all $x, y \in \mathbb{R}$.

Japanese Mathematical Olympiad 2006

R 135. Find all functions $f : \mathbb{R} \to \mathbb{R}$ such that

$$(f(x) + f(y))(f(z) + 1) = f(xz - y) + f(x + yz)$$

for all $x, y, z \in \mathbb{R}$.

Középiskolai Matematikai és Fizikai Lapok, January 2006, A. 390

R 136. Find all real functions $f : \mathbb{R} \to \mathbb{R}$ such that

$$f(f(x) + y) = f(f(x) - y) + 4f(x)y$$

for all $x, y \in \mathbb{R}$.

Balkan Mathematical Olympiad 2007

R 137. Determine all functions $f : \mathbb{R} - \{0\} \to \mathbb{R} - \{1\}$ satisfying the following conditions
- $f(xy) = f(x)f(-y) - f(x) + f(y)$ for all $x, y \in \mathbb{R} - \{0\}$.
- $f(f(x)) = \frac{1}{f(\frac{1}{x})}$ for all $x, y \in \mathbb{R} - \{0\}$.

Baltic Way 2007

R 138. Find all functions $f : \mathbb{R}^+ \to \mathbb{R}^+$ such that
$$f(x + f(y)) = f(x + y) + f(y)$$
for all $x, y > 0$.

IMO Shortlist 2007 A4
Proposed by Paisan Nakmahachalasint (Thailand)

R 139. Find all functions $f : (0, \infty) \to (0, \infty)$ such that
$$\frac{(f(w))^2 + (f(x))^2}{f(y^2) + f(z^2)} = \frac{w^2 + x^2}{y^2 + z^2}$$
for all $w, x, y, z > 0$ with $wx = yz$.

IMO Shortlist 2008 A1, IMO 2008 Problem 4
Proposed by Hojoo Lee (Korea)

R 140. Determine all functions $f : \mathbb{R}^+ \to \mathbb{R}^+$ such that
$$f\left(\frac{f(x)}{yf(x) + 1}\right) = \frac{x}{xf(y) + 1}$$
for all $x, y > 0$.

Germany Team Selection Test 2007

R 141. Find all functions $f : \mathbb{R}^+ \to \mathbb{R}$ satisfying the following conditions.
- $|f(x) - f(y)| \leq |x - y|$ for all $x, y > 0$.
- There exist constants $a, b, c \in \mathbb{R}$ such that $f(ax) + f\left(\frac{b}{x}\right) = f(cx)$ for all $x > 0$.

Középiskolai Matematikai és Fizikai Lapok, February 2007, A. 420
Proposed by Edgár Dobribán

R 142. Find all continuous functions $f : \mathbb{R} \to \mathbb{R}$ such that
$$f(x) = f\left(x^2 + \frac{x}{3} + \frac{1}{9}\right)$$
for all $x \in \mathbb{R}$.

Vietnam Team Selection Test 2007

R 143. Find all $f : \mathbb{R} \to \mathbb{R}$ such that
$$f(xy + f(x)) = xf(y) + f(x)$$
for all $x, y \in \mathbb{R}$.

Italy Team Selection Test 2007

R 144. Find all functions $f : \mathbb{R} \to \mathbb{R}$ satisfying the following conditions.

- $f(x) + f(y) \leq \frac{f(x+y)}{2}$ for all $x, y \in \mathbb{R}$.
- $\frac{f(x)}{x} + \frac{f(y)}{y} \geq \frac{f(x+y)}{x+y}$ for all $x, y \in \mathbb{R}$.

Japanese Mathematical Olympiad 2007

R 145. Find all nondecreasing functions $f : \mathbb{R} \to \mathbb{R}$ such that

$$f(x + f(y)) = f(f(x)) + f(y)$$

for all $x, y \in \mathbb{R}$.

Amer. Math. Monthly 2008 Problem 11345
Proposed by Roger Cuculière

$f(x) = 0$, $f(x) = x$, $f(x) = a\left\lfloor \frac{x+b}{a} \right\rfloor$, $f(x) = a\left\lceil \frac{x-b}{a} \right\rceil$ for some $b \in [0, a)$

R 146. Let $f : [0, 1] \to [0, \infty)$ be a continuous function such that $f(0) = f(1) = 0$ and $f(x) > 0$ for all $0 < x < 1$. Show that there exists a square with two vertices in the interval $(0, 1)$ on the x-axis and the other two vertices on the graph of f.

Amer. Math. Monthly 2008 Problem 11402
Proposed by Doru Catalin Barboianu

R 147. Let $f : \mathbb{R} \to \mathbb{N}$ be a function such that

$$f\left(x + \frac{1}{f(y)}\right) = f\left(y + \frac{1}{f(x)}\right)$$

for all $x, y \in \mathbb{R}$. Prove that there is a positive integer which is not a value of f.

IMO Shortlist 2008 A6
Proposed by Žymantas Darbėnas (Lithuania)

R 148. Determine all functions $f : \mathbb{R} \mapsto \mathbb{R}$ such that

$$f(x - f(y)) = f(x + y) + f(y)$$

for all $x, y \in \mathbb{R}$.

Germany Team Selection Test 2008

R 149. Find all functions $f : \mathbb{R} \to \mathbb{R}$ such that

$$f(xf(y)) + y + f(x) = f(x + f(y)) + yf(x)$$

for all $x, y \in \mathbb{R}$.

Iran Team Selection Test 2008

R 150. Find all functions $f : \mathbb{R} \to \mathbb{R}$ such that

$$f(x) + f(x + f(y)) = 2x + y$$

for all $x, y \in \mathbb{R}$.

Középiskolai Matematikai és Fizikai Lapok, January 2008, B. 4060

R 151. Find all functions $f : \mathbb{R} \to \mathbb{R}$ such that

$$xf(x + xy) = xf(x) + f\left(x^2\right) f(y)$$

for all $x, y \in \mathbb{R}$.

Middle European Mathematical Olympiad 2008

R 152. Let $f : \mathbb{R}^+ \to \mathbb{R}$ be a continuous function such that the sequences $\{f(nx)\}_{n\geq 1}$ are nondecreasing for all $x \in \mathbb{R}$. Prove that f is nondecreasing.

Romanian Mathematical Olympiad 2008

R 153. Let $f : \mathbb{R}^2 \to \mathbb{R}$ be a function such that

$$f(x, y) + f(y, z) + f(z, x) = 0$$

for all $x, y, z \in \mathbb{R}$. Prove that there exists a function $g : \mathbb{R} \to \mathbb{R}$ such that

$$f(x, y) = g(x) - g(y)$$

for all $x, y \in \mathbb{R}$.

William Lowell Putnam Competition 2008

R 154. Find all continuously differentiable functions $f : \mathbb{R} \to \mathbb{R}$ such that for every rational number q, the number $f(q)$ is rational and has the same denominator as q. (The denominator of a rational number q is the unique positive integer b such that $q = a/b$ for some integer a with $\gcd(a, b) = 1$.)

William Lowell Putnam Competition 2008

R 155. Let $f : \mathbb{R} \to \mathbb{R}$ be a continuous function such that

$$f(x + y) = f\left(f(x)f(y)\right)$$

for all $x, y \in \mathbb{R}$. Prove that the function f is constant.

Crux Mathematicorum 2009 Problem 3431
Proposed by Michel Bataille

R 156. Find all functions $f : \mathbb{R} \to \mathbb{R}$ such that

$$x^2y^2\left(f(x + y) - f(x) - f(y)\right) = 3(x + y)f(x)f(y)$$

for all $x, y \in \mathbb{R}$.

Crux Mathematicorum 2009 Problem 3477
Proposed by Michel Bataille

R 157. Find all functions $f : \mathbb{R}^+ \to \mathbb{R}^+$ such that

$$\Big(1 + yf(x)\Big)\Big(1 - yf(x + y)\Big) = 1$$

for all $x, y > 0$.

Czech-Polish-Slovak Match 2009

R 158. Find all functions $f : \mathbb{R} \to \mathbb{R}$ such that, for all $x, y, z \in \mathbb{R}$, if we have
$$x^3 + xf(y) + f(z) = 0,$$
then
$$f(x)^3 + yf(x) + z = 0.$$

Germany Team Selection Test 2009

R 159. Let $f : \mathbb{R} \to \mathbb{R}$ be a function. Show that there exist $x, y \in \mathbb{R}$ such that
$$f\left(x - f(y)\right) > yf(x) + x.$$

IMO Shortlist 2009 A5, Germany Team Selection Test 2010
Proposed by Igor Voronovich (Belarus)

R 160. Find all functions $f : \mathbb{R} \to \mathbb{R}$ such that
$$f\left(xf(x+y)\right) = f\left(yf(x)\right) + x^2$$
for all $x, y \in \mathbb{R}$.

IMO Shortlist 2009 A7, Germany Team Selection Test 2010
Proposed by Japan

R 161. Find all functions $f : [0, \infty) \to [0, \infty)$ such that
$$f\left(x^2\right) + f(y) = f\left(x^2 + y + xf(4y)\right)$$
for all $x, y \geq 0$.

Japanese Mathematical Olympiad 2009

R 162. Find all functions $f : \mathbb{R} \to \mathbb{R}$ such that
$$f(xf(y)) + f(f(x) + f(y)) = yf(x) + f(x + f(y))$$
for all $x, y \in \mathbb{R}$.

Middle European Mathematical Olympiad 2009

R 163. Find all functions $f : [0, \infty) \to [0, \infty)$ such that
$$f(x + y - z) + f\left(2\sqrt{xz}\right) - f\left(2\sqrt{yz}\right) = f(x + y + z)$$
for all $x, y, z \geq 0$ with $x + y \geq z$.

Moldova Team Selection Test 2009

R 164. Find all functions $f : \mathbb{R} \to \mathbb{R}$ such that
$$f\left(x + xy + f(y)\right) = \left(f(x) + \frac{1}{2}\right)\left(f(y) + \frac{1}{2}\right)$$
for all $x, y \in \mathbb{R}$.

Ukrainian Mathematical Olympiad 2009

R 165. Let f be a real-valued function on the plane such that for every square $ABCD$ in the plane, $f(A)+f(B)+f(C)+f(D)=0$. Does it follow that $f(P)=0$ for all points P in the plane?

William Lowell Putnam Competition 2009

R 166. Find all functions $f:\mathbb{R}\to\mathbb{R}$ such that

$$f(f(x)+f(y)+f(z))=f(f(x)-f(y))+f(2xy+f(z))+2f(xz-yz)$$

for all $x,y,z\in\mathbb{R}$.

Asian Pacific Mathematics Olympiad 2010

R 167. Prove that there is no function $f:\mathbb{R}^3\to\mathbb{R}^2$ with the property that

$$|\,f(x)-f(y)|\geq|\,x-y\,|$$

for all $x,y\in\mathbb{R}^3$.

Amer. Math. Monthly 2010 Problem 11526
Proposed by Marius Cavachi

R 168. Find all functions $f:\mathbb{R}\to\mathbb{R}$ such that

$$f\left(x^2\right)+f(xy)=f(x)f(y)+yf(x)+xf(x+y)$$

for all $x,y\in\mathbb{R}$.

Baltic Way 2010

R 169. Find all functions $f:\mathbb{R}\to\mathbb{R}$ such that

- $f(x)>1$ for all $x>0$.
- $f(x)f(y)=(x+y+1)^2f\left(\frac{xy-1}{x+y+1}\right)$ for all $x,y\in\mathbb{R}$ with $x+y+1\neq 0$.

Germany Team Selection Test 2010

R 170. Find all functions $f:\mathbb{R}^+\to\mathbb{R}^+$ satisfying the following conditions.

- $f\left(\frac{f(x)}{f(y)}\right)=\frac{f(f(x))}{y}$ for all $x,y>0$.
- the function f is strictly monotone.

Greece Team Selection Test 2010

R 171. Find all function $f:\mathbb{R}\to\mathbb{R}$ such that

$$f(\lfloor x\rfloor\,y)=f(x)\,\lfloor f(y)\rfloor$$

for all $x,y\in\mathbb{R}$. Here, $\lfloor a\rfloor$ denotes the greatest integer not greater than a.

IMO Shortlist 2010 A1, IMO 2010 Problem 1
Proposed by Pierre Bornsztein (France)

R 172. Find all non-decreasing functions $f : [0, \infty) \to [0, \infty)$ such that

$$f\left(\frac{x + f(x)}{2} + y\right) = 2x - f(x) + f(f(y))$$

for all $x, y \geq 0$.

Iran Team Selection Test 2010

R 173. The function $f : [0, \infty) \longrightarrow [0, \infty)$ satisfies the following properties for all $a, b \in [0, \infty)$:

- $f(a) = 0$ if and only if $a = 0$.
- $f(ab) = f(a)f(b)$ for all $a, b \geq 0$.
- $f(a + b) \leq 2 \max\{f(a), f(b)\}$ for all $a, b \geq 0$.

Prove that $f(a + b) \leq f(a) + f(b)$ for all $a, b \geq 0$.

Iran Team Selection Test 2010
Proposed by Masoud Shafaei

R 174. Is there a function defined on the points of the plane, which is not identically zero but the sum of its values at the vertices of every regular pentagon in the plane is zero?

Középiskolai Matematikai és Fizikai Lapok, March 2010, B. 4254

R 175. Determine all continuous functions $f : \mathbb{R} \to \mathbb{R}$ such that

$$f(f(f(x))) - 3f(x) + 2x = 0$$

for all $x \in \mathbb{R}$.

Amer. Math. Monthly 2011 Problem 11549
Proposed by Marian Tetiva

R 176. Determine all functions $f : \mathbb{R} \to \mathbb{R}$ satisfying the following conditions.

- There exists a real number M such that $xf(x) < M$ for all $x \in \mathbb{R}$.
- $f(xf(y)) + yf(x) = xf(y) + f(xy)$ for all $x, y \in \mathbb{R}$.

Asian Pacific Mathematics Olympiad 2011

R 177. Let $f : \mathbb{R} \to \mathbb{R}$ be a function such that

$$f(f(x)) = x^2 - x + 1$$

for all $x \in \mathbb{R}$. Determine $f(0)$.

Baltic Way 2011

R 178. Let $n \geq 2$ be an integer. Find all functions $f : \mathbb{R} \to \mathbb{R}$ such that

$$f(x - f(y)) = f(x + y^n) + f(f(y) + y^n)$$

for all $x, y \in \mathbb{R}$.

China Team Selection Test 2011

R 179. Let $a > 0$ be a constant. Find all functions $f : \mathbb{R}^+ \to \mathbb{R}^+$ such that
$$(x + a)f(x + y) = af(yf(x))$$
for all $x, y > 0$.

Crux Mathematicorum 2011 Problem 3695
Proposed by Michel Bataille

R 180. Prove that there exist two functions $f, g : \mathbb{R} \to \mathbb{R}$, such that $f \circ g$ is strictly decreasing and $g \circ f$ is strictly increasing.

Romanian Masters In Mathematics 2011
Proposed by Andrzej Komisarski and Marcin Kuczma

R 181. Let $f : \mathbb{R} \to \mathbb{R}$ be a real-valued function defined on the set of real numbers that satisfies
$$f(x + y) \leq yf(x) + f(f(x))$$
for all $x, y \in \mathbb{R}$. Prove that $f(x) = 0$ for all $x \leq 0$.

IMO 2011 Problem 3
Proposed by Igor Voronovich (Belarus)

R 182. Determine all functions $f, g : \mathbb{R} \to \mathbb{R}$ such that
$$g(f(x + y)) = f(x) + (2x + y)g(y)$$
for all $x, y \in \mathbb{R}$.

IMO Shortlist 2011 A3
Proposed by Japan

R 183. Let $f : \mathbb{R} \to \mathbb{R}$ be a function such that
$$f(x + y) \leq yf(x) + f(f(x))$$
for all $x, y \in \mathbb{R}$. Prove that $f(x) = 0$ for all $x \leq 0$.

IMO Shortlist 2011 A5
Proposed by Igor Voronovich (Belarus)

R 184. Find all functions $f : \mathbb{R} \to \mathbb{R}$ such that
$$f(f(x) - f(y)) = f(f(x)) - 2x^2f(y) + f\left(y^2\right)$$
for all $x, y \in \mathbb{R}$.

Japanese Mathematical Olympiad 2011, Indian National Mathematical Olympiad Program 2016

R 185. Find all functions $f : \mathbb{R} \to \mathbb{R}$ such that
$$y^2f(x) + x^2f(y) + xy = xyf(x + y) + x^2 + y^2$$
for all $x, y \in \mathbb{R}$.

Middle European Mathematical Olympiad 2011

R 186. Find all functions $f : \mathbb{R} \to \mathbb{R}$ such that
$$f(f(x) + y) = f(x^2 - y) + 4yf(x)$$
for all $x, y \in \mathbb{R}$.

Nordic Mathematical Contest 2011

R 187. Determine all pairs of functions $f, g : \mathbb{R} \to \mathbb{R}$ such that

$$f(x)f(y) = g(x)g(y) + g(x) + g(y)$$

for all $x, y \in \mathbb{R}$.

Polish Mathematical Olympiad 2011

R 188. Determine all functions $f : \mathbb{R} \to \mathbb{R}$ such that

$$2f(x) = f(x+y) + f(x+2y)$$

for all $x, y \in \mathbb{R}$ with $y \geq 0$.

Romania Team Selection Test 2011

R 189. Find all functions $f : \mathbb{R} \to \mathbb{R}^+$ such that

$$(1-z)f(x) = f\left(\frac{1-z}{z} f(xz)\right)$$

for all $x > 0$ and $z \in (0,1)$.

Amer. Math. Monthly 2012 Problem 11661
Proposed by Giedrius Alkauskas

R 190. Find all functions $f : \mathbb{R} \to \mathbb{R}$ such that

$$f(x+y) = f(x-y) + f(f(1-xy))$$

for all $x, y \in \mathbb{R}$.

Baltic Way 2012

R 191. Find all functions $f : \mathbb{R} \to \mathbb{R}$ such that

$$f(x + f(y)) - f(x) = (x + f(y))^4 - x^4$$

for all $x, y \in \mathbb{R}$.

Czech-Polish-Slovak Match 2012

R 192. Find all functions $f : \mathbb{R} \to \mathbb{R}$ such that

$$f(yf(x+y) + f(x)) = 4x + 2yf(x+y)$$

for all $x, y \in \mathbb{R}$.

European Girls' Mathematical Olympiad 2012
Proposed by Netherlands (Birgit van Dalen)

R 193. Find all functions $f : \mathbb{R} \to \mathbb{R}$ satisfying the following conditions.

- $f(-1) \neq 0$.
- $f(1+xy) - f(x+y) = f(x)f(y)$ for all $x, y \in \mathbb{R}$.

IMO Shortlist 2012 A5

R 194. Find all functions $f:\mathbb{R}\to\mathbb{R}$ such that

$$f\left(f(x+y)f(x-y)\right)=x^2-yf(y)$$

for all $x,y\in\mathbb{R}$.

Japanese Mathematical Olympiad 2012

R 195. Let $f:\mathbb{R}^+\to\mathbb{R}$ be a function such that

$$f\left(\sqrt{xy}\right)=f\left(\frac{x+y}{2}\right)$$

for all $x,y>0$. Prove that f is a constant function.

Középiskolai Matematikai és Fizikai Lapok, May 2012, B. 4456
Z. Daróczy

R 196. Find all functions $f:\mathbb{R}^+\to\mathbb{R}^+$ such that

$$f\left(x+f(y)\right)=yf(xy+1)$$

for all $x,y>0$.

Middle European Mathematical Olympiad 2012

R 197. Find all functions $f:\mathbb{R}\to\mathbb{R}$ with the following property that, for any open bounded interval $\mathbf{I}$, the set $f(\mathbf{I})$ is an open interval having the same length with $\mathbf{I}$.

Romanian Mathematical Olympiad 2012

R 198. Find all functions $f:\mathbb{R}\to\mathbb{R}$ such that

$$(x-2)f(y)+f\left(y+2f(x)\right)=f\left(x+yf(x)\right)$$

for all $x,y\in\mathbb{R}$.

Spanish Mathematical Olympiad 2012

R 199. Determine all functions $f:\mathbb{R}\to\mathbb{R}$ such that

$$f\left(x+y^2\right)=f(x)+|\,yf(y)\,|$$

for all $x,y\in\mathbb{R}$.

USA Team Selection Test 2012

R 200. Find all functions $f:\mathbb{R}\to\mathbb{R}$ satisfying the following conditions.

- For each $a\in\mathbb{R}$, there exist $b\in\mathbb{R}$ with $f(b)=a$.
- $f(x)>f(y)$ for $x>y$.
- $f(f(x))=f(x)+12x$ for all $x\in\mathbb{R}$.

Vietnamese Mathematical Olympiad 2012

R 201. Let a and b be real, with $1<a<b$, and let m and n be real with $m\neq 0$. Find all continuous functions $f:[0,1)\to[0,\infty)$ such that

$$f\left(a^x\right)+f\left(b^x\right)=mx+n$$

for all $x\geq 0$.

Amer. Math. Monthly 2013 Problem 11732
Proposed by Marcel Chirita

R 202. Find all functions $f : \mathbb{R} \to \mathbb{R}$ such that

$$f(xf(y) + y) + f(-f(x)) = f(yf(x) - y) + y$$

for all $x, y \in \mathbb{R}$.

Baltic Way 2013

R 203. Find all functions $f : \mathbb{R}^+ \times \mathbb{R}^+ \times \mathbb{R}^+ \to \mathbb{R}^+$ satisfying the following conditions.

- $xf(x, y, z) = zf(z, y, x)$ for all $x, y, z > 0$.
- $f\left(x, ky, k^2 z\right) = kf(x, y, z)$ for all $k, x, y, z > 0$.
- $f(1, k, k + 1) = k + 1$ for all $k > 0$.

Balkan Mathematical Olympiad 2013

R 204. Find all functions $f : \mathbb{R} \to \mathbb{R}$ such that

$$f(x + y) + y \le f(f(f(x)))$$

for all $x, y \in \mathbb{R}$.

Benelux Mathematical Olympiad 2013

R 205. Find all functions $f : \mathbb{R} \to \mathbb{R}$ satisfying the following conditions.

- $f(0) \in \mathbb{Q}$.
- $f\left(x + f(y)^2\right) = f(x + y)^2$ for all $x, y \in \mathbb{R}$.

Iranian Mathematical Olympiad 2013

R 206. Find all functions $f : \mathbb{R} \to \mathbb{R}$ such that

$$f\left(xf(x) + 2y\right) = f\left(x^2\right) + f(y) + x + y - 1$$

for all $x, y \in \mathbb{R}$.

Middle European Mathematical Olympiad 2013

R 207. Let $f : \mathbb{R} \to \mathbb{R}$ be a continuous and strictly increasing function satisfying

$$f^{-1}\left(\frac{f(x) + f(y)}{2}\right)(f(x) + f(y)) = (x + y)f\left(\frac{x + y}{2}\right)$$

for all $x, y \in \mathbb{R}$. Here, the function f^{-1} denotes the inverse of the function f. Prove that there exist real constants $a \neq 0$ and b such that $f(x) = ax + b$ for all $x \in \mathbb{R}$.

Miklós Schweitzer Competition 2013
Proposed by Zoltán Daróczy

R 208.

(1) Prove that there exists a function $f : \mathbb{R}^+ \to \mathbb{R}^+$ which is nowhere continuous and
$$f\left(\left(\frac{x^\alpha + y^\alpha}{2}\right)^{\frac{1}{\alpha}}\right) \le \left(\frac{f(x)^\alpha + f(y)^\alpha}{2}\right)^{\frac{1}{\alpha}}$$
for all $x, y > 0$ and $\alpha \in \mathbb{Q}$.

(2) Is there such a function if instead the above relation holds for all $x, y > 0$ and for all irrational number α?

Miklós Schweitzer Competition 2013
Proposed by Maksa Gyula, Zsolt Páles

R 209. Determine all functions $f : \mathbf{R} \to \mathbf{R}^+$ such that for all real numbers x, y the following conditions hold:

- $f\left(x^2\right) = f(x)^2 - 2xf(x)$.
- $f(-x) = f(x-1)$.
- $f(x) < f(y)$ for $1 < x < y$.

Turkey Team Selection Test 2013

R 210. Find all functions $f : \mathbb{R} \to \mathbb{R}$ satisfying

- $f(0) = 0$.
- $f(1) = 2013$.
- $(x-y)\left(f\left(f(x)^2\right) - f\left(f(y)^2\right)\right) = (f(x) - f(y))\left(f(x)^2 - f(y)^2\right)$.

Vietnamese Mathematical Olympiad 2013

R 211. Find all functions $f : \mathbb{R} \to \mathbb{R}$ such that
$$f(f(y)) + f(x-y) = f(xf(y) - x)$$
for all $x, y \in \mathbb{R}$.

Baltic Way 2014

R 212. Determine all functions $f : \mathbb{R} \to \mathbb{R}$ satisfying the condition
$$f(y^2 + 2xf(y) + f(x)^2) = (y + f(x))(x + f(y))$$
for all $x, y \in \mathbb{R}$.

European Girls' Mathematical Olympiad 2014

R 213. Find all functions $f : \mathbb{R}^+ \to \mathbb{R}^+$ such that
$$f\left(\frac{y}{f(x+1)}\right) + f\left(\frac{x+1}{xf(y)}\right) = f(y)$$
for all $x, y \in \mathbb{R}^+$.

Iran Team Selection Test 2014

R 214. Determine all functions $f : \mathbb{R} \to \mathbb{R}$ such that
$$f\left(xf(x) + f(x)f(y) + y - 1\right) = f\left(xf(x) + xy\right) + y - 1$$
for all $x, y \in \mathbb{R}$.

Korean Mathematical Olympiad 2014

R 215. Find all functions $f : \mathbb{R} \to \mathbb{R}$ such that
$$xf(y) + f(xf(y)) - xf(f(y)) - f(xy) = 2x + f(y) - f(x + y)$$
for all $x, y \in \mathbb{R}$.

Middle European Mathematical Olympiad 2014

R 216. Find all functions $f : \mathbb{R} \to \mathbb{R}$ such that
$$f(xf(y) + y) + f(xy + x) = f(x + y) + 2xy$$
for all $x, y \in \mathbb{R}$.

Moldova Team Selection Test 2014

R 217. Find all functions $f : \mathbb{R} \to \mathbb{R}$ such that
$$f\left(xf(y) - yf(x)\right) = f(xy) - xy$$
for all $x, y \in \mathbb{R}$.

Serbian Mathematical Olympiad 2014
Proposed by Dusan Djukic

R 218. Find all functions $f : \mathbb{R} \to \mathbb{R}$ such that
$$f(f(y) + x^2 + 1) + 2x = y + (f(x + 1))^2$$
for all $x, y \in \mathbb{R}$.

Turkey Team Selection Test 2014

R 219. Find all functions $f : \mathbb{R} \to \mathbb{R}$ such that
$$f\left(x^3\right) + f\left(y^3\right) = (x + y)\ \left(f\left(x^2\right) + f\left(y^2\right) - f(xy)\right)$$
for all $x, y \in \mathbb{R}$.

Uzbekistan Mathematical Olympiad 2014

R 220. Let $f : [0, 1] \to \mathbb{R}$ be a function for which there exists a constant $K > 0$ such that
$$|f(x) - f(y)| \le K\,|x - y|$$
for all $x, y \in [0, 1]$. Suppose also that for each rational number $r \in [0, 1]$, there exist integers a and b such that $f(r) = a + br$. Prove that there exist finitely many intervals $I_1, \cdots, I_n$ such that f is a linear function on each I_i and $[0, 1] = \bigcup_{i=1}^{n} I_i$.

William Lowell Putnam Competition 2014

R 221. Find all functions $f : \mathbb{R} \to \mathbb{R}$ such that
$$|x|\,f(y) + yf(x) = f(xy) + f\left(x^2\right) + f(f(y))$$
for all $x, y \in \mathbb{R}$.

Baltic Way 2015

R 222. Find all functions $f : \mathbb{R} \to \mathbb{R}$ such that

$$yf(x) + f(y) \geq f(xy)$$

for all $x, y \in \mathbb{R}$.

Greece Team Selection Test 2015

R 223. Determine all functions $f : \mathbb{R} \to \mathbb{R}$ that satisfy the equation

$$f(x + f(x + y)) + f(xy) = x + f(x + y) + yf(x)$$

for all $x, y \in \mathbb{R}$.

IMO Shortlist 2015 A4, IMO 2015/5
Proposed by Dorlir Ahmeti (Albania)

R 224. Find all functions $f : \mathbb{R} \to \mathbb{R}$ such that

$$f(x + y) + f(x)f(y) = x^2y^2 + 2xy$$

for all $x, y \in \mathbb{R}$.

Középiskolai Matematikai és Fizikai Lapok, September 2015, B. 4727

R 225. Find all surjective functions $f : \mathbb{N} \to \mathbb{N}$ such that, for all $a, b \in \mathbb{N}$, exactly one of the following two equations is true:

- $f(a) = f(b)$,
- $f(a + b) = \min\{f(a), f(b)\}$.

Middle European Mathematical Olympiad 2015

R 226. Let $f : \mathbb{R}^+ \to \mathbb{R}$ be a continuous function such that

$$(f(x) - f(y))\left(f\left(\frac{x + y}{2}\right) - f\left(\sqrt{xy}\right)\right) = 0$$

for all $x, y > 0$. Show that f is a constant function.

Miklós Schweitzer Competition 2015

R 227. Find all functions $f : \mathbb{R} \to \mathbb{R}$ such that

$$f\left(x^2\right) + 4y^2f(y) = \left(f(x - y) + y^2\right)\left(f(x + y) + f(y)\right)$$

for all $x, y \in \mathbb{R}$.

Turkey Team Selection Test 2015

R 228. Find all functions $f : \mathbb{R}^+ \to \mathbb{R}^+$ such that

$$(z + 1)f(x + y) = f\left(xf(z) + y\right) + f\left(yf(z) + x\right),$$

for all $x, y, z > 0$.

Asian Pacific Mathematics Olympiad 2016
Proposed by Fajar Yuliawan (Indonesia)

R 229. Find all injective functions $f : \mathbb{R} \to \mathbb{R}$ such that
$$\left| \sum_{i=1}^{n} i\left(f(x+i+1) - f(f(x+i))\right) \right| < 2016$$
for all $x \in \mathbb{R}$ and $n \in \mathbb{N}$.

Balkan Mathematical Olympiad 2016

R 230. Find all functions $f : \mathbb{R} \to \mathbb{R}$ such that
$$f(f(y) - x)^2 + f(x)^2 + f(y)^2 = f(y)\left(1 + 2f(f(y))\right)$$
for all $x, y \in \mathbb{R}$.

Benelux Mathematical Olympiad 2016

R 231. Find all functions $f : \mathbb{R}^+ \to \mathbb{R}^+$ satisfying that, for any three distinct numbers $a, b, c > 0$, a triangle can be formed with side lengths a,b,c, if and only if a triangle can be formed with side lengths $f(a)$,$f(b)$,$f(c)$.

China Team Selection Test 2016

R 232. Find all functions $f : \mathbb{R} \to \mathbb{R}$ such that
$$f(xf(y) - yf(x)) = f(xy) - xy.$$
for all $x, y \in \mathbb{R}$.

Indian National Mathematical Olympiad Program 2016

R 233. Find all functions $f : (0, \infty) \to (0, \infty)$ such that
$$xf(x^2)f(f(y)) + f(yf(x)) = f(xy)\left(f(f(x^2)) + f(f(y^2))\right)$$
for all $x, y > 0$.

IMO Shortlist 2016 A4

R 234. Find all functions $f : \mathbb{R} \to \mathbb{R}$ satisfying the following conditions.

- $f(0) \neq 0$.
- $f(x+y)^2 = 2f(x)f(y) + \max\left\{f(x^2+y^2), f(x^2) + f(y^2)\right\}$ for all $x, y \in \mathbb{R}$.

IMO Shortlist 2016 A7

R 235. Find all functions $f : \mathbb{R} \to \mathbb{R}$ such that
$$f\left(yf(x) - x\right) = f(x)f(y) + 2x$$
for all $x, y \in \mathbb{R}$.

Japanese Mathematical Olympiad 2016

R 236. Find all $a \in \mathbb{R}$ for which there exists a function $f : \mathbb{R} \to \mathbb{R}$ satisfying the following conditions.

- $f(f(x)) = f(x) + x$ for all $x \in \mathbb{R}$.
- $f(f(x) - x) = f(x) + ax$ for all $x \in \mathbb{R}$

Nordic Mathematical Contest 2016

R 237. Find all functions $f:\mathbb{R}\to\mathbb{R}$ such that

$$(f(x)+xy)\,f(x-3y)+(f(y)+xy)\,f(3x-y)=f(x+y)^2.$$

for all $x,y\in\mathbb{R}$.

United States of America Mathematical Olympiad 2016

R 238. Find all $a\in\mathbb{R}$ such that there is function $f:\mathbb{R}\to\mathbb{R}$ satisfying the following conditions.

- $f(1)=2016$.
- $f(x+y+f(y))=f(x)+ay$ for all $x,y\in\mathbb{R}$.

Vietnamese Mathematical Olympiad 2016

R 239. Find all functions $f:(1,\infty)\to(1,\infty)$ such that

$$[\,f(x)\,]^2\le f(y)\le[\,f(x)\,]^3.$$

for all $x,y\in(1,\infty)$ with $x^2\le y\le x^3$.

William Lowell Putnam Competition 2016

R 240. Determine all functions $f:\mathbb{R}\to\mathbb{R}$ such that

$$f(f(x)f(y))+f(x+y)=f(xy).$$

for all $x,y\in\mathbb{R}$.

IMO 2017 Problem 2

R 241. Find all functions $f:\mathbb{R}^+\times\mathbb{R}^+\to\mathbb{R}^+$ satisfying the following conditions.

- $f\left(f(x,y),z\right)=x^2y^2f(x,z)$ for all $x,y,z>0$.
- $f\left(x,1+f(x,y)\right)\ge x^2+xyf(x,x)$ for all $x,y>0$.

Iran Team Selection Test 2017
Proposed by Mojtaba Zare and Ali Daei Nabi

R 242. Find all functions $f:\mathbb{R}^+\to\mathbb{R}^+$ such that

$$\frac{x+f(y)}{xf(y)}=f\left(\frac{1}{y}+f\left(\frac{1}{x}\right)\right)$$

for all $x,y>0$.

Iranian Mathematical Olympiad 2017

R 243. Let $f:[0,1]\to\mathbb{R}^+$ be a bounded function. Prove that there exist $x_1,x_2\in[0,1]$ such that

$$(x_2-x_1)\,\frac{\left[f\left(x_1\right)\right]^2}{f\left(x_2\right)}>\frac{f(0)}{4}.$$

Középiskolai Matematikai és Fizikai Lapok, January 2017, B. 4847
O. Reutter

R 244. Find all continuous functions $f : \mathbb{R} \to \mathbb{R}$ such that

$$f(xy) = f\left(\frac{x^2+y^2}{2}\right) + (x-y)^2$$

for all $x, y \in \mathbb{R}$.

Moldova Team Selection Test 2017

R 245. Find all functions $f : \mathbb{R} \to \mathbb{R}$ such that

$$f(xf(y) - f(x)) = 2f(x) + xy$$

for all $x, y \in \mathbb{R}$.

Vietnamese Mathematical Olympiad 2017

(after Ramanujan completes a mathematical equation on the board in his class)

Fellow: **I hadn't completed that proof. How do you know?**
Srinivasa Ramanujan: **I don't know. I just do.**

The Man Who Knew Infinity (2016)

8. Shortlisted problems for the IMOs (2001 ∼ 2016)

Each problem that I solved became a rule, which served afterwards to solve other problems.

– Rene Descartes

IMO 2001 United States of America.

A 1. Let $\mathcal{T} = \mathbb{N}_0 \times \mathbb{N}_0 \times \mathbb{N}_0$ denote the set of all ordered triples (p, q, r) of nonnegative integers. Find all functions $f : \mathcal{T} \to \mathbb{R}$ such that

$$f(p,q,r) = \begin{cases} 0 & \text{if } pqr = 0, \\ 1 + \frac{1}{6}(f(p+1,q-1,r) + f(p-1,q+1,r) & \\ +f(p-1,q,r+1) + f(p+1,q,r-1) & \text{otherwise,} \\ +f(p,q+1,r-1) + f(p,q-1,r+1)) & \end{cases}$$

for all nonnegative integers p, q, r.

IMO Shortlist 2001 A1

A 2. Let $a_0, a_1, a_2, \cdots$ be an arbitrary infinite sequence of positive numbers. Show that the inequality

$$1 + a_n > 2^{\frac{1}{n}} a_{n-1}$$

holds for infinitely many positive integers n.

IMO Shortlist 2001 A2

A 3. Let $x_1, x_2, \cdots, x_n$ be arbitrary real numbers. Prove the inequality

$$\frac{x_1}{1+x_1^2} + \frac{x_2}{1+x_1^2+x_2^2} + \cdots + \frac{x_n}{1+x_1^2+\cdots+x_n^2} < \sqrt{n}.$$

IMO Shortlist 2001 A3

A 4. Find all functions $f : \mathbb{R} \to \mathbb{R}$ such that

$$f(xy)(f(x) - f(y)) = (x-y)f(x)f(y)$$

for all $x, y \in \mathbb{R}$.

IMO Shortlist 2001 A4

A 5. Find all positive integers $a_0 = 1, a_1, a_2, \cdots, a_n$ satisfying the following conditions.

- $(a_{k+1} - 1)a_{k-1} \geq {a_k}^2(a_k - 1)$ for all $k \in \{1, 2, \cdots, n-1\}$.
- $\frac{99}{100} = \frac{a_0}{a_1} + \frac{a_1}{a_2} + \cdots + \frac{a_{n-1}}{a_n}$.

IMO Shortlist 2001 A5

A 6. Prove that, for all $a, b, c > 0$,

$$\frac{a}{\sqrt{a^2+8bc}} + \frac{b}{\sqrt{b^2+8ca}} + \frac{c}{\sqrt{c^2+8ab}} \geq 1.$$

IMO Shortlist 2001 A6, IMO 2001 Problem 2
Proposed by Hojoo Lee (Korea)

IMO 2002 United Kingdom.

A 7. Find all functions $f : \mathbb{R} \to \mathbb{R}$ such that

$$f\left(f(x) + y\right) = 2x + f\left(f(y) - x\right)$$

for all $x, y \in \mathbb{R}$.

IMO Shortlist 2002 A1, Italy Team Selection Test 2003, Germany Team Selection Test 2003

A 8. Let $a_1, a_2, \cdots$ be an infinite sequence of real numbers, for which there exists a real number c with $0 \le a_i \le c$ for all i, such that

$$|a_i - a_j| \ge \frac{1}{i+j} \quad \text{for all } i,\ j \text{ with } i \neq j.$$

Prove that $c \ge 1$.

IMO Shortlist 2002 A2

A 9. Let P be a cubic polynomial given by $P(x) = ax^3 + bx^2 + cx + d$, where a, b, c, d are integers and $a \neq 0$. Suppose that $xP(x) = yP(y)$ for infinitely many pairs x, y of integers with $x \neq y$. Prove that the equation $P(x) = 0$ has an integer root.

IMO Shortlist 2002 A3

A 10. Find all functions $f : \mathbb{R} \to \mathbb{R}$ such that

$$\left(f(x) + f(z)\right)\left(f(y) + f(t)\right) = f(xy - zt) + f(xt + yz)$$

for all $x, y, z, t \in \mathbb{R}$.

IMO Shortlist 2002 A4, IMO 2002 Problem 5

A 11. Let n be a positive integer that is not a perfect cube. Define real numbers a, b, c by

$$a = \sqrt[3]{n}, \qquad b = \frac{1}{a - [a]}, \qquad c = \frac{1}{b - [b]},$$

where $[x]$ denotes the integer part of x. Prove that there are infinitely many such integers n with the property that there exist integers r, s, t, not all zero, such that $ra + sb + tc = 0$.

IMO Shortlist 2002 A5

A 12. Let A be a non-empty set of positive integers. Suppose that there are positive integers $b_1, \cdots b_n$ and $c_1, \cdots, c_n$ such that

- for each i the set $b_iA + c_i = \{b_ia + c_i \colon a \in A\}$ is a subset of A, and
- the sets $b_iA + c_i$ and $b_jA + c_j$ are disjoint whenever $i \neq j$

Prove that

$$\frac{1}{b_1} + \cdots + \frac{1}{b_n} \le 1.$$

IMO Shortlist 2002 A6

IMO 2003 Japan.

A 13. Let a_{ij} (with the indices i and j from the set $\{1,\ 2,\ 3\}$) be real numbers such that

- $a_{ij} > 0$ for $i = j$;
- $a_{ij} < 0$ for $i \neq j$.

Prove the existence of positive real numbers c_1, c_2, c_3 such that the numbers

$$a_{11}c_1 + a_{12}c_2 + a_{13}c_3, \quad a_{21}c_1 + a_{22}c_2 + a_{23}c_3, \quad a_{31}c_1 + a_{32}c_2 + a_{33}c_3$$

are either all negative, or all zero, or all positive.

IMO Shortlist 2003 A1
Proposed by Kiran Kedlaya (USA)

A 14. Find all nondecreasing functions $f : \mathbb{R} \to \mathbb{R}$ satisfying the following conditions.

- $f(0) = 0$, $f(1) = 1$.
- $f(a) + f(b) = f(a)f(b) + f(a + b - ab)$ for all real numbers a, b such that $a < 1 < b$.

IMO Shortlist 2003 A2
Proposed by A. Di Pisquale, D. Matthews (Australia)

A 15. Consider two monotonically decreasing sequences (a_k) and (b_k), where $k \geq 1$, and a_k and b_k are positive real numbersfor all $k \in \mathbb{N}$. Now, define the sequences

$$c_k = \min\left(a_k, b_k\right), \quad A_k = a_1 + a_2 + \cdots + a_k, \quad B_k = b_1 + b_2 + \cdots + b_k, \quad C_k = c_1 + c_2 + \cdots + c_k, k \in \mathbb{N}.$$

(1) Do there exist two monotonically decreasing sequences (a_k) and (b_k) of positive real numbers such that the sequences (A_k) and (B_k) are not bounded, while the sequence (C_k) is bounded?

(2) Does the answer to problem (1) change if we stipulate that the sequence (b_k) must be $b_k = \dfrac{1}{k}$ for all $k \in \mathbb{N}$?

IMO Shortlist 2003 A3

A 16. Let n be a positive integer and let $x_1 \leq x_2 \leq \cdots \leq x_n$ be real numbers. Prove that

$$\left(\sum_{i,j=1}^{n} |x_i - x_j|\right)^2 \leq \frac{2(n^2-1)}{3} \sum_{i,j=1}^{n} (x_i - x_j)^2.$$

Show that the equality holds if and only if $x_1, \cdots, x_n$ is an arithmetic sequence.

IMO Shortlist 2003 A4, IMO 2003 Problem 5

A 17. Find all functions $f : \mathbb{R}^+ \to \mathbb{R}^+$ satisfying the following conditions.

- $f(xyz) + f(x) + f(y) + f(z) = f(\sqrt{xy})f(\sqrt{yz})f(\sqrt{zx})$ for all $x, y, z \in \mathbb{R}^+$.
- $f(x) < f(y)$ for $1 \leq x < y$.

IMO Shortlist 2003 A5
Proposed by Hojoo Lee (Korea)

A 18. Let $n \in \mathbb{N}$ and let $x_1, \cdots, x_n, y_1, \cdots, y_n > 0$. Suppose $(z_2, \cdots, z_{2n})$ is a sequence of positive real numbers such that ${z_{i+j}}^2 \geq x_i y_j$ for all $1 \leq i, j \leq n$. Let $M = \max\{z_2, \cdots, z_{2n}\}$. Prove that

$$\left(\frac{M + z_2 + \cdots + z_{2n}}{2n}\right)^2 \geq \left(\frac{x_1 + \cdots + x_n}{n}\right)\left(\frac{y_1 + \cdots + y_n}{n}\right).$$

IMO Shortlist 2003 A6
Proposed by Reid Barton (USA)

IMO 2004 Greece.

A 19. Let $n \geq 3$ be an integer. Let t_1, t_2, ..., t_n be positive real numbers such that
$$n^2+1 > (t_1+t_2+\cdots+t_n)\left(\frac{1}{t_1}+\frac{1}{t_2}+\cdots+\frac{1}{t_n}\right).$$
Show that t_i, t_j, t_k are side lengths of a triangle for all i, j, k with $1 \leq i < j < k \leq n$.

IMO Shortlist 2004 A1, IMO 2004 Problem 4
Proposed by Hojoo Lee (Korea)

A 20. Let a_0, a_1, a_2, ... be an infinite sequence of real numbers satisfying the equation $a_n = |a_{n+1} - a_{n+2}|$ for all $n \geq 0$, where a_0 and a_1 are two different positive reals. Can this sequence a_0, a_1, a_2, ... be bounded?

IMO Shortlist 2004 A2
Proposed by Mihai Bălună (Romania)

A 21. Does there exist a function $s\colon \mathbb{Q} \to \{-1, 1\}$ such that if x and y are distinct rational numbers satisfying $xy = 1$ or $x + y \in \{0, 1\}$, then $s(x)s(y) = -1$?

IMO Shortlist 2004 A3
Proposed by Dan Brown (Canada)

A 22. Find all polynomials $f \in \mathbb{R}[x]$ such that
$$f(a-b)+f(b-c)+f(c-a) = 2f(a+b+c)$$
for all $a, b, c \in \mathbb{R}$ with $ab + bc + ca = 0$.

IMO Shortlist 2004 A4, IMO 2004 Problem 2
Proposed by Hojoo Lee (Korea)

A 23. Prove that, for all $a, b, c > 0$ with $ab + bc + ca = 1$,
$$\sqrt[3]{\frac{1}{a}+6b}+\sqrt[3]{\frac{1}{b}+6c}+\sqrt[3]{\frac{1}{c}+6a} \leq \frac{1}{abc}.$$

IMO Shortlist 2004 A5

A 24. Find all functions $f : \mathbb{R} \to \mathbb{R}$ such that
$$f(x^2+y^2+2f(xy)) = (f(x+y))^2.$$
for all $x, y \in \mathbb{R}$.

IMO Shortlist 2004 A6

A 25. Let $a_1, a_2, \cdots, a_n$ be positive real numbers, $n > 1$. Denote by $g_n = (a_1 a_2 \cdots a_n)^{\frac{1}{n}}$ their geometric mean, and by $A_1, A_2, \cdots, A_n$ the sequence of arithmetic means defined by
$$A_k = \frac{a_1+a_2+\cdots+a_k}{k}, \qquad k = 1, 2, \cdots, n.$$
Let G_n be the geometric mean of $A_1, A_2, \cdots, A_n$. Prove the inequality
$$n\left(\frac{G_n}{A_n}\right)^{\frac{1}{n}} + \frac{g_n}{G_n} \leq n+1.$$
and establish the cases of equality.

IMO Shortlist 2004 A7
Proposed by Finbarr Holland (Ireland)

IMO 2005 Mexico.

A 26. Find all pairs of integers a, b for which there exists a polynomial $P(x) \in \mathbb{Z}[X]$ such that

$$(x^2 + ax + b)\,P(x) = x^n + c_{n-1}x^{n-1} + \cdots + c_1x + c_0,$$

where $c_0, c_1, \cdots, c_{n-1} \in \{-1, 1\}$.

IMO Shortlist 2005 A1

A 27. Find all functions $f : \mathbb{R}^+ \to \mathbb{R}^+$ such that

$$f(x)f(y) = 2f(x + yf(x))$$

for all $x, y > 0$.

IMO Shortlist 2005 A2, Germany Team Selection Test 2006
Proposed by Nikolai Nikolov (Bulgaria)

A 28. Four real numbers p, q, r, s satisfy $p + q + r + s = 9$ and $p^2 + q^2 + r^2 + s^2 = 21$. Prove that there exists a permutation (a, b, c, d) of (p, q, r, s) such that $ab - cd \geq 2$.

IMO Shortlist 2005 A3

A 29. Find all functions $f : \mathbb{R} \to \mathbb{R}$ such that

$$f(x + y) + f(x)f(y) = f(xy) + 2xy + 1$$

for all $x, y \in \mathbb{R}$.

IMO Shortlist 2005 A4, Germany Team Selection Test 2006
Proposed by B. J. Venkatachala (India)

A 30. Prove that, for all $x, y, z > 0$ with $xyz \geq 1$,

$$\frac{x^5 - x^2}{x^5 + y^2 + z^2} + \frac{y^5 - y^2}{x^2 + y^5 + z^2} + \frac{z^5 - z^2}{x^2 + y^2 + z^5} \geq 0.$$

IMO Shortlist 2005 A5, IMO 2005 Problem 3
Proposed by Hojoo Lee (Korea)

IMO 2006 Slovenia.

A 31. A sequence of real numbers $a_0,\ a_1,\ a_2,\cdots$ is defined by the formula

$$a_{i+1} = \lfloor a_i \rfloor \langle a_i \rangle$$

for all $i \in \mathbb{N}_0$. Here, a_0 is an arbitrary real number, $\lfloor a_i \rfloor$ denotes the greatest integer not exceeding a_i, and $\langle a_i \rangle = a_i - \lfloor a_i \rfloor$. Prove that $a_i = a_{i+2}$ for all sufficiently large integers i.

IMO Shortlist 2006 A1
Proposed by Harmel Nestra (Estionia)

A 32. The sequence of real numbers $a_0, a_1, a_2, \ldots$ is defined recursively by

$$a_0 = -1, \qquad \sum_{k=0}^{n} \frac{a_{n-k}}{k+1} = 0 \quad \text{for} \quad n \geq 1.$$

Show that $a_n > 0$ for all $n \geq 1$.

IMO Shortlist 2006 A2
Proposed by Mariusz Skalba (Poland)

A 33. The sequence $c_0, c_1, ..., c_n, ...$ is defined by $c_0 = 1, c_1 = 0$, and $c_{n+2} = c_{n+1} + c_n$ for $n \geq 0$. Consider the set S of ordered pairs (x, y) for which there is a finite set J of positive integers such that $x = \sum_{j \in J} c_j$, $y = \sum_{j \in J} c_{j-1}$. Prove that there exist real numbers α, β, and M with the following property: An ordered pair of nonnegative integers (x, y) satisfies the inequality

$$m < \alpha x + \beta y < M$$

if and only if $(x, y) \in S$.

IMO Shortlist 2006 A3

A 34. Prove that

$$\sum_{i<j} \frac{a_i a_j}{a_i + a_j} \leq \frac{n}{2(a_1 + a_2 + \cdots + a_n)} \sum_{i<j} a_i a_j$$

for all $a_1, a_2, \cdots, a_n > 0$.

IMO Shortlist 2006 A4
Proposed by Dusan Dukic (Serbia)

A 35. Let a, b, c be the lengths of the sides of a triangle. Prove that

$$\frac{\sqrt{b+c-a}}{\sqrt{b}+\sqrt{c}-\sqrt{a}} + \frac{\sqrt{c+a-b}}{\sqrt{c}+\sqrt{a}-\sqrt{b}} + \frac{\sqrt{a+b-c}}{\sqrt{a}+\sqrt{b}-\sqrt{c}} \leq 3.$$

IMO Shortlist 2006 A5
Proposed by Hojoo Lee (Korea)

A 36. Determine the least real number M such that the inequality

$$|ab(a^2 - b^2) + bc(b^2 - c^2) + ca(c^2 - a^2)| \leq M(a^2 + b^2 + c^2)^2$$

holds for all $a, b, c \in \mathbb{R}$.

IMO Shortlist 2006 A6, IMO 2006 Problem 3

IMO 2007 Vietnam.

A 37. Given $a_1, a_2, \cdots, a_n \in \mathbb{R}$, define $d_i = \max\{a_j \mid 1 \le j \le i\} - \min\{a_j \mid i \le j \le n\}$ and $d = \max\{d_i \mid 1 \le i \le n\}$.

(1) Prove that, for all $x_1, x_2, \cdots, x_n \in \mathbb{R}$ with $x_1 \le x_2 \le \cdots \le x_n$,

$$\max\{\, |x_i - a_i| \mid 1 \le i \le n\} \ge \frac{d}{2}.$$

(2) Show that there exist $x_1, x_2, \cdots, x_n \in \mathbb{R}$ such that $x_1 \le x_2 \le \cdots \le x_n$ and

$$\max\{|x_i - a_i| \mid 1 \le i \le n\} = \frac{d}{2}.$$

IMO Shortlist 2007 A1
Proposed by Michael Albert (New Zealand)

A 38. Consider those functions $f : \mathbb{N} \to \mathbb{N}$ such that

$$f(m+n) \ge f(m) + f(f(n)) - 1$$

for all $m, n \in \mathbb{N}$. Find all possible values of $f(2007)$.

IMO Shortlist 2007 A2, Germany Team Selection Test 2008
Proposed by Nikolai Nikolov (Bulgaria)

A 39. Let n be a positive integer, and let x and y be a positive real number such that $x^n + y^n = 1$. Prove that

$$\left(\sum_{k=1}^{n} \frac{1+x^{2k}}{1+x^{4k}}\right)\left(\sum_{k=1}^{n} \frac{1+y^{2k}}{1+y^{4k}}\right) < \frac{1}{(1-x)(1-y)}.$$

IMO Shortlist 2007A3
Proposed by Juhan Aru (Estonia)

A 40. Find all functions $f : \mathbb{R}^+ \to \mathbb{R}^+$ such that

$$f\left(x + f\left(y\right)\right) = f\left(x+y\right) + f\left(y\right)$$

for all $x, y > 0$.

IMO Shortlist 2007 A4
Proposed by Paisan Nakmahachalasint (Thailand)

A 41. Let $a_1, a_2, \cdots$ be a sequence of nonnegative real numbers such that $a_{m+n} \le 2a_m + 2a_n$ for all $n \in \mathbb{N}$. Suppose that there exists a constant $c > 2$ such that $a_{2^k} \le \frac{1}{(k+1)^c}$ for all $k \in \mathbb{N}_0$. Prove that the sequence $a_1, a_2, \cdots$ is bounded.

IMO Shortlist 2007 A5

A 42. Let $a_1, a_2, \cdots, a_{100}$ be nonnegative real numbers such that ${a_1}^2 + {a_2}^2 + \cdots + {a_{100}}^2 = 1$. Prove that

$${a_1}^2 a_2 + {a_2}^2 a_3 + \cdots + {a_{100}}^2 a_1 < \frac{12}{25}.$$

IMO Shortlist 2007 A6
Proposed by Marcin Kuczma (Poland)

A 43. Let n be a positive integer. Consider $S = \{\, (x, y, z) \mid x, y, z \in \{0, 1, \cdots, n\}, x+y+z > 0 \,\}$ as a set of $(n+1)^3 - 1$ points in the three-dimensional space. Determine the smallest possible number of planes, the union of which contains S but does not include $(0, 0, 0)$.

IMO Shortlist 2007 A7
Proposed by Gerhard Wöginger (Netherlands)

IMO 2008 Spain.

A 44. Find all functions $f : \mathbb{R}^+ \to \mathbb{R}^+$ such that, for all $w, x, y, z > 0$ with $wx = yz$,

$$\frac{(f(w))^2 + (f(x))^2}{f(y^2) + f(z^2)} = \frac{w^2 + x^2}{y^2 + z^2}.$$

IMO Shortlist 2008 A1, IMO 2008 Problem 4
Proposed by Hojoo Lee (Korea)

A 45. Prove that, for all $x, y, z \in \mathbb{R} - \{1\}$ such that $xyz = 1$,

$$\frac{x^2}{(x-1)^2} + \frac{y^2}{(y-1)^2} + \frac{z^2}{(z-1)^2} \geq 1.$$

Prove that equality holds for infinitely many triples of numbers $x, y, z \in \mathbb{Q} - \{1\}$ with $xyz = 1$.

IMO Shortlist 2008 A2
Proposed by Walther Janous (Austria)

A 46. Let $S \subseteq \mathbb{R}$ be a set of real numbers. We say that a pair (f, g) of functions from S into S is a *Spanish Couple* on S, if they satisfy the following conditions.

(1) Both functions are strictly increasing, i.e. $f(x) < f(y)$ and $g(x) < g(y)$ for all $x,\, y \in S$ with $x < y$.
(2) The inequality $f\left(g\left(g\left(x\right)\right)\right) < g\left(f\left(x\right)\right)$ holds for all $x \in S$.

Decide whether there exists a Spanish Couple (1) on the set $S = \mathbb{N}$ and (2) on the set $S = \{a - \frac{1}{b} \mid a, b \in \mathbb{N}\}$.

IMO Shortlist 2008 A3, Germany Team Selection Test 2009
Proposed by Hans Zantema (Netherlands)

A 47. For an integer m, denote by $t(m)$ the unique number in $\{1, 2, 3\}$ such that $m + t(m)$ is a multiple of 3. When a function $f : \mathbb{Z} \to \mathbb{Z}$ satisfies the following conditions, Prove that $f(3p) \geq 0$ for all integers $p \geq 0$.

- $f(-1) = 0,\ f(0) = 1,\ f(1) = -1.$
- $f\left(2^n + m\right) = f\left(2^n - t(m)\right) - f(m)$ for all integers $m, n \in \mathbb{Z}$ with $n \geq 0$ and $2^n > m$.

IMO Shortlist 2008 A4
Proposed by Gerhard Wöginger (Austria)

A 48. Let a, b, c, d be positive real numbers such that $abcd = 1$ and $a + b + c + d > \dfrac{a}{b} + \dfrac{b}{c} + \dfrac{c}{d} + \dfrac{d}{a}$. Prove that

$$a + b + c + d < \frac{b}{a} + \frac{c}{b} + \frac{d}{c} + \frac{a}{d}.$$

IMO Shortlist 2008 A5
Proposed by Pavel Novotný (Slovakia)

A 49. Let $f : \mathbb{R} \to \mathbb{N}$ be a function such that $f\left(x + \dfrac{1}{f(y)}\right) = f\left(y + \dfrac{1}{f(x)}\right)$ for all $x, y \in \mathbb{R}$. Prove that there is a positive integer which is not a value of f.

IMO Shortlist 2008 A6
Proposed by Žymantas Darbėnas (Lithuania)

A 50. Prove that, for all a, b, c, $d > 0$,

$$\frac{(a-b)(a-c)}{a+b+c} + \frac{(b-c)(b-d)}{b+c+d} + \frac{(c-d)(c-a)}{c+d+a} + \frac{(d-a)(d-b)}{d+a+b} \geq 0.$$

IMO Shortlist 2008 A7
Proposed by Darij Grinberg, Christian Reiher (Germany)

IMO 2009 Germany.

A 51. Find the largest possible integer k, such that the following statement is true: Let 2009 arbitrary non-degenerated triangles be given. In every triangle the three sides are colored, such that one is blue, one is red and one is white. Now, for every color separately, let us sort the lengths of the sides. We obtain

$$\begin{array}{rll} & b_1 \le b_2 \le \cdots \le b_{2009} & \text{the lengths of the blue sides} \\ & r_1 \le r_2 \le \cdots \le r_{2009} & \text{the lengths of the red sides} \\ \text{and} & w_1 \le w_2 \le \cdots \le w_{2009} & \text{the lengths of the white sides} \end{array}$$

Then there exist k indices j such that we can form a non-degenerated triangle with side lengths b_j, r_j, w_j.

IMO Shortlist 2009 A1
Proposed by Michal Rolinek (Czech Republic)

A 52. Let a, b, c be positive real numbers such that $\frac{1}{a}+\frac{1}{b}+\frac{1}{c}=a+b+c$. Prove that

$$\frac{1}{(2a+b+c)^2}+\frac{1}{(a+2b+c)^2}+\frac{1}{(a+b+2c)^2}\le\frac{3}{16}.$$

IMO Shortlist 2009 A2
Proposed by Juhan Aru (Estonia)

A 53. Determine all functions $f:\mathbb{N}\to\mathbb{N}$ such that, for all positive integers a and b, there exists a non-degenerate triangle with sides of lengths a, $f(b)$ and $f(b+f(a)-1)$. (A triangle is non-degenerate if its vertices are not collinear.)

IMO Shortlist 2009 A3, IMO 2009 Problem 5
Proposed by Bruno Le Floch (France)

A 54. Prove that, for all $a,b,c>0$ with $ab+bc+ca\le 3abc$,

$$\sqrt{\frac{a^2+b^2}{a+b}}+\sqrt{\frac{b^2+c^2}{b+c}}+\sqrt{\frac{c^2+a^2}{c+a}}+3\le\sqrt{2}\left(\sqrt{a+b}+\sqrt{b+c}+\sqrt{c+a}\right).$$

IMO Shortlist 2009 A4

A 55. Let $f:\mathbb{R}\to\mathbb{R}$ be a function. Show that there exist $x,y\in\mathbb{R}$ such that

$$f\left(x-f(y)\right)>yf(x)+x.$$

IMO Shortlist 2009 A5, Germany Team Selection Test 2010
Proposed by Igor Voronovich (Belarus)

A 56. Suppose that $s_1,s_2,s_3,\cdots$ is a strictly increasing sequence of positive integers such that the sub-sequences

$$s_{s_1},\, s_{s_2},\, s_{s_3},\,\cdots \qquad \text{and} \qquad s_{s_1+1},\, s_{s_2+1},\, s_{s_3+1},\,\cdots$$

are both arithmetic progressions. Prove that the sequence $s_1,s_2,s_3,\cdots$ is itself an arithmetic progression.

IMO Shortlist 2009 A6, IMO 2009 Problem 3
Proposed by Gabriel Carroll (USA)

A 57. Find all functions $f:\mathbb{R}\to\mathbb{R}$ such that

$$f\left(xf(x+y)\right)=f\left(yf(x)\right)+x^2$$

for all $x,y\in\mathbb{R}$.

IMO Shortlist 2009 A7, Germany Team Selection Test 2010
Proposed by Japan

IMO 2010 Kazakhstan.

A 58. Find all function $f : \mathbb{R} \to \mathbb{R}$ such that

$$f(\lfloor x \rfloor y) = f(x) \lfloor f(y) \rfloor$$

for all $x, y \in \mathbb{R}$. Here, $\lfloor a \rfloor$ denotes the greatest integer not greater than a.

IMO Shortlist 2010 A1, IMO 2010 Problem 1
Proposed by Pierre Bornsztein (France)

A 59. Let $a, b, c, d \in \mathbb{R}$ with $a + b + c + d = 6$ and $a^2 + b^2 + c^2 + d^2 = 12$. Prove that

$$36 \le 4\left(a^3 + b^3 + c^3 + d^3\right) - \left(a^4 + b^4 + c^4 + d^4\right) \le 48.$$

IMO Shortlist 2010 A2
Proposed by Nazar Serdyuk (Ukraine)

A 60. Let $x_1, \cdots, x_{100}, x_{101} = x_1, x_{102} = x_2$ be nonnegative real numbers such that $x_i + x_{i+1} + x_{i+2} \le 1$ for all $i = 1, \cdots, 100$. Find the maximal possible value of the sum $S = \sum_{i=1}^{100} x_i x_{i+2}$.

IMO Shortlist 2010 A3
Proposed by Sergei Berlov, Ilya Bogdanov (Russia)

A 61. A sequence $x_1 = 1, x_2, \cdots$ satisfies $x_{2k} = -x_k, \quad x_{2k-1} = (-1)^{k+1} x_k$ for all $k \ge 1$. Prove that, for all $n \in \mathbb{N}$,

$$x_1 + x_2 + \cdots + x_n \ge 0.$$

IMO Shortlist 2010 A4
Proposed by Gerhard Wöginger (Austria)

A 62. Determine all functions $f : \mathbb{Q}^+ \mapsto \mathbb{Q}^+$ such that $f\left(f(x)^2 y\right) = x^3 f(xy)$ for all $x, y \in \mathbb{Q}^+$.

IMO Shortlist 2010 A5
Proposed by Thomas Huber (Switzerland)

A 63. Suppose that f and g are two functions defined on the set of positive integers and taking positive integer values. Suppose also that the equations $f(g(n)) = f(n) + 1$ and $g(f(n)) = g(n) + 1$ hold for all positive integers. Prove that $f(n) = g(n)$ for all $n \in \mathbb{N}$.

IMO Shortlist 2010 A6
Proposed by Alex Schreiber (Germany)

A 64. Let $a_1, a_2, a_3, \cdots$ be a sequence of positive real numbers, and s be a positive integer, such that

$$a_n = \max\{a_k + a_{n-k} \mid 1 \le k \le n-1\} \quad \text{for all} \quad n > s.$$

Prove there exist positive integers $\ell \le s$ and Nsuch that $a_n = a_\ell + a_{n-\ell}$ for all $n \ge N$.

IMO Shortlist 2010 A7, IMO 2010 Problem 6
Proposed by Morteza Saghafiyan (Iran)

A 65. Given six positive numbers a, b, c, d, e, f such that $a < b < c < d < e < f$. Let $a + c + e = S$ and $b + d + f = T$. Prove that

$$2ST > \sqrt{3(S+T)\left(S(bd + bf + df) + T(ac + ae + ce)\right)}.$$

IMO Shortlist 2010 A8
Proposed by Sungyoon Kim (Korea)

IMO 2011 Netherlands.

A 66. Given any set $A = \{a_1, a_2, a_3, a_4\}$ of four distinct positive integers, we denote the sum $a_1 + a_2 + a_3 + a_4$ by s_A. Let n_A denote the number of pairs (i, j) with $1 \le i < j \le 4$ for which $a_i + a_j$ divides s_A. Find all sets A of four distinct positive integers which achieve the largest possible value of n_A.

IMO Shortlist 2011 A1
Proposed by Fernando Campos (Mexico)

A 67. Determine all sequences $(x_1, x_2, \cdots, x_{2011})$ of positive integers, such that for every positive integer n, there exists an integer a with

$$\sum_{j=1}^{2011} j\, {x_j}^n = a^{n+1} + 1$$

IMO Shortlist 2011 A2
Proposed by Warut Suksompong (Thailand)

A 68. Determine all functions $f, g : \mathbb{R} \to \mathbb{R}$ such that

$$g(f(x + y)) = f(x) + (2x + y)g(y)$$

for all $x, y \in \mathbb{R}$.

IMO Shortlist 2011 A3
Proposed by Japan

A 69. Determine all functions $f, g : \mathbb{N} \to \mathbb{N}$ such that

$$f^{g(n)+1}(n) + g^{f(n)}(n) = f(n + 1) - g(n + 1) + 1$$

for all $n \in \mathbb{N}$. Here, $f^k(n)$ means $\underbrace{f(f(\cdots f)}_{k}(n)\cdots))$.

IMO Shortlist 2011 A4
Proposed by Bojan Bašić (Serbia)

A 70. Let $f : \mathbb{R} \to \mathbb{R}$ be a function such that

$$f(x + y) \le yf(x) + f(f(x))$$

for all $x, y \in \mathbb{R}$. Prove that $f(x) = 0$ for all $x \le 0$.

IMO Shortlist 2011 A5
Proposed by Igor Voronovich (Belarus)

A 71. Prove that for every positive integer n, the set $\{2, 3, 4, \cdots, 3n + 1\}$ can be partitioned into n triples in such a way that the numbers from each triple are the lengths of the sides of some obtuse triangle.

IMO Shortlist 2011 A6
Proposed by Canada

A 72. Let a, b and c be positive real numbers satisfying $\min(a+b, b+c, c+a) > \sqrt{2}$ and $a^2 + b^2 + c^2 = 3$. Prove that

$$\frac{a}{(b + c - a)^2} + \frac{b}{(c + a - b)^2} + \frac{c}{(a + b - c)^2} \ge \frac{3}{(abc)^2}.$$

IMO Shortlist 2011 A7
Proposed by Titu Andreescu (Saudi Arabia)

IMO 2012 Argentina.

A 73. Find all functions $f : \mathbb{Z} \to \mathbb{Z}$ such that
$$f(a)^2 + f(b)^2 + f(c)^2 = 2f(a)f(b) + 2f(b)f(c) + 2f(c)f(a)$$
for all integers $a, b, c \in \mathbb{Z}$ with $a + b + c = 0$.

IMO Shortlist 2012 A1, IMO 2012 Problem 4
Proposed by Liam Baker (South Africa)

A 74. Let $\mathbb{Z}$ and $\mathbb{Q}$ be the sets of integers and rationals respectively.

(1) Does there exist a partition of $\mathbb{Z}$ into three non-empty subsets A, B, C such that the sets $A+B, B+C, C+A$ are disjoint?
(2) Does there exist a partition of $\mathbb{Q}$ into three non-empty subsets A, B, C such that the sets $A+B, B+C, C+A$ are disjoint?

Here, $X + Y$ denotes the set $\{x + y : x \in X, y \in Y\}$, for $X, Y \subseteq \mathbb{Z}$ and for $X, Y \subseteq \mathbb{Q}$.

IMO Shortlist 2012 A2

A 75. Let $n \geq 3$ be an integer, and let $a_2, a_3, \cdots, a_n$ be positive real numbers such that $a_2 a_3 \cdots a_n = 1$. Prove that
$$(1 + a_2)^2 (1 + a_3)^3 \cdots (1 + a_n)^n > n^n.$$

IMO Shortlist 2012 A3
Proposed by Angelo Di Pasquale (Australia)

A 76. Let f and g be two nonzero polynomials with integer coefficients and $\deg f > \deg g$. Suppose that for infinitely many primes p the polynomial $pf + g$ has a rational root. Prove that f has a rational root.

IMO Shortlist 2012 A4

A 77. Find all functions $f : \mathbb{R} \to \mathbb{R}$ satisfying the following conditions.

- $f(-1) \neq 0$.
- $f(1 + xy) - f(x + y) = f(x)f(y)$ for all $x, y \in \mathbb{R}$.

IMO Shortlist 2012 A5

A 78. Let $f : \mathbb{N} \to \mathbb{N}$ be a function. Suppose that for every $n \in \mathbb{N}$ there exists a $k \in \mathbb{N}$ such that $f^{2k}(n) = n + k$, and let k_n be the smallest such k. Prove that the sequence $k_1, k_2, \cdots$ is unbounded.

IMO Shortlist 2012 A6

A 79. We say that a function $f : \mathbb{R}^k \to \mathbb{R}$ is a metapolynomial if, for some positive integers m and n, it can be represented in the form
$$f(x_1, \cdots, x_k) = \max_{i=1,\cdots,m} \min_{j=1,\cdots,n} P_{i,j}(x_1, \cdots, x_k),$$
where $P_{i,j}$ are multivariate polynomials. Prove that the product of two metapolynomials is also a metapolynomial.

IMO Shortlist 2012 A7

IMO 2013 Colombia.

A 80. Let n be a positive integer and let $a_1, \cdots, a_{n-1}$ be arbitrary real numbers. Define the sequences $u_0, \cdots, u_n$ and $v_0, \cdots, v_n$ inductively by $u_0 = u_1 = v_0 = v_1 = 1$, and $u_{k+1} = u_k + a_k u_{k-1}$, $v_{k+1} = v_k + a_{n-k} v_{k-1}$ for $k = 1, \cdots, n-1$.

IMO Shortlist 2013 A1

A 81. Prove that in any set of 2000 distinct real numbers there exist two pairs $a > b$ and $c > d$ with $a \neq c$ or $b \neq d$, such that

$$\left| \frac{a-b}{c-d} - 1 \right| < \frac{1}{100000}.$$

IMO Shortlist 2013 A2

A 82. Let $f : \mathbb{Q}^+ \to \mathbb{R}$ be a function satisfying the following conditions.

- $f(x)f(y) \geq f(xy)$ for all $x, y \in \mathbb{Q}^+$.
- $f(x+y) \geq f(x) + f(y)$ for all $x, y \in \mathbb{Q}^+$.
- There exists a rational number $a > 1$ such that $f(a) = a$.

Prove that $f(x) = x$ for all $x \in \mathbb{Q}^+$.

IMO Shortlist 2013 A3, IMO 2013 Problem 5
Proposed by Bulgaria

A 83. Let n be a positive integer, and consider a sequence $a_1, a_2, \cdots, a_n$ of positive integers. Extend it periodically to an infinite sequence $a_1, a_2, \cdots$ by defining $a_{n+i} = a_i$ for all $i \geq 1$. If

$$a_1 \leq a_2 \leq \cdots \leq a_n \leq a_1 + n$$

and

$$a_{a_i} \leq n + i - 1 \quad \text{for} \quad i = 1, 2, \cdots, n,$$

prove that

$$a_1 + \cdots + a_n \leq n^2.$$

IMO Shortlist 2013 A4

A 84. Find all functions $f : \mathbb{N}_0 \to \mathbb{N}_0$ such that

$$f(f(f(n))) = f(n+1) + 1$$

for all $n \in \mathbb{N}_0$.

IMO Shortlist 2013 A5

A 85. Let $m \neq 0$ be an integer. Find all polynomials $P(x) \in \mathbb{R}[x]$ such that

$$\left(x^3 - mx^2 + 1\right) P(x+1) + \left(x^3 + mx^2 + 1\right) P(x-1) = 2\left(x^3 - mx + 1\right) P(x)$$

for all $x \in \mathbb{R}$.

IMO Shortlist 2013 A6

IMO 2014 South Africa.

A 86. Let $a_0 < a_1 < a_2 \cdots$ be an infinite sequence of positive integers. Prove that there exists a unique integer $n \geq 1$ such that

$$a_n < \frac{a_0 + a_1 + a_2 + \cdots + a_n}{n} \leq a_{n+1}.$$

sourceIMO Shortlist 2014 A1
Proposed by Gerhard Wöginger (Austria)

A 87. Define the function $f : (0,1) \to (0,1)$ by

$$f(x) = \begin{cases} x + \frac{1}{2} & \text{if } x < \frac{1}{2} \\ x^2 & \text{if } x \geq \frac{1}{2} \end{cases}$$

Let a and b be two real numbers such that $0 < a < b < 1$. We define the sequences a_n and b_n by $a_0 = a, b_0 = b$, and $a_n = f(a_{n-1})$, $b_n = f(b_{n-1})$ for $n > 0$. Show that there exists a positive integer n such that

$$(a_n - a_{n-1})(b_n - b_{n-1}) < 0.$$

IMO Shortlist 2014 A2
Proposed by Denmark

A 88. For a sequence $x_1, x_2, \cdots, x_n$ of real numbers, we define its *price* as

$$\max_{1 \leq i \leq n} |x_1 + \cdots + x_i|.$$

Given n real numbers, Dave and George want to arrange them into a sequence with a low price. Diligent Dave checks all possible ways and finds the minimum possible price D. Greedy George, on the other hand, chooses x_1 such that $|x_1|$ is as small as possible; among the remaining numbers, he chooses x_2 such that $|x_1 + x_2|$ is as small as possible, and so on. Thus, in the i-th step he chooses x_i among the remaining numbers so as to minimise the value of $|x_1 + x_2 + \cdots x_i|$. In each step, if several numbers provide the same value, George chooses one at random. Finally he gets a sequence with price G. Find the least possible constant c such that for every positive integer n, for every collection of n real numbers, and for every possible sequence that George might obtain, the resulting values satisfy the inequality $G \leq cD$.

IMO Shortlist 2014 A3
Proposed by Georgia

A 89. Determine all functions $f : \mathbb{Z} \to \mathbb{Z}$ such that

$$f\big(f(m) + n\big) + f(m) = f(n) + f(3m) + 2014$$

for all $m, n \in \mathbb{Z}$.

IMO Shortlist 2014 A4
Proposed by Netherlands

A 90. Consider all polynomials $P(x) \in \mathbb{R}[x]$ that have the following property: for all $x, y \in \mathbb{R}$, one has

$$|y^2 - P(x)| \leq 2|x| \quad \text{if and only if} \quad |x^2 - P(y)| \leq 2|y|.$$

Determine all possible values of $P(0)$.

IMO Shortlist 2014 A5
Proposed by Belgium

A 91. Find all functions $f : \mathbb{Z} \to \mathbb{Z}$ such that

$$n^2 + 4f(n) = f(f(n))^2$$

for all $n \in \mathbb{Z}$.

IMO Shortlist 2014 A6
Proposed by Sahl Khan (United Kingdom)

IMO 2015 Thailand.

A 92. Suppose that a sequence $a_1, a_2, \cdots$ of positive real numbers satisfies

$$a_{k+1} \geq \frac{ka_k}{{a_k}^2 + (k-1)}$$

for all $k \in \mathbb{N}$. Prove that $a_1 + a_2 + \cdots + a_n \geq n$ for all $n \geq 2$.

IMO Shortlist 2015 A1

A 93. Determine all functions $f : \mathbb{Z} \to \mathbb{Z}$ such that

$$f(x - f(y)) = f(f(x)) - f(y) - 1$$

for all $x, y \in \mathbb{Z}$.

IMO Shortlist 2015 A2, Greece Team Selection Test 2016

A 94. Let n be a fixed positive integer. Find the maximum possible value of

$$\sum_{1 \leq r < s \leq 2n} (s - r - n)x_r x_s,$$

where $-1 \leq x_i \leq 1$ for all $i = 1, \cdots, 2n$.

IMO Shortlist 2015 A3

A 95. Determine all functions $f : \mathbb{R} \to \mathbb{R}$ that satisfy the equation

$$f(x + f(x + y)) + f(xy) = x + f(x + y) + yf(x)$$

for all $x, y \in \mathbb{R}$.

IMO Shortlist 2015 A4, IMO 2015/5
Proposed by Dorlir Ahmeti (Albania)

A 96. Let $2\mathbb{Z} + 1$ denote the set of odd integers. Find all functions $f : \mathbb{Z} \to 2\mathbb{Z} + 1$ such that

$$f\left(x + f(x) + y\right) + f\left(x - f(x) - y\right) = f(x + y) + f(x - y)$$

for all $x, y \in \mathbb{Z}$.

IMO Shortlist 2015 A5

A 97. Let n be a fixed integer with $n \geq 2$. We say that two polynomials P and Q with real coefficients are *block-similar* if for each $i \in \{1, 2, \cdots, n\}$ the sequences

$$P(2015i), P(2015i - 1), \cdots, P(2015i - 2014) \quad \text{and}$$
$$Q(2015i), Q(2015i - 1), \cdots, Q(2015i - 2014)$$

are permutations of each other.

(1) Prove that there exist distinct block-similar polynomials of degree $n + 1$.
(2) Prove that there do not exist distinct block-similar polynomials of degree n.

IMO Shortlist 2015 A6
Proposed by David Arthur (Canada)

IMO 2016 Hong Kong.

A 98. Let a, b, c be positive real numbers such that $\min(ab, bc, ca) \geq 1$. Prove that

$$\sqrt[3]{(a^2+1)(b^2+1)(c^2+1)} \leq \left(\frac{a+b+c}{3}\right)^2 + 1.$$

IMO Shortlist 2016 A1

A 99. Find the smallest constant $C > 0$ for which the following statement holds: among any five positive real numbers a_1, a_2, a_3, a_4, a_5 (not necessarily distinct), one can always choose distinct subscripts i, j, k, l such that

$$\left|\frac{a_i}{a_j} - \frac{a_k}{a_l}\right| \leq C.$$

IMO Shortlist 2016 A2

A 100. Find all positive integers n such that the following statement holds: Suppose real numbers a_1, a_2, $\cdots$, a_n, b_1, b_2, $\cdots$, b_n satisfy $|a_k| + |b_k| = 1$ for all $k = 1, \cdots, n$. Then there exists ε_1, ε_2, $\cdots$, ε_n, each of which is either -1 or 1, such that

$$\left|\sum_{i=1}^{n} \varepsilon_i a_i\right| + \left|\sum_{i=1}^{n} \varepsilon_i b_i\right| \leq 1.$$

IMO Shortlist 2016 A3

A 101. Find all functions $f : (0, \infty) \to (0, \infty)$ such that

$$xf(x^2)f(f(y)) + f(yf(x)) = f(xy)\left(f(f(x^2)) + f(f(y^2))\right)$$

for all $x, y > 0$.

IMO Shortlist 2016 A4

A 102. Consider fractions $\frac{a}{b}$ where a and b are positive integers. (a) Prove that for every positive integer n, there exists such a fraction $\frac{a}{b}$ such that $\sqrt{n} \leq \frac{a}{b} \leq \sqrt{n+1}$ and $b \leq \sqrt{n}+1$. (b) Show that there are infinitely many positive integers n such that no such fraction $\frac{a}{b}$ satisfies $\sqrt{n} \leq \frac{a}{b} \leq \sqrt{n+1}$ and $b \leq \sqrt{n}$.

IMO Shortlist 2016 A5

A 103. The equation

$$(x-1)(x-2)\cdots(x-2016) = (x-1)(x-2)\cdots(x-2016)$$

is written on the board, with 2016 linear factors on each side. What is the least possible value of k for which it is possible to erase exactly k of these 4032 linear factors so that at least one factor remains on each side and the resulting equation has no real solutions?

IMO Shortlist 2016 A6, IMO 2016 Problem 5

A 104. Find all functions $f : \mathbb{R} \to \mathbb{R}$ satisfying the following conditions.

- $f(0) \neq 0$.
- $f(x+y)^2 = 2f(x)f(y) + \max\left\{f(x^2+y^2), f(x^2) + f(y^2)\right\}$ for all $x, y \in \mathbb{R}$.

IMO Shortlist 2016 A7

A 105. Find the largest real constant a such that for all $n \geq 1$ and for all real numbers $x_0, x_1, \cdots, x_n$ satisfying $0 = x_0 < x_1 < x_2 < \cdots < x_n$ we have

$$\frac{1}{x_1 - x_0} + \frac{1}{x_2 - x_1} + \cdots + \frac{1}{x_n - x_{n-1}} \geq a\left(\frac{2}{x_1} + \frac{3}{x_2} + \cdots + \frac{n+1}{x_n}\right)$$

IMO Shortlist 2016 A8

It is impossible to be a mathematician without being a poet in soul. – Sofia Kovalevskaya